

Animal Health

Level-III

**Based on March 2022, Version IV Occupational
standard**



**Module Title: - Identifying and Handling Infectious
Diseases of Animals**

LG Code: AGR ANH3 M01LO (1-4) LG (1-4)

TTLM Code: AGR ANH3 M01 TTLM 0523v1

May, 2023

Addis Ababa, Ethiopia

Table of Contents

Introduction to the Module.....	1
LO#1- Develop basic knowledge on veterinary microbiology	2
Instruction sheet 1	2
Information Sheet 1	3
Self-check 1.....	9
LO #2-Identify and handle Bacterial disease of animals.....	10
Instruction sheet 2	10
Information Sheet 2	12
Self-Check 2.....	70
LO3 # - Identify and handle viral disease of animals	72
Instruction sheet 3	72
Information Sheet 3	73
Self-Check – 3.....	114
LO4 # Identify and handle fungal disease of animals.....	115
Instruction sheet 4	115
Information sheet 4.....	116
Self-Check – 4.....	125
Reference Materials	127

Introduction to the Module

This module covers the knowledge and skills and attitude required to develop basic understanding on biology, characteristic feature and, clinical/economic/public health impacts of infectious microorganisms. It also requires the trainees' ability to classify, characterize and identify microorganisms which are threat of animal health through the use clinical examination and microbiological laboratory examination. In addition this competency covers trainees' skill to treat and prevent/control important and more prevalent microbial diseases animals.

Page 1 of 132	Ministry of Labor and Skills Author/Copyright	Animal Health Level -3	Version -1
			May, 2023

LG #1	LO#1- Develop basic knowledge on veterinary microbiology
-------	---

Instruction sheet 1
This learning guide is developed to provide you the necessary information regarding the following content coverage and topics: • Introduction to microbiology and infectious microorganisms • Categorization of infectious microorganisms • Current status and economic importance of infectious animal disease • Public health significance of zoonotic microbial agents This guide will also assist you to attain the learning outcomes stated in the cover page. Specifically, upon completion of this learning guide, you will be able to: • Define the term microbiology and group infectious micro-organisms based on their biological similarity. • Explain sub classes of infectious microorganisms and their characteristics • Explain Statistical information on the current distribution and its implication of the disease in Ethiopian livestock. • Explain Public health significance of zoonotic microbial agents
Learning Instructions:
1. Read the specific objectives of this Learning Guide. 2. Follow the instructions described below and try to understand what are being discussed. Ask your teacher for assistance if you have hard time understanding them. 3. Read the information written in the information Sheets 4. Accomplish the Self-checks

Information Sheet 1

1.1 Introduction to microbiology and infectious microorganisms

Microbiology: is the study of microorganisms (also known as microbes), a collection of organisms visible only with a microscope.

Infectious microorganisms: Are microbes that Cause Infectious Diseases.

Microorganism is simply very, very small organism that you cannot see with your naked eye, but with a microscope Microorganisms are the scientific name for what most people refer to as “germs.” The term microorganism has a neutral connotation, whereas “germ” has a negative connotation and generally refers to a pathogen (something capable of causing disease).

No one is sure who made the first observations of microorganisms, but the microscope was available by the mid- 1600s, and an English scientist named **Robert Hooke** made observations of cells in slices of cork tissue. He also observed strands of fungi among the specimens he viewed. Beginning in the 1670s, a Dutch merchant named Anton van **Leeuwenhoek** made careful observations of microscopic organisms which he Called **animalcules**. Among his descriptions were those of protozoa, fungi, and various kinds of bacteria. **Leeuwenhoek** is regarded as one of the first to provide accurate descriptions of the world of microorganisms. Not all microorganisms involved in infectious disease. Some microorganisms produce many of the foods we eat, including fermented dairy products (sour cream, yogurt, and buttermilk), as well as fermented foods such as pickles, sauerkraut, breads, and alcoholic beverages.

1.2 Biological base categorization of infectious microorganisms

Terminology

- **Eukaryotic:** are cells containing membrane-bound organelles and are the basis for both unicellular and multicellular organisms.
- **Prokaryotic:** are cells do not have any membrane-bound organelles and are always parts of unicellular organisms.
- **Tropism:** refers to the ability of a given pathogen to infect a specific location.

- **Bacteria:** Bacteria are single-celled microorganisms that can exist in a variety of environments.
- **Virus:** Viruses are small infectious agents that can only replicate inside the living cells of other organisms.
- **Fungus:** Fungi are a diverse group of eukaryotic organisms that include yeasts, molds, and mushrooms.
- **Helminth:** Helminths are parasitic worms that can infect both humans and animals.
- **Protozoa** are a diverse group of single-celled eukaryotic microorganisms.

Microorganisms have a set of characteristics that place them in either of the two major groups of organisms: prokaryotes and eukaryotes. Certain microorganisms such as bacteria and archaea are prokaryotes because of their cellular properties, while other microorganisms such as fungi, protozoa, and unicellular algae are eukaryotes. There are **five** major types of infectious agents: bacteria, viruses, fungi, helminthes, and protozoa. In addition, a new class of infectious agents, the **prions**, has recently been recognized.

I. Bacteria

- **Biological and genetic makeup:** Bacteria are single-celled organisms that lack a nucleus and other membrane-bound organelles. They have a circular strand of DNA as genetic material in their cytoplasm.
- **Reproduction and growth characteristics:** Most bacteria reproduce through binary fission which results in two identical daughter cells. They require suitable environmental conditions such as sufficient nutrients, oxygen, and suitable temperature to reproduce.
- **Nutritional requirement:** Bacteria can be autotrophic or heterotrophic. Some can produce their own food from inorganic substances. Others require organic compounds as nutrients. **Morphology:** Bacteria can have different shapes such as spheres (cocci), rods (bacilli), or spirals (spirilla).
- **Tissue tropism and host preference:** Bacteria can infect a wide range of host including humans, plants, and animals and can affect different tissues in their hosts.
- **Disease development:** Bacteria can cause a variety of diseases

II. Virus

- **Biological and genetic makeup:** Viruses are tiny infectious agents consisting of genetic material (DNA or RNA) surrounded by a protein coat called a capsid. They can only replicate inside their host cells.
- **Reproduction and growth characteristics:** Viruses use their host cell's metabolic machinery to reproduce themselves by hijacking the host's DNA replication machinery. They require a suitable host to replicate.
- **Nutritional requirement:** Viruses do not metabolize or produce their own nutrients.
Morphology: Virus structures can vary, but they usually have a spherical or long cylindrical shape.
- **Tissue tropism and host preference:** Viruses have strict host ranges that usually determine their tissue tropism. Different viruses have different preferred hosts; for example, the influenza virus affects humans, birds, and pigs.
- **Disease development:** Viruses can cause various diseases such as the common cold, influenza or
- The study of viruses is called virology.

III. Fungi

- **Biological and genetic makeup:** Fungi are eukaryotic organisms with a nucleus and membrane-bound organelles. They have cell walls made of chitin and use hyphae to grow.
- **Reproduction and growth characteristics:** Can reproduce both sexually and asexually by producing spores. They require a suitable substrate or host to grow.
- **Nutritional requirement:** Fungi are heterotrophs and usually digest their food externally by releasing enzymes that break down complex molecules into simple compounds that they can absorb
- **Morphology:** Fungi have various shapes, ranging from single-celled yeasts to multicellular mushrooms.
- **Tissue tropism and host preference:** Fungi can infect a wide range of hosts including humans, animals, and plants and can affect different tissues in their hosts, depending on their species.

- **Disease development:** Fungal infections can cause a wide range of diseases from mild skin infections (such as ringworm) to lethal systemic infections (such as invasive aspergillosis). The study of fungi is called Mycology

IV. Helminth:

- Are parasitic worms that can infect both humans and animals. There are many different types of helminths, including nematodes (roundworms), cestodes (tapeworms), and trematodes (flukes). Helminth infections can cause a range of symptoms, from mild discomfort to severe illness

V. Protozoa

- Are unicellular, eukaryotic, heterotrophic organisms. They are either free-living or parasites. They lack a cell wall. There are many different cell organelles that perform various tasks performed by different organs in higher animals, e.g. mouth, anus, intestinal tract.

1.3 Economic importance of infectious animal disease

Infectious diseases are caused by microorganisms such as viruses, bacteria, fungi or parasites and can spread between individuals. Microorganisms that cause disease are collectively called pathogens. Ethiopia is endemic to a number of livestock diseases, including OIE-listed diseases such as contagious bovine pleuropneumonia (CBPP), lumpy skin disease (LSD), foot and mouth disease (FMD), Newcastle disease (NCD), Peste des petits ruminants (PPR), sheep and goat pox (SGP), and African horse sickness (AHS).

Infectious animal diseases can have a **significant economic impact** on agriculture and the broader economy. Here are a few ways in which these diseases can affect economic activity:

- **Loss of livestock:** Infectious animal diseases can lead to the death of livestock or require them to be culled to prevent further spread of the disease. This can have a significant impact on agricultural production, especially for farms that rely on livestock as their primary source of income.
- **Reduced productivity:** Even if animals don't die from an infectious disease, they may still suffer reduced productivity due to illness or the need for quarantine. This can lead

to lower yields and increased costs for farmers who need to provide additional care and treatment.

- **Trade restrictions:** When an infectious animal disease is detected in a country or region, other countries may impose trade restrictions on that area's livestock or animal products. This can lead to reduced export earnings and decreased demand for local products, which can hurt farmers and related businesses.
- **Control measures:** Controlling the spread of an infectious disease often requires significant resources, including testing, surveillance, treatment, vaccination, and public awareness campaigns. These measures can be costly for governments and private businesses alike.
- **Zoonotic potential:** Some infectious animal diseases can also infect humans, posing a public health threat. This can lead to increased healthcare costs and lost productivity if workers become ill or need to care for sick family members. Overall, infectious animal diseases can have a substantial economic impact, both in terms of direct losses to agriculture and related industries as well as indirect costs such as trade restrictions and public health measures. Preventing and controlling these diseases is therefore an important priority for governments, farmers, and other stakeholders

1.3.1. Statistical information of Ethiopian livestock

Ethiopia has a diverse livestock population that includes over 65 million cattle, 40 million sheep, and 51million goats million goats, 8 million camels and 49 million chickens in 2020. According to the OIE, some of the most prevalent animal diseases in Ethiopia include peste des petits ruminants (PPR), foot-and-mouth disease (FMD), contagious bovine pleuropneumonia (CBPP), and Newcastle disease. African swine fever is also a significant concern in Ethiopia, especially in areas with high pig populations. Ethiopia has several government agencies and research institutions dedicated to animal health, including the Ministry of Agriculture and the National Animal Health Diagnostic and Investigation Center (NAHDIC).

1.4 Public health significance of zoonotic microbial agents

Zoonotic microbial agents are pathogenic microorganisms that can be transmitted from animals to humans causing disease. These agents pose a significant public health threat as they can cause

outbreaks of infectious diseases that can affect large human populations, leading to significant morbidity and mortality rates. Some of the most common zoonotic microbial agents include bacteria such as Salmonella and E. coli, as well as various viruses and parasites. The transmission routes for these agents can vary depending on the pathogen and the host animal involved. For example, some of these agents can be transmitted through direct contact with infected animals or their bodily fluids, while others can be transmitted through ingestion of contaminated food or water.

One important aspect of the public health significance of zoonotic microbial agents is their potential emergence and spread in new areas. As global transportation and movement of animals and animal products increases, the risk for introduction of new zoonotic agents increases. This is particularly relevant in areas with poor infrastructure or limited capacity for surveillance and control, where an outbreak of zoonotic disease can quickly become a public health emergency. Another important consideration is the impact that zoonotic disease outbreaks can have on livestock production and animal agriculture. When animal populations are affected by zoonotic disease outbreaks, it can lead to significant economic losses for farmers and producers.

To address the public health significance of zoonotic microbial agents, it is important to implement effective surveillance systems for early detection of outbreaks, as well as to develop targeted prevention and control strategies. This includes measures such as vaccination programs, implementation of biosecurity measures to prevent disease spread between animals, and appropriate use of antibiotics to prevent the development of antibiotic resistance. By taking a comprehensive approach to preventing and controlling zoonotic disease outbreaks, we can minimize the risk to human and animal health, and protect public health worldwide.

Self-check 1	Written test
--------------	--------------

Name_____ ID_____ signature_____ date_____

Directions: Answer all the questions accordingly.

Part I: write **true** if the statement is correct and false if the statement is incorrect

1. **Not** all microorganisms involved in infectious disease.
2. Zoonotic microbial agents are pathogenic microorganisms that can be transmitted from animals to humans causing disease.

Part II: Choose the best answer (2 point)

1. Which one is **not** true about bacteria?
 - A. it is unicellular
 - B. it is microscopic
 - C. it is Prokaryotic
 - D. it is Eukaryotic
2. among the first scientists to see microorganisms was
 - A. Robert Hooke
 - B. Louis Pasteur
 - C. Joseph Lister
 - D. James T. Watson

Part III: Short Answer Questions (4 pts each)

1. List types of microorganism with their characteristics
2. Define Microbiology, Prokaryotic, Eukaryotic,
3. Explain impact of livestock disease in Ethiopia

Note: Satisfactory rating –14 points

Unsatisfactory – 14 below points

LG #2	LO #2-Identify and handle Bacterial disease of animals
--------------	---

Instruction sheet 2

This learning guide is developed to provide you the necessary information regarding the following content coverage and topics:

- General characteristics of bacteria
- Aetiology, host, pathogenesis, clinical sign, diagnosis, treatment and prevention of endo-spore forming bacterias
- Aetiology, host, pathogenesis, clinical sign, diagnosis, treatment and prevention of Gram positive bacteria
- Aetiology, host, pathogenesis, clinical sign, diagnosis, treatment and prevention of Gram negative bacteria
- Aetiology, host, pathogenesis, clinical sign, diagnosis, treatment and prevention of Enterobacteriaceae
- Aetiology, host, pathogenesis, clinical sign, diagnosis, treatment and prevention of Mycobacterium
- Aetiology, host, pathogenesis, clinical sign, diagnosis, treatment and prevention of Mycoplasma and Rickettsia
- Identifying and handling cases of mastitis

This guide will also assist you to attain the learning outcomes stated in the cover page. Specifically, upon completion of this learning guide, you will be able to:

- Explain General characteristics of bacteria
- Identify and implement Etiology, transmission, host preference, clinical sign, diagnosis, treatment and prevention/control measures of economically important and prevalent diseases of farm animals' caused by endo-spore forming bacteria
- Identify and implement Etiology, transmission, host preference, clinical sign,

diagnosis, treatment and prevention/control measures of important and prevalent diseases of farm animals' caused by gram positive bacteria

- Identify and implement Aetiology, transmission, host preference, clinical sign, diagnosis, treatment and prevention/control measures of economically important and prevalent diseases of animals' caused by gram negative bacteria are
- Identify and implement Aetiology, transmission, host preference, clinical sign, diagnosis, treatment and prevention/control measures of economically important and prevalent diseases of animals' caused by Enterobacteriaceae
- Identify and implement Etiology, transmission, host preference, clinical sign, diagnosis, treatment and prevention/control measures of diseases of farm animals' caused by mycobacterium
- Identify and implement Etiology, transmission, host preference, clinical sign, diagnosis, treatment and prevention/control measures of disease of animals' caused by mycoplasma and rickettsia.
- Identify and apply Etiology, classification, transmission, clinical sign, diagnosis, treatment and prevention/control measures mastitis

Learning Instructions:

1. Read the specific objectives of this Learning Guide.
2. Follow the instructions described below.
3. Read the information written in the information Sheets
4. Accomplish the Self-checks

Information Sheet 2

2.1. General characteristics of bacteria

Bacteria are single-celled organisms that can be found in a wide range of environments, including soil, water, and the human body. While many bacteria are harmless or even beneficial to humans, some bacteria can cause a wide range of infections and diseases. Some of the most common bacterial infections that infect humans include strep throat, urinary tract infections, and pneumonia. Treatment for bacterial infections varies depending on the specific bacteria and the severity of the infection. Prevention measures include good hygiene practices and vaccination.

The bacteria may appear as rods (bacilli), spheres (cocci), or spirals (spirilla or spirochetes) in shape. Bacteria reproduce by binary fission, have unique cell walls, and exist in most environments on earth. They live at temperatures ranging from 0 to over 100°C and in conditions that are oxygen- rich or oxygen free.

Many bacteria are aerobic; that is, they require oxygen to grow and perform the biochemical reactions of their metabolism. There are, however, a large number of bacterial species that are anaerobic and are cultivated in a special apparatus where the oxygen has been removed. These bacteria (known as anaerobes) live in the absence of oxygen, and they are found in the muddy bottom of swamps, landfills, and soil. Some anaerobes are obligate anaerobes, while others are facultative anaerobes, meaning that they live with or without oxygen.

2.1.1 Bacterial structure

- **Cell wall:** is tough and rigid structure surrounding the bacterium like a shell containing peptidoglycan. Is structure that gives bacteria their shape. The only bacteria that have no cell wall are the mycoplasma. The bacterial cell wall contains a large molecule called peptidoglycan. **Peptidoglycan** is a polysaccharide containing **N- acetylmuramic acid** and **N- acetylglucosamine**, with numerous chains of amino acids. In Gram positive bacteria, the cell wall contains many layers of **peptidoglycan** together with another organic substance called **echoic acid**. In Gram- negative bacteria, there is only a thin layer of peptidoglycan, surrounded by an outer membrane and an inner

membrane. The peptidoglycan in Gram- negative bacteria is bonded to **lipoproteins**, which are not found in the Gram- positive cell wall. This difference in structure is responsible for the differential reaction of Gram- positive and Gram- negative cells to the Gram stain.

- **Plasma membrane:** (cytoplasmic membrane): is thin elastic semi-permeable layer and lies beneath cell wall separating from cytoplasm. It has a lipid bilayer. The cell membrane has several important functions, one of which is to provide a semipermeable barrier between the cytoplasm and the external environment. For this reason, certain substances pass through the membrane and others are held back. For example, substances that are small in size or that dissolve in lipids enter the cell more easily than ions, which do not dissolve in lipids. In addition, the cell membrane is the site of enzymes that participate in the respiration of prokaryotic cells.
- **Spores (endospores):** formed by certain bacteria when they are starved. It is commonly found in genus bacillus and clostridium.
- **Flagella:** is mobile or locomotory organelle in some bacteria. The form of flagellation varies. Some bacteria, called monotrichous bacteria, have a single flagellum, while other bacteria, called amphitrichous bacteria, have a flagellum at each end of the cell. Lophotrichous bacteria have multiple flagella at the ends of the cells, and peritrichous bacteria have flagella distributed over the entire body of the cell.
- **Capsule:** outer covering of thick jelly like material surrounding bacterial cell wall. It serves as protection and enhances virulence
- **Pilli (fimbriae):** thin short filamentous appendage extruding from cytoplasmic membrane. Fimbriae found only in some gram negative bacteria.

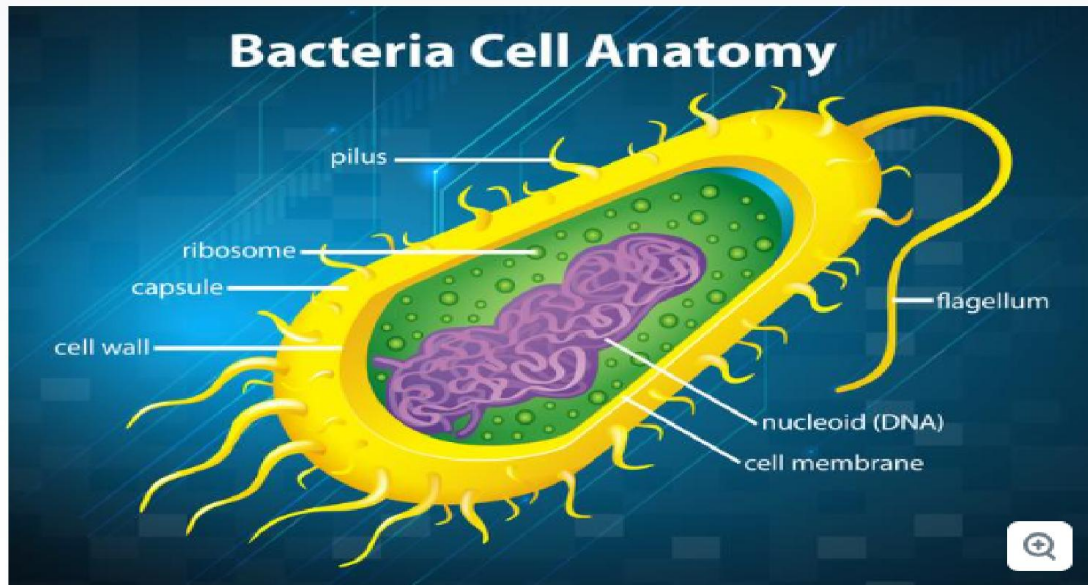


Figure 2.1. Bacterial structure

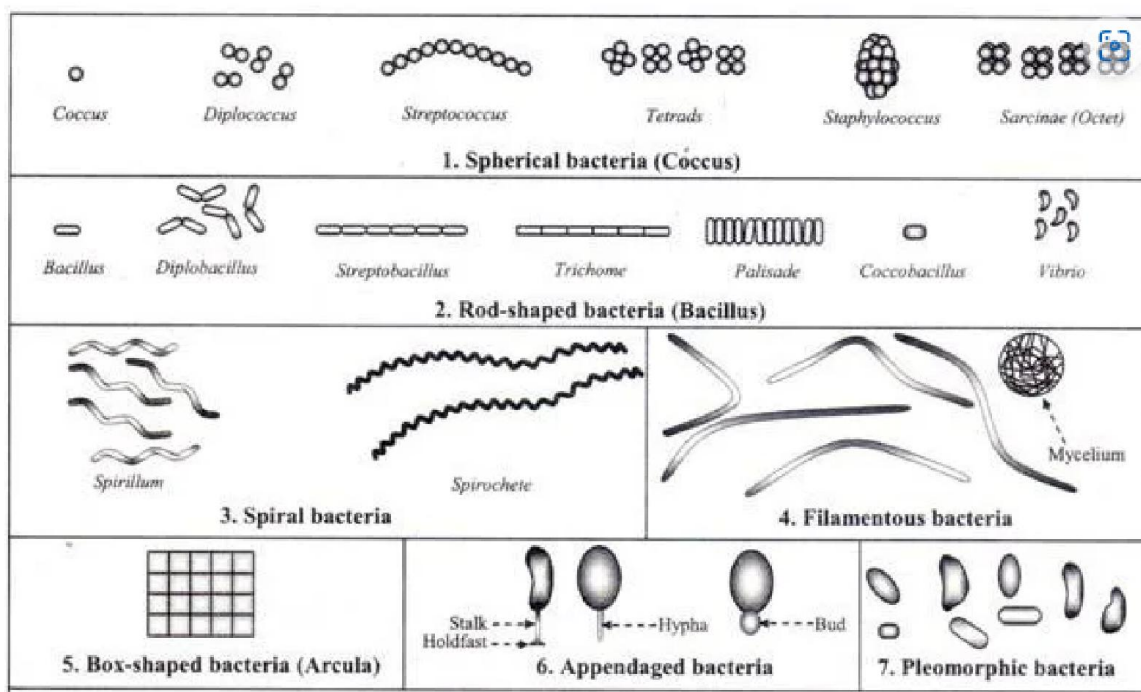


Figure 2.2. Bacterial Morphology

2.1.2 Types of bacteria

Bacteria are a type of microorganism that can be found in almost every environment on Earth. Bacteria are divided into various types based on their structure, characteristics, morphology, reproduction physiology, and metabolism. Here are some of the most common types of bacteria.

- I. Gram-positive bacteria:** These bacteria have a thick outer layer made up of a peptidoglycan layer, which makes them appear purple when stained with a crystal violet dye. Some examples of gram-positive bacteria include *Streptococcus* and *Staphylococcus*.
- II. Gram-negative bacteria:** These bacteria have a thinner outer layer of peptidoglycan, which makes them appear pink when stained with a counter stain like safranin. Some examples of gram-negative bacteria include *Escherichia coli* (E.coli) and *Salmonella*.
- III. Aerobic bacteria:** These bacteria require oxygen to grow and survive. Examples include *Pseudomonas aeruginosa* and *Mycobacterium tuberculosis*.
- IV. Anaerobic bacteria:** These bacteria can survive without oxygen and may even be killed by it. Examples include *Clostridium botulinum* and *Bacteroides fragilis*.
- V. Acid-fast bacteria:** These bacteria have a unique cell wall that makes them resistant to staining with traditional methods. Examples include *Mycobacterium leprae* and *Mycobacterium tuberculosis*.
- VI. Spore-forming bacteria:** These bacteria can form highly resistant spores that allow them to survive in harsh environments. Examples include *Bacillus anthracis* and *Clostridium tetani*.
- VII. Non-spore-forming bacteria:** These are bacteria that do not produce spores but can still survive in harsh conditions. Examples include *Neisseria gonorrhoeae* and *Streptococcus pyogenes*.

2.1.3 Bacterial diagnostic techniques

There are several diagnostic techniques used for identifying and characterizing bacterial infections. These include:

- I. Microscopy:** This technique involves using a microscope to examine samples of body fluids, tissues, or secretions for the presence of bacteria. Microscopy can help identify the shape and structure of bacteria, as well as their arrangement (e.g. clustered or chains).

II. Culture: Culturing involves growing bacteria from a sample on a nutrient-rich medium. Culturing allows for determination of the type or strain of bacteria causing the infection, and also helps identify antibiotic susceptibility

III. Polymerase Chain Reaction (PCR): PCR is a molecular technique that amplifies specific DNA sequences of bacteria to detect their presence in a sample. PCR can identify bacteria even at low levels and is useful for rapid diagnosis in cases where culturing may not be possible.

IV. Serology: This type of testing measures the immune response to a bacterial infection by detecting antibodies in a patient's blood sample. Serology is useful for diagnosing acute or chronic bacterial infections and can help determine whether a patient has already been exposed to a specific type of bacteria.

V. Biochemical tests: Biochemical tests involve measuring the metabolic capabilities of bacteria in culture to determine their species or strain. These tests measure enzyme activity, carbon source utilization, and other metabolic pathways unique to specific types of bacteria.

A combination of techniques such as microscopy, culture, and serology may be used for diagnosis of complex bacterial infections. Early detection and accurate diagnosis are crucial for appropriate antibiotic treatment and control of bacterial infections.

2.1.4 Types of Media for Bacterial Growth

There are several different types of media that can be used to support the growth of bacteria:

- **Nutrient agar:** This is a general-purpose medium that can support the growth of a wide range of bacteria. It contains nutrients such as proteins, carbohydrates, vitamins, and minerals
- **Blood agar:** This is a nutrient-rich medium that also includes red or white blood cells. It is often used to grow fastidious bacteria such as *Streptococcus pneumoniae*, *Streptococcus pyogenes*, and *Streptococcus agalactiae*.
- **MacConkey agar:** This medium contains bile salts and crystal violet, which inhibit the growth of gram-positive bacteria while allowing the growth of gram-negative bacteria. It is commonly used to isolate and identify enteric bacteria such as *Escherichia coli* and *Salmonella*.

- **Sabouraud agar:** This medium is used for the selective cultivation of fungi and yeasts due to its low pH.
- **Thioglycollate broth:** This is a liquid medium that can be used to grow both anaerobic and aerobic bacteria by creating varying levels of oxygen concentration throughout the broth.
- **Chocolate agar:** This medium is made by heating blood agar to lyse the red blood cells, releasing nutrients that enhance bacterial growth. It is often used to grow *Haemophilus influenzae*.

2.2. Diseases caused by Endospore forming bacteria

Certain species of bacteria are able to form highly resistant structures called **endospores**, also known as **spores**. Members of the genera *Bacillus* and *Clostridium* are notable for their ability to form spores. The organisms normally grow and multiply as vegetative cells, but when the environmental conditions become unfavorable, each cell reverts to a spore. Bacterial spores are among the most resistant forms of life known. They can resist boiling water temperatures, extensive drying, and the effects of disinfectants, antibiotics, and many other chemicals. To destroy spores, it is necessary to treat them in an autoclave with steam under pressure.

2.2.1 Clostridium

Clostridia are relatively large, anaerobic, spore-forming, rod-shaped, gram-positive organisms. The spores are oval, sometime spherical. The forms of Clostridia in tissue fluids occur singly, in pairs, or rarely in chains. Clostridia organisms are all potent producers of *exotoxins* upon which their pathogenicity depends on species. Pathogenic clostridia are commonly present in soils. They are also found in the intestinal contents of normal animals and cause disease only in special circumstances. Diseases caused by clostridium species are black leg (*cl. chavoei*), tetanus (*cl. tetani*), botulism (*cl. botulinum*), bacillary hemoglobinuria (*cl. septicum*)

- **Blackleg**

✓ **Synonyms:** Black quarter; symptomatic Anthrax

Blackleg is an acute infectious but not contagious disease of cattle, goat and sheep and characterized by inflammation of the muscles, severe toxemia and high mortality.

- ✓ **Etiology:** caused by *Clostridium chauvoei*, a gram-positive, spore forming and toxin producing anaerobe. The spores are highly resistant to environmental changes and disinfectants and persist in the soil for many years.
- ✓ **Occurrence:** Worldwide prevalence of disease has been recorded, and may be enzootic in certain areas, especially when they are subject to flooding. It affects cattle but occasional infection occurs in sheep and horses. The disease is common in well built cattle of below 2 years age. Blackleg is common in Ethiopia during dry periods of the year and is locally known as '*abagorba*' or '*koreba*'
- ✓ **Species Affected:** bovine are the most susceptible animals but infection may spread to sheep and goats following trauma of muscles. Horse and pig can also be infected in a very rare occasion.
- ✓ **Source of infection:** Blackleg is a soil-borne infection but the portal by which the organism enters the body is still in dispute. It is presumed that the portal of entry is through the alimentary mucosa after ingestion of contaminated feed or associated with erupting teeth. The bacteria may be found in the spleen, liver, and alimentary tract of normal animals, and contamination of the soil and pasture may occur from infected feces or decomposition of carcasses of animals dying of the disease.
- ✓ **Transmission:** In cattle the disease usually occurs without a history of trauma. The organisms enter the host through ingestion of contaminated feed and water. The disease is largely confined to young stock between the ages of 6 months and 2 years, in the field, risk factors include rapidly growing cattle and a high plane of nutrition but in sheep is almost always a wound infection. Infection of skin wounds at shearing, docking and of the navel at birth and vulva and vagina of ewe at lambing. There is no restriction of age groups in sheep.
- ✓ **Pathogenesis:** Following ingestion the latent spores they are carried by blood circulation and lodge in the skeletal muscles and visceral organs of susceptible host. In the presence of external influence of anaerobic environment like injections with irritating drugs, walking of long distances that facilitate the formation anaerobic environment in the infected muscle the spores germinate in to vegetative form and during the replication of the vegetative form the bacteria produce α -toxin. Toxins

formed by the organism produce a severe necrotizing myositis locally in skeletal muscles, and systemic toxemia which is usually fatal.

✓ **Clinical Findings:** Incubation period is usually 2 to 5 days.

- ✚ In cattle the first symptom is fever (41 °c).
- ✚ Anorexia, ruminal stasis and there is lameness in one of the limb.
- ✚ Characteristic swelling develops in one of the thick layers of muscles. Most commonly the lesions are located on the thigh, shoulder, neck and lumbar region.
- ✚ Swellings are hot and painful in the early stage and become cold and painless later.
- ✚ The muscle tissues are swollen, dark in colour and turn dry.
- ✚ On palpation swellings emit **cracking or crepitation** sound due to emphysema.
- ✚ There is also laboured breathing and abdominal pain.

✓ **Diagnosis:** Tentative diagnosis is based on history, clinical observation and post-mortem findings. In the laboratory the disease can be diagnosed by microscopic examination of smear made from the affected tissues or fluid of the swellings. Gram-positive rods with sub terminal spores will be seen.

✓ **Differential Diagnosis:** Anthrax, Malignant oedema and bacillary haemoglobinuria.

✓ **Treatment:** Satisfactory response has been reported from the use of penicillin and oxytetracycline. The antibiotic may be injected into the affected muscles. Penicillin at the dose of 10,000 units / kg body weight / day is extensively used and considered as drug of choice.

✓ **Control and prevention:** In endemic areas annual vaccination of all cattle between 6 months and 2 years of age should be carried out prior to the anticipated danger period. Immunity following vaccination doesn't develop for 14 days and deaths may continue for some days if vaccination is carried out during an outbreak. Therefore, to avoid new cases during this time antibiotics (penicillin at a dose of 6000 unit / kg body weight intramuscularly) should be administered at the time of vaccination. The carcasses of animals dying of blackleg are destroyed by burning or deep burial to limit soil contamination.

- **Tetanus**

- ✓ **Synonym:** lock jaw

It is a highly fatal, non-contagious, non-febrile infectious disease of mammals caused by toxin and characterized by **spasmodic** contraction of skeletal muscles.

- ✓ **Etiology:** *Clostridium tetani*. The organism is anaerobic and it is gram-positive bacteria. The bacteria are spore forming and are capable of persisting in the soil for many years. The spores are also resistant to many standard disinfectants.
- ✓ **Occurrence:** Tetanus occurs in all parts of the world and in all farm animals as sporadic case. The portal of entry is usually through deep puncture of wounds. A high incidence of tetanus may occur following castration, shearing, docking and vaccination for other diseases in lambs.
- ✓ **Species Affected:** There is a considerable variation in susceptibility between species, the horse being the most susceptible and cattle the least. Human beings are extremely susceptible.
- ✓ **Transmission:** Organisms may continue to live in the faeces for a long period of time and thus remain as a potential source of infection to man and other animals. The organism gains entrance through deep punctured wound contaminated with bacterial spores.
- ✓ **Pathogenesis:** Organisms may gain access during parturition and manual handling of the genitalia with contaminated hands. The tetanus bacilli remain localized at their sites of introduction and don't invade surrounding tissues. They proliferate and produce tetanolysin and tetanospasmin at lowering of the local tissue oxygen tension. Tetanolysin promotes local tissue necrosis. Tetanospasmin absorbed from the site, disseminated in the blood to produce neurotoxic effect through absorption by peripheral nerve endings. It affects centrally the spinal cord and brain and peripherally on motor nerve to produce **spastic paralysis**. Death occurs by asphyxiation due to fixation of the muscles of respiration
- ✓ **Clinical Findings:** The clinical picture is similar in all animal species.
 - ✚ A generalized increase in muscle stiffness is observed first and is accompanied by muscle tremor.
 - ✚ Prolapse of the third eyelid.

- ✚ Restriction of jaw movements - the animal may continue to eat and drink in the early stage but mastication is soon prevented by tetany of the masseter muscle and saliva may drool from the mouth.
- ✚ Stiffness of the hind legs
- ✚ Anxious and alert expression contributed by an erection of the ears, retraction of the eyelids and dilation of the nostrils and exaggerated responses to normal stimuli.
- ✚ The temperature and pulse rate are within the normal range in the early stages but may rise later when muscular tone are further increased.
- ✚ As the disease progresses, muscular tetany increases and the animal adopts a 'Sawhorse' posture.

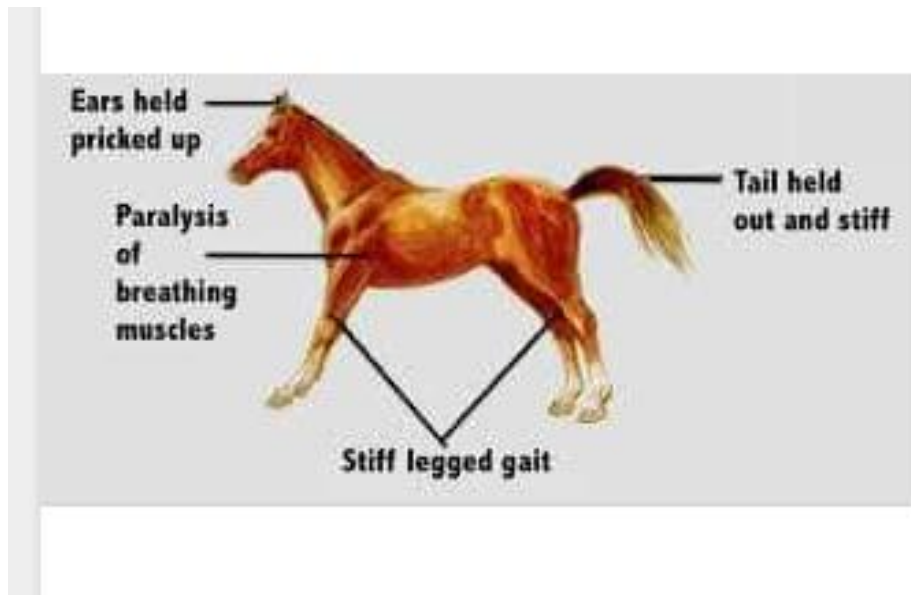


Figure 2.3. Symptoms of tetanus in equine



Figure 2.4. Protrusion of third eye lid

- ✓ **Diagnosis:** Fully developed tetanus is so characteristic clinically that it is seldom confused with other diseases. The muscular spasms, prolapse of the third eyelid and a recent history of accidental injury or surgery are characteristic findings.
- ✓ **Differential Diagnosis:** strychnine poisoning, grass tetany, milk fever, and rabies may be confused with tetanus.
- ✓ **Treatment** - The principle of treatment directs to:
 - ✚ Destruction of tetanus organisms - for this penicillin is the drug of choice and it should be given in massive doses. It may be given locally to minimize further multiplication of the bacteria. 1% H₂O₂ may be used to clean the wound.
 - ✚ Neutralization of circulating toxin - Antitoxins is used for this purpose. 3000 to 7000 i.u. of antitoxin may be used depending on the size of the animal at 12hrs interval.
 - ✚ Relaxation of muscle - muscle relaxants like chlorpromazine can be used for this purpose. This is important to prevent asphyxiation
 - ✚ Control: Proper skin and instrument disinfection during castration, docking and shearing time.

- ✚ For short term prophylaxis, passive immunity can be achieved by the injection of antitoxin subcutaneously. In endemic areas all susceptible animals should be actively immunized with toxoid formalin treated toxin. Vaccinating their mothers at late pregnancy can protect young foals and lambs.

- **Botulism**

Botulism is a rapidly fatal **motor paralysis** caused by ingestion of preformed toxin of *Clostridium botulinum*.

- ✓ **Etiology:** The causative organism is *Cl. botulinum*, a spore forming anaerobe, which proliferates only in decaying animal and plant material. Under favourable conditions of warmth and moisture the spores multiply rapidly elaborating a stable and lethal toxin and when ingested causes the disease.
- ✓ **Occurrence:** Botulism has no geographical limitations and outbreaks are most likely to occur during drought periods when feed is scarce, phosphorus intake is low and carrion is plentiful.
- ✓ **Species Affected:** Botulism is most common in birds. Cattle, sheep and horses are also susceptible, but pigs, dogs and cats appear to be resistant.
- ✓ **Transmission:** The spores are occurs commonly in soils in affected areas and soil and water contamination occurs from faeces and decomposing carcasses. The spore is extremely resistant and survives for long periods in most environmental circumstances. The toxin is also capable of surviving for long periods. The source of infection is ingestion of preformed, long surviving toxin in decayed animal or plant organic matter. Possibly, implantation of *Cl. botulinum* in tissue.
- ✓ **Pathogenesis :** preformed toxins are ingested through contaminated feed and water, absorbed in gastrointestinal tract and carried via blood circulation and finally reach nerve tissues without producing any histo-pathological change cause motor paralysis. And the animals die of respiratory paralysis
- ✓ **Clinical Findings:** The disease is not accompanied by fever Progressive muscular paralysis affecting particularly the muscle of the limbs, jaw throat head and neck. Restlessness, incoordination, stumbling and ataxia are followed by inability to rise or to lift the head. In some cases the tongue becomes paralysed and hangs from the

mouth, the animal is unable to chew or swallow and drools saliva. Paralysis of the chest muscles results in abdominal type of respiration.

- ✓ **Diagnosis:** Demonstrating the presence of toxin in the suspected food. In ruminants the occurrence of pica has a diagnostic importance.
- ✓ **Differential diagnosis:** Parturient paresis in cattle Hypocalcaemia and louping ill in sheep Paralytic rabies and plant poisoning.
- ✓ **Treatment:** No reliable treatment is available. Antitoxin, if available can be tried.
- ✓ **Control:** Prevention is the best method of control. For this, the following measures can be carried out.
 - ✚ Contaminated feeds and carcasses should be avoided.
 - ✚ Cattle and sheep grazing on phosphorus deficient pasture should be provided with adequate phosphorus.
 - ✚ When the above methods are difficult to follow, vaccination is to be resorted. Toxoids can be used for vaccination.

- **Malignant Oedema**

- ✓ **Synonyms:** gas gangrene

It is an acute wound infection and characterized by toxemia, fever, edematous and emphysematous swelling around the wound.

- ✓ **Etiology:** *Cl. Septicum*. It is an anaerobe, gram-positive, spore-forming organism. Though *Cl. septicum* has been considered as a primary reason of edematous and emphysematous condition but other organisms like *Cl. chauvoei*, *Cl. Perfringens* and *Cl. novyi* are also associated with malignant edema.
- ✓ **Occurrence:** The disease is distributed throughout the world and the prevalence is high in those regions where soil is heavily contaminated with bacterial spore.
- ✓ **Species Affected:** All ages and species of animals are affected.
- ✓ **Transmission:** It is a soil borne infection. The organisms are common inhabitants of the alimentary tract. Organisms gain access through surgical or accidental wounds. Castration, docking and parturition injuries may augment the flare up of infection.

- ✓ **Pathogenesis:** The clostridia organism enters through the wound, potent neurotoxins are produced in the local lesion and locally the exotoxins cause extensive edema and necrosis followed by gangrene and cause death when absorbed into the bloodstream.
- ✓ **Clinical Findings:** Clinical signs appear within 12-48 hours of infection.
 - ✚ There is always a local lesion at the site of infection consisting of a soft, doughy swelling with marked local erythema accompanied by severe pain on palpation. At a later stage the swelling becomes tense and dark.
 - ✚ High fever (41-42°C)
 - ✚ Depression, weakness, muscular tremor and usually stiffness and lameness.
- ✓ **Diagnosis:** diagnosis of the disease is based on history of injury and wound and clinical signs
- ✓ **Differential Diagnosis:** Blackleg and anthrax
- ✓ **Treatment:**
 - ✚ Administration of penicillin or a broad-spectrum antibiotic.
 - ✚ Antitoxin for the control of toxemia but too expensive.
 - ✚ Local treatment consists of surgical incision to provide drainage, and irrigation with hydrogen peroxide.
- ✓ **Control:** Both Surgical and accidental wounds should be properly treated
 - ✚ The dead carcasses should be disposed properly.
 - ✚ Vaccination should be done as a routine measure in endemic areas.

- **Bacillary hemoglobin urea**

- ✓ **Synonym:** Red water disease

This is acute, highly fatal toxemia of cattle and sheep and characterized clinically by high fever, hemoglobinuria and jaundice and the presence of necrotic infarcts in the liver.

- ✓ **Etiology:** *Cl. hemolyticum*.
- ✓ **Occurrence:** The disease is highly prevalent in those areas where fasciolosis is very common. The highest incidence of bacillary haemoglobinuria is on irrigated or poorly drained pasture especially if the soil is alkaline in reaction.
- ✓ **Species affected:** This is a primary disease of cattle. Cattle of all ages are affected. It can also affect sheep and Pig.

- ✓ **Transmission:** The disease is spread from infected to non- infected areas by flooding, natural drainage, by contaminated hay from infected areas or by carrier animals. The carriage of bones or meat by dogs or other carnivores could also affect spread of the infection.
- ✓ **Pathogenesis:** In endemic areas invasion occurs from the alimentary tract after ingestion of contaminated material. The bacteria are carried to the liver and lodge there until damage to the parenchyma of the liver. Migrating flukes are the primary factor leading to liver necrosis and the establishment of anaerobic conditions. The development of an organized thrombus in a sub terminal branch of the portal vein produces the large anemic infarct that is characteristic of the disease.
- ✓ **Clinical findings:** The animal may be found lying dead without showing any pre-monitory signs. There is sudden appearance of the disease characterized by high-rise of body temperature, reduced appetite, and depression, suppression of rumination, defecation, and lactation. There is also arched back posture and reluctant to movement. Grunting may be evident on walking. Respiration is shallow and labored and the pulse is weak and rapid. Edema of the brisket is a common finding. The faeces are dark brown, and the urine is dark red. Pregnant cows often abort. Severe dyspnoea is evident just before death.
- ✓ **Necropsy findings:** Rigor mortis develops quickly.
 - ✚ Posterior part is soiled with red coloured urine.
 - ✚ Signs of generalized jaundice and subcutaneous haemorrhages
 - ✚ Liver is enlarged and anaemic.
 - ✚ Presences of infarcts in the liver.
 - ✚ Pleural and peritoneal cavities are filled with blood-tinged fluid.
- ✓ **Diagnosis -** This is based on the following considerations:
 - ✚ Clinical signs- haemolysis and hemoglobinurea.
 - ✚ Blood Picture - low red blood cell count and low hemoglobin percentage.
 - ✚ Lesions - Pathgnomonic liver infarcts.
 - ✚ Isolation of the organism through culture.
- ✓ **Differential diagnosis:** include Leptospirosis, Anaplasmosis, Babesiosis, Post-parturient haemoglobinurea, Enzootic hematuria, and myoglobinurea.

- ✓ **Treatment:** Antibiotics like penicillin or tetracycline are to be used at large doses in 12 hours interval. Antiserum may be given at a dose rate of 500 - 1000 ml depending on body weight. Supportive treatment includes blood transfusion, administration of physiological solution and iron compound.
- ✓ **Control:** immunization of animal. The vaccine is killed vaccine and is carried out 4-6 weeks before the expected occurrence of the disease. Annual vaccination of all animals over 6 months of age is necessary in enzootic areas. The carcasses of animals dying of the disease should be disposed of by burning or deep burial

2.2.2 Bacillus

Gram-positive rods that form endospores under unfavorable conditions, Non-motile, Aerobic or facultative anaerobic, Found in soil, water, and gastrointestinal tracts of animals and humans, and can produce a number of useful enzymes or toxins.

- **Anthrax**

Anthrax is a serious zoonotic disease that can affect most mammals and several species of birds, but is particularly important in herbivores. Carnivores are more resistant and birds rarely affected. The disease is characterized by sudden death and black tarry exudates from the natural orifices.

- ✓ **Etiology:** caused by *Bacillus anthracis*, *Bacillus anthracis* is gram positive, aerobic, rod shape, spore forming bacteria that can persist in the soil in warm moist climate for long periods. *Bacillus anthracis* can take two forms: the vegetative bacilli and the spore within the infected host the spores germinate to produce the vegetative forms which multiply, eventually killing the host.
- ✓ **Occurrence and Environmental risk factors:** Anthrax has worldwide distribution outbreaks originating from a soil-borne infection always occur after a major climate change.
- ✓ **Source of infection and transmission :** Animal can get infection from
 - 🚧 Contaminated soil, fodder grown on infected soil,
 - 🚧 Contaminated bone meal or protein concentrates, or
 - 🚧 Infected excreta, blood, or other discharges from infected animals.

- ✚ B. anthracis gains entrance to the body by ingestion, by inhalation, or through the skin (flies bite).
- ✓ **Pathogenesis:** After wound inoculation, ingestion, or inhalation, spores infect macrophages, germinate, and proliferate. After proliferation in the lymph node, the bacilli pass via the lymphatic vessels into the blood stream and septicemia take place followed by massive invasion of all body tissue. Lethal toxin and edema toxin are produced by *B. anthracis* and respectively cause local necrosis and extensive edema, which is a frequent characteristic of the disease. With the increase in toxin production, the potential for disseminated tissue destruction and organ failure increases.
- ✓ **Clinical signs:** Virtually all mammals can contract anthrax. Clinical signs in animals differ by the species, with ruminants being the most at risk. In ruminants,
 - ✚ **Per acute** systemic disease is common; sudden death may be the only sign. Staggering, trembling, and dyspnea may be seen in some animals, followed by rapid collapse, terminal convulsions, and death.
 - ✚ **Acute form** will affect ruminants, as well as horses. There is rise in body temperature, excitement, depression, staggering, and rumination ceases, milk production is reduced, pregnant animals may abort and there is bloody discharge from natural body orifices. Alimentary tract involvement is usual and is characterized by diarrhea and dysentery.
 - ✚ **Sub acute** to chronic infections occur in less susceptible species such as pigs, but can also be seen in cattle, horses, dogs, and cats.
- ✓ **Post-mortem Findings:** Absence of rigor mortis and the carcass undergoes decomposition. All natural orifices usually exudates dark tarry blood, which doesn't clot, and putrefaction and bloating are rapid.

WARNING: If there is a good reason to suspect the existence of anthrax, the carcass should not be opened. The failure of the blood to clot, the presence of blood stained serous fluid in the body cavities, severe enteritis and gross enlargement of the spleen are almost certain indications of anthrax.



Figure 2.5. Animal dead due to anthrax (Absence of rigor mortis)

- ✓ **Diagnosis:** To confirm the diagnosis on an opened carcass, smears of peripheral blood or local oedema fluid should be collected by needle puncture. Then, the smear stained with Methylene blue or Geimsa stain. Positive blood film stained shows the presence of large square ended blue rods in short chains with pink capsules.
- ✓ **Differential Diagnosis:** Anthrax should be differentiated from lightening stroke, black quarter, hemorrhagic septicemia and snakebite.
- ✓ **Treatment:** Antibiotics and anti-anthrax serum are most commonly used. Severely ill animals are unlikely to recover but in early stages, particularly when fever is detected before other signs are evident, recovery can be anticipated. Anthrax bacilli are sensitive to Penicillin and other broad spectrum antibiotics.
 - ✚ Penicillin - 10,000 units / kg body weight twice daily.
 - ✚ Streptomycin - 8-10 mg/kg / day in two doses IM for cattle
 - ✚ Oxytetracycline- 5 mg /kg body weight / day has also proved superior to penicillin in the treatment of clinical cases after vaccination in cattle.
 - ✚ Anti-anthrax serum- 100-250 ml daily through intravenous route along with a course of antibiotic may be given.

Note: In all cases treatment has to be continued at least for 5-7 days.

- ✓ **Control:** Careful disposal of infected material is most important. Infected carcasses should not be opened but immediately burned or buried, together with beddings and

soil contaminated by discharges. Burial should be at least 2m deep with an ample supply of quicklime added. Strict quarantine arrangement should be made in anthrax prone areas. Adjacent areas of the dead animals should be thoroughly disinfected by applying 10% caustic soda or 10% formaline. Fly repellent may be used to control the insect population. Immunization- Anthrax vaccine should be given annually one to two months prior to anthrax season. *Sterne* vaccine is an avirulent vaccine that does not have risk of causing anthrax following vaccination and produce a strong immunity. Isolation of sick animal and removal of well animals from the contaminated areas. Disinfection of tables, pens, milking barns, and equipments used on infected livestock. Control of scavengers that feed on animals dead of the disease. Observation of general sanitary procedures for persons who contact diseased animals, for their own safety and to prevent spread of the disease.

2.3. Diseases caused by Gram positive bacteria

Gram-positive bacteria are bacteria that have a thick peptidoglycan layer in their cell walls that stains purple when subjected to the Gram stain test. These bacteria can cause a wide range of infections in animals and humans,

2.3.1 Streptococcus

Gram-positive cocci arranged in chains or pairs, Non-motile, Facultative anaerobes, some species are normal flora of the human body while others cause disease, Sensitive to drying, heat, and disinfectants. Streptococcus species are a group of Gram-positive bacteria that are found in humans and many animal species, including dogs, cats, horses, and livestock.

There are several species of Streptococcus that have been identified in animals, but the most commonly isolated species are Streptococcus equi subsp. zooepidemicus and Streptococcus suis. *Streptococcus equi subsp. zooepidemicus* is a common commensal organism in horses that is also a significant cause of respiratory tract infections, skin infections, and septicemia. Infections can spread easily between horses and in crowded stable environments, and young horses or horses with compromised immune systems are particularly susceptible.

Treatment typically involves antibiotics and supportive care. *Streptococcus suis* are an important pathogen in pigs that can cause a range of clinical signs including meningitis, septicemia, pneumonia, and arthritis. The bacteria can be transmitted between pigs through nasal secretions and contact with infected tissues, and can also be spread to humans who come into contact with infected pigs or pig products. Treatment of *S. suis* infections in pigs typically involves antibiotics and vaccination programs to prevent future outbreaks. In addition to the above-mentioned species, some other *Streptococcus* species that affect animals include *Streptococcus agalactiae*, which can cause mastitis in dairy cattle, *Streptococcus dysgalactiae* subspecies *equisimilis*, which is associated with skin infections and tonsillitis in horses, and *Streptococcus canis* in dogs and cats.

Diagnosis of streptococcal infections in animals is typically based on identification of clinical signs, bacterial culture and sensitivity testing, and occasionally molecular diagnostic techniques such as PCR. Prevention and control measures for streptococcal infections in animals include good hygiene practices such as frequent hand washing, disinfection of animal care surfaces and equipment, implementation of infection control protocols in veterinary clinics, and appropriate use of antibiotics to prevent the spread of antibiotic-resistant strains.

- **Strangles**

- ✓ **Synonym:** Equine Distemper

It is an acute infectious disease of equines characterized by catarrhal inflammation of the upper respiratory tract with suppuration and abscessation of the associated lymph nodes.

- ✓ **Etiology:** The disease is caused by *Streptococcus equi*. It is a gram-positive coccus.

- ✓ **Occurrence:** The distribution is worldwide. Although, with the decline in horse numbers and improvement in therapy, it has become a minor importance in most countries. Outbreaks can occur at any time of the year but are most likely to happen in cold, wet weather.

- ✓ **Species affected:** this is essentially a disease of equines. Horses suffer much more severely than donkeys and mules. Young horses within the age group of 6 months to 3 years are most susceptible.

- ✓ **Transmission:** nasal discharges of the infected animals are the most important source of transmission. Organisms usually get entry through ingestion of contaminated materials. Inhalation of infected droplet is also a possible source of infection. Transmission may also take place from contaminated fomites.
- ✓ **Clinical Findings:** The incubation period ranges from 2 to 5 days. The mortality rate of the disease is 9-10% but the morbidity can reach 30-100%.
 - ✚ High-rise of temperature, profound depression, reluctance to move, eat or drink.
 - ✚ Nasal discharge which range from serous to mucopurulent and finally purulent.
 - ✚ Cough is evident due to pharyngitis and laryngitis.
 - ✚ In most cases, there is formation of abscess, which burst out liberating large quantities of thick yellowish or creamy pus.
- ✓ **Diagnosis:** based on clinical sing *i.e.* upper respiratory tract infection with purulent nasal discharge and enlargement of the lymph nodes of the throat region are diagnostic of strangles. Epidemic nature of the disease in young horse. Detection of streptococcus equi in supportive material by microscopic and serological tests.
- ✓ **Differential Diagnosis:** Glanders
- ✓ **Treatment:** Infected horse should be isolated and treatment started early as soon as possible. In the early stage penicillin is quite adequate but at a later stage intravenous injection of tetracycline (10mg/kg body wt) will be more effective but must be continued for 4-5 days.
- ✓ **Control and prevention**
 - ✚ The infected animals should be kept in isolation.
 - ✚ Contaminated premises should be thoroughly disinfected
 - ✚ In contact animals may be passively immunized by injecting immune serum.
 - ✚ Freeze dried strangles vaccine may be given to susceptible animals for active immunization.

- **Dermatophilosis**

- ✓ **Synonyms:** Mycotic dermatitis; cutaneous streptothricosis; cutaneous actinomycosis.

This is a superficial skin infection of animals and man characterized by exudation and matting of hairs and wools and formation of crusts and scab.

- ✓ **Etiology:** *Dermatophilus congolensis*. The organism is a gram positive. The organisms invade hair follicles, sweat glands and other epidermal structures.
- ✓ **Occurrence:** The disease is prevalent throughout the tropical and temperate regions of the world. The disease appears to be most common under low moist climatic conditions.
- ✓ **Species Affected:** The disease is most commonly encountered in cattle, sheep, and horse. But, it has been also reported in goat, donkey, dog, cat and pigs. Man is readily infected with dermatophilosis.
- ✓ **Transmission:** All ages of animals are susceptible and the disease is usually transmitted through direct contact with infected animals or infected materials. Mechanical transmission through contaminated instrument, brush, bedding materials is possible. The disease may be transmitted through arthropod vectors like flies, ticks and sheep kids. Moisture and cold weather are predisposing factors to transmit the disease and most of the outbreaks have been found to occur during rainy season. Skin abrasion is also another predisposing factor of dermatophilosis.
- ✓ **Clinical Findings:** In cattle, lesions occur on the neck, body or back of the udder and may extend over the sides and down the legs and the ventral surface of the body. In adult cattle the characteristic lesions are thick, horny crusts, varying in color from cream to brown and they are 2-5 cm in diameters. In the early stages the crusts are very tenacious and attempts to lift them cause pain. Beneath the crusts there is granulation tissue and some pus. In the later stages, the dermatitis heals and the crusts separate from the skin but are held in place by penetrating hairs or wool fibers and are easily removed. In young calves crust formation does not occur. There is extensive hair loss with tufting of the fibers, heavy dandruff and thickening and folding of the skin in later stages.
- ✓ **Diagnosis:** Microscopic examinations of stained exudates with Giemsa or Gram's stain. The causative organism may be isolated from skin scrapings or a biopsy section and is much easier to isolate from an acute case than a chronic one
- ✓ **Differential Diagnosis:** Photosensitization

✓ **Treatment:** There is no completely satisfactory treatment for cases that show very extensive involvement. In general terms, better results are obtained during dry hot weather and in dry climate.

- ✚ Penicillin and streptomycin at a very heavy dose rates (70mg streptomycin and procaine penicillin G 70000 Iu/kg body weight) is recommended as being 100% effective in animals.

- ✚ Tetracycline (5mg /kg body weight repeated weekly as required is recommended and long acting tetracycline (20mg/kg body weight) in one injection has been reported to give excellent result in cattle.

- ✚ Topical application of astringent or antibacterial solution as spray or dip has been considered as rational approach. For this, copper sulphate, Zinc sulphate are useful.

✓ **Control**

- ✚ Contact with infected animals and grooming materials are to be avoided.

- ✚ Control of tick and fly population are to be made with appropriate insecticide application.

- ✚ Quarantine of the affected animals may be made to prevent possible spread.

2.3.2 Staphylococcus

Common inhabitant of the skin and mucous membranes, Spherical cells arranged in irregular clusters, Gram-positive, Lack spores and flagella and May have capsules. Staphylococcus; frequently involved in nosocomial and opportunistic infections. Staphylococcus species are a group of Gram-positive bacteria that can be found in both humans and animals, including dogs, cats, horses, and livestock.

These bacteria can cause a variety of infections in animals, ranging from mild skin infections to more severe systemic infections. There are several species of Staphylococcus that have been identified in animals, including Staphylococcus aureus, Staphylococcus hyicus, and Staphylococcus pseudintermedius. Of these species, Staphylococcus pseudintermedius is the most commonly isolated species in dogs and cats. Staphylococcus infections in animals can be transmitted through direct contact with infected animals or their secretions, or through contaminated environmental surfaces. Animals that have compromised immune systems are at a higher risk of developing infections from these organisms.

- **Clinical signs** of staphylococcal infections in animals can vary depending on the severity of the infection and the underlying health status of the animal. Common clinical signs include skin lesions, abscesses, fever, lethargy, anorexia, and respiratory or urinary tract symptoms.
- **Diagnosis** of staphylococcal infections in animals is typically based on clinical signs, bacterial culture and sensitivity testing, and occasionally on molecular diagnostic techniques such as PCR.
- **Treatment** of staphylococcal infections in animals typically involves medications such as antibiotics or antiseptics depending on the severity of the infection. In severe cases or when an animal is immunocompromised, hospitalization may be necessary.
- **Prevention and control measures** for staphylococcal infections in animals include good hygiene practices such as frequent hand washing, disinfection of animal care surfaces and equipment, implementation of infection control protocols in veterinary clinics, and appropriate use of antibiotics to prevent the spread of antibiotic-resistant strains.

2.3.3 Listeriosis

Listeriosis usually results from infection by *Listeria monocytogenes*, a Gram positive rod in the family Listeriaceae. *L. monocytogenes* is found worldwide and is widely distributed in the environment. Is relatively resistant to freezing, drying and heat. It will grow at temperatures from 1°C to 45°C and can proliferate at refrigeration temperatures on contaminated foods.

✓ **Species Affected:** A wide variety of domestic and wild mammals, birds, fish and crustaceans Clinical disease is seen most often in ruminants. Occasional cases occur in animals. Listeriosis in humans is usually a serious problem

✓ **Transmission:**

- ✚ Infected animals can shed *L. monocytogenes* in the feces, milk and uterine discharges.
- ✚ It is also found in aborted fetuses and occasionally in the nasal discharges and urine of symptomatic animals.
- ✚ Most infections are acquired by ingestion, but inhalation or direct contact is also possible.
- ✚ Venereal transmission might also be possible.

- ✚ In ruminants, listeriosis typically occurs after the consumption of contaminated silage or other feed.
- ✚ For humans, contaminated food sources include raw meat and fish, unpasteurized dairy products and uncooked vegetables.
- ✚ Processed foods that have become contaminated after processing, particularly soft cheeses and ice cream are also source of infection.
- ✚ Humans can also be infected by direct contact with infected animals during calving, lambing or necropsies.

✓ Clinical Signs

- ✚ Listeriosis is usually characterized by encephalitis, abortions or septicemia.
- ✚ Listeria can cause encephalitis, abortions and septicemia in sheep, cattle and goats.
- ✚ In the encephalitic form, the initial symptoms of depression and anorexia are followed by neurologic signs:
- ✚ Abortions and stillbirths mainly occur late in gestation.
- ✚ Septicemia occurs most often in newborns and young ruminants; older animals are rarely affected.
- ✚ Typical symptoms may include fever, depression, inappetence and death.
- ✚ Localized infections can also be seen, including subclinical, acute or chronic mastitis in cattle, and ophthalmitis in sheep.

✓ Diagnosis

- ✚ Listeriosis can be diagnosed by isolating *L. monocytogenes* from the placenta, fetus or uterine discharges after an abortion.
- ✚ Blood or cerebrospinal fluid (CSF), respectively, can be cultured in animals with septicemia or encephalitis.
- ✚ Liver, kidneys and spleen can be cultured in septicemia animals, or the pons and medulla in animals with encephalitis.

✓ Treatment

Listeriosis can be treated with a variety of antibiotics.

- ✚ IV injection of chlortetracycline (10 mg/kg BW per day for 5 d) is reasonably effective in meningoencephalitis of cattle but less so in sheep.

- ✚ Penicillin at a dosage of 44000 IU/kg BW given IM daily for 7 days, and in many cases for 10-14 days can also be used.
- ✚ High doses and early treatment are required for animals with encephalitis; animals with severe neurologic signs usually die despite treatment.
- ✚ Supportive treatment may also be required.

✓ Prevention

- ✚ The risk of listeriosis can be decreased in ruminants by feeding good quality silage with a low PH.
- ✚ Spoiled or moldy silage should be avoided, as well as silage from the superficial few inches exposed to air.
- ✚ Any leftover silage should be removed after feeding.
- ✚ Rodents should be controlled.
- ✚ New animals added to the herd should be quarantined, and animals with clinical listeriosis isolated.
- ✚ The placenta and fetus should be removed after an abortion.
- ✚ There is no vaccine for listeriosis.

2.3.4 Corynebacterium

Corynebacteria are a type of gram-positive bacteria that can be found in various animal species. These bacteria are known to cause a range of infections in animals such as mastitis, pneumonia, and skin infections. In cows, there are two types of Corynebacteria that are commonly found: *Corynebacterium bovis* and *Corynebacterium pseudotuberculosis*. *C. bovis* is typically associated with mastitis, which is an inflammation of the udder caused by bacterial infection. Infected animals may experience swollen and painful udders, as well as a reduced milk yield and quality. Meanwhile, *C. pseudotuberculosis* is associated with caseous lymphadenitis, which is a chronic and often fatal disease that affects sheep and goats.

In pigs, there is a species of Corynebacteria called *Corynebacterium suis*, which can cause respiratory disease, such as pneumonia, in piglets. Additionally, *C. suis* can also cause skin and soft tissue infections in pigs. In terms of diagnosis, veterinarians typically use bacterial culture to isolate the specific strain of Corynebacteria causing the infection. It's important to accurately identify the bacteria causing the infection in order to select the appropriate treatment.

Page 37 of 132	Ministry of Labor and Skills Author/Copyright	Animal Health Level -3	Version -1
			May, 2023

Treatment for *Corynebacteria* infections typically involves antibiotics such as penicillin or tetracycline, along with supportive care such as intravenous fluids and wound care. In order to prevent the spread of these infections, good hygiene practices such as proper cleaning and disinfection of livestock facilities are important. Additionally, appropriate use of antibiotics and vaccination programs may also be used for prevention and control of these infections in animals.

- **Ulcerative lymphangitis**

Ulcerative lymphangitis is a mildly contagious disease of horses and cattle characterized by lymphangitis of the lower limbs.

- ✓ **Etiology:** caused by *corynebacterium pseudotuberculosis*. This is pleomorphic and Gram-positive rod.
- ✓ **Occurrence:** The disease is worldwide in distribution.
- ✓ **Species affected:** It is primarily a disease of horse but it has also been recorded in cattle.
- ✓ **Transmission:** Disease is related to soil contamination. The organisms may remain on the skin as contaminants and may invade the tissues through abrasion and other injuries. Passive transmission also occur by fly
- ✓ **Clinical findings:** The disease is sporadic in nature and develops slowly.
 - ✚ It begins as moderate diffuse swelling on one or both limbs causing lameness.
 - ✚ Nodules formed on the legs break down leading to the formation of ulcers.
 - ✚ Thick greenish exudates mixed with blood appear from such ulcers.
 - ✚ Lesions heal up in one week. Systemic reactions are not a feature.
 - ✚ There is also enlargement of lymphatic vessel
- ✓ **Diagnosis:** This is based on clinical signs and isolation and identification of the organism from discharging lesions.
- ✓ **Differential Diagnosis:** Epizootic lymphangitis and Glanders
- ✓ **Treatment:** Local treatment of ulcers is the usual procedure but parenteral injections of penicillin or tetracycline may be necessary in severe cases.
- ✓ **Control:** Good hygiene in stables and careful disinfections of injuries to the lower limbs usually afford adequate protection against the disease.

- **Actionomycosis**

Page 38 of 132	Ministry of Labor and Skills Author/Copyright	Animal Health Level -3	Version -1
			May, 2023

✓ **Synonym:** lumpy jaw

It is a chronic infectious disease, primarily affect cattle but also affect pig, horse and man. The disease is characterized by a rarefying osteomyelitis of bones of the head, particularly the mandible and maxilla.

✓ **Etiology:** *Actinomyces bovis* is a Gram-positive, non-spore-forming, non-motile, and anaerobic to microaerophilic bacterium. It is a normal inhabitant of the oral cavity and upper respiratory tract of cattle and sheep.

✓ **Occurrence: and transmission:** *A. bovis* is a common inhabitant of bovine oral cavity and infection occurs through wounds to the buccal mucosa caused by sharp pieces of feed or foreign material. Infection may also occur when the teeth are erupting. Infection of the alimentary tract is probably related to laceration by sharp foreign bodies.

✓ **Clinical Findings:** Initially it is a painless bony swelling, which appears on the mandible and maxilla. The swellings are very hard and discharge small sticky, honey-like pus containing minute, hard, yellow-white granules. Difficulty in mastication and dyspnoea may be noted.



Figure 2.6. Bony swelling on mandible and maxilla (Lumpy jaw)

Source: <https://www.google.com/search?q=actinomycosis+in+bovine&client=firefox-b-d&source=lnms&tbn=isch&sa=X&ved=2ahUKEwicm>

✓ **Diagnosis:** Smears of the discharging pus stained with Gram's stain will reveal gram-positive filaments of the organism

✓ **Differential diagnosis:** Localized abscesses in mandibular region, Nocardiosis and actinobacillosis

- ✓ **Treatment:** Surgical Drainage of the abscess should be done and dress with iodine.
 - ✚ Iodine may be fed orally KI 6-10 gm/day for 7-10 days
 - ✚ Intravenous injection of Sodium iodide (NaI) (10-20%) 70mg/kg body weight, slow IV injection. Repeat after 7 to 10 days.
 - ✚ Antibiotics like streptomycin 5mg/day for 3 to 5 day may be given. The organism is also sensitive to oxyetracycline and chloramphenicol.

- **Actionobacillosis**

- ✓ **Synonyms:** Wooden tongue

It is a sporadic chronic infectious disease of cattle and sheep characterized by inflammation of the tongue and soft tissues around the neck, head, oral cavity, pharyngeal lymph nodes, and nasal cavity.

- ✓ **Etiology:** *Actionbacillus lignieresii*. The organism produces sulphur granules and is susceptible to ordinary environmental influences and does not survive for more than 5 days on hay or straw. **Species Affected:** Cattle and sheep are susceptible to the disease
- ✓ **Occurrence:** It is worldwide in distribution and occurs sporadically.
- ✓ **Transmission:** Infected discharges are the source of infection and transmission is effected by the ingestion of contaminated pasture or feed. Infection of the soft tissues occur through wound and abrasion caused by foreign bodies like sharp objects. Injury of the oral mucosa due to abrasive feedstuffs may help the process of transmission. Eruption of permanent teeth of calves and stomatitis caused by FMD creates a conducive condition to the organism to penetrate the sub epithelial tissues.
- ✓ **Clinical Findings:** The onset of glossal actinobacillosis is usually acute and the affected animal being unable to eat.
 - ✚ There is excessive salivation and gentle chewing of the tongue as though a foreign body were present in the mouth.
 - ✚ The tongue is swollen and hard, particularly at the base. Manipulation of the tongue causes pain.

- ✚ Nodules and ulcers are present on the surface of the tongue. In the later stages when the acute inflammation is replaced by fibrous tissue, the tongue becomes shrunken and immobile and there is considerable interference with prehension.
- ✚ Enlargement of the sub maxillary and parotid lymph nodes. Enlargement of the retropharyngeal lymph nodes causes interference with swallowing and loud snoring respiration.
- ✚ The pus contains numerous granules, which appears as and known as sulphur granules.
- ✓ **Diagnosis:** It is based on clinical findings, examination of granules and identification of the causative organism
- ✓ **Treatment:**
 - ✚ Surgical Drainage of the abscess should be done and dress with iodine.
 - ✚ Iodine may be fed orally KI 6-10 gm/day for 7-10 days
 - ✚ Intravenous injection of NaI (10-20%) 70mg/kg body weight, slow IV injection. Repeat after 7 to 10 days.
 - ✚ Antibiotics like streptomycin 5mg/day for 3 to 5 day may be given. The organism is also sensitive to oxyetracyclin and chloramphenicol.
- ✓ **Control:** quick treatment of affected animals and the preventions of contamination of pasture and feed troughs. Isolation or disposal of animals with discharging lesions is essential, although the disease does not spread readily unless predisposing environmental factors cause a high incidence of oral lesions.

2.4. Diseases caused by Gram negative bacteria

Gram-negative bacteria are bacteria that have a thin peptidoglycan layer in their cell walls that stains pink when subjected to the Gram stain test. These bacteria can cause a wide range of infections in animals and humans. Some of the most common Gram-negative bacteria that infect animals include Salmonella, Brucella, Pasturella, Moraxella and others. Treatment for Gram-negative bacterial infections varies depending on the specific bacteria and the severity of the infection. Prevention measures include good hygiene practices and vaccination.

2.4.1 Pasteurella

Page 41 of 132	Ministry of Labor and Skills Author/Copyright	Animal Health Level -3	Version -1
			May, 2023

Pasteurella is a genus of small, gram-negative, facultative anaerobic bacteria that are known to cause a variety of infections in both humans and animals. Some species of *Pasteurella* are commonly found in the oral cavity and upper respiratory tract of various animal species, including cats and dogs. They are usually non-motile, although some species may have flagella which allow for some movement. *Pasteurella* bacteria can grow on various types of media, including blood agar and MacConkey agar.

Pasteurellosis

This is an acute septicemic disease occurring most often in cattle, camel, sheep, goat and pig and usually occurs following some form of stress such as transportation, shipping to feedlots or inclement weather. In cattle there are two types of pasteurellosis: pneumonic pasteurellosis and Septicemic pasteurellosis.

- **Pneumonic pasteurellosis**

- ✓ **Synonyms:** Shipping fever pneumonia

It is associated with infection by *pasteurella hemolytica* and occasionally *pasteurella multocida*.

It is characterized clinically by acute bronchopneumonia with toxemia.

- ✓ **Occurrence** - In most cases, both of these *pasteurella* species are part of the normal flora of the upper respiratory tract. Following stress or viral infection these appears to be an increase in the total numbers and virulence of the organism in the nasopharynx, which are then inhaled in to the alveoli. Young growing cattle within the age group of 6 months to 2 years are most often affected. Feedlot cattle may suffer when they are put under stress (transportation, temporary starvation, rapid fluctuations in ambient temperature, mixing of cattle from different origins)
 - ✓ **Transmission:** occurs by the inhalation of infected droplets. This may be from clinical cases or recovered carriers in which the infection persists in the upper respiratory tract. The disease may spread very quickly in closely confined animals but may be much slower in animals at pasture.
 - ✓ **Clinical finding:** The disease usually develops in cattle within 10 to 14 days after they have been stressed. The affected cattle are usually depressed, anorectic, have a fever (40 -41° c) and a serous to mucopurulent nasal discharge with rapid, shallow breathing and often coughing if they move. In severe cases, pleurisy may develop, which is

characterized by an irregular breathing pattern and grunting on expiration. Lung auscultation may reveal moist rales, pleuritic frictional rubs, and crackling. As lung consolidation progresses, lung sounds may be decreased.

✓ **Diagnosis:** is based on history of movement or any stress, clinical findings and necropsy findings. Confirmatory diagnosis made by the isolation and identification of the organism.

✓ **Differential diagnosis:** Infectious Bovine Rhinotrachitis (IBR), Lungworm (verminous) pneumonia and CBPP

✓ **Treatment**

✚ Oxytetracycline 10 mg/Kg body weight (10%), Iv or IM daily for 3 days

✚ Trimethoprim-sulfamethoxazole 3-5 ml/45kg body weights IV or IM for 3 days

✚ Penicillin 20000-30000 IU/kg body wt. IM or SC daily for 3 days

✚ Sulfamethazine (liquid preparation) 150 mg/kg b. wt IV or orally daily for 3 days.

✓ **Control and prevention**

✚ Stressful procedures such as castration, dehorning, branding, deworming should be distributed over a period of time rather than concentrated at once. Vaccination at least 2 weeks before shipment is also recommended.

✚ During transportation bedding are necessary and cattle should not be without feed and water for more than 24 -30 hours. Ventilation is very important.

✚ In dairy calves, vaccination of the dam may be beneficial to provide passive immunity to the calves

• **Septicemic Pasteurellosis**

✓ **Synonym:** hemorrhagic septicemia

✓ **Etiology:** The disease is an acute septicemia caused by *pasteurella multocida*.

✓ **Occurrence:** it occurs in an outbreak during periods of environmental stress, the causative organism in the intervening periods persisting on the tonsillar and nasopharyngeal mucosa of carrier animals.

✓ **Species affected:** cattle, camel, sheep, goat and pig

✓ **Transmission:** spread occurs by ingestion of contaminated foodstuff, the infection originating from clinically normal carriers or clinical cases. The saliva of affected

animals contains large numbers of *pasteurella* during the early stages of the disease. Although infection occurs by ingestion, the organism does not survive on pasture for more than 24 hours.

- ✓ **Clinical signs:** Clinically the disease is characterized by a sudden onset of fever (41-42°C), profuse salivation, and submucosal petechiation. Severe depression and death within 24 hours. Localization may occur in subcutaneous tissue, resulting in the development of warm, painful swellings about the throat, dewlap, brisket or perineum, and severe dyspnoea may occur if the respiration is obstructed.
- ✓ **Diagnosis:** tentatively, the disease is diagnosed based on history, clinical findings and necropsy findings but confirmatory diagnosis is done by isolation and identification of the causative bacteria.
- ✓ **Differential Diagnoses:** anthrax, blackleg, and acute leptospirosis
- ✓ **Treatment:** Those drugs listed for pneumonic pasterellosis are also effective for septicemic pasterellosis
- ✓ **Prevention and control:** Vaccination of susceptible group is effective to protect for at least 12 months. The effective control lies with institution of adequate management, rational executions of vaccines and care in the transportation of animals

- **Avian Pasteurellosis**

- ✓ **Synonyms:** fowl cholera

It is a per acute infectious disease that affect all poultry species and wild birds.

- ✓ **Etiology:** caused by *pasteurella multocida*. It is a gram-negative bacteria
- ✓ **Occurrence:** it has worldwide distribution and notifiable in many countries. Adult birds are more frequently affected than younger stock.
- ✓ **Species affected:** all species of bird are susceptible
- ✓ **Transmission:** source of infection include carrier birds, clinically diseased poultry and their excretion and carcass of birds, which have died of the infection. It transmitted by inhalation and ingestion of contaminated feed and water. Infection also occur through conjunctival and wound
- ✓ **Clinical sign:** the disease may occur in many forms: Per acute, acute, chronic and localized disease.

- ✚ **In per acute form:** there may be no premonitory sign and large numbers of birds in the flock are found dead but in a good body condition.
- ✚ **In acute form:** marked depression, anorexia mucus discharge from the orifices and fetid diarrhoea may be seen.
- ✚ **In chronic form** depression, conjunctivitis, dyspnoea as well as lameness, torticollis and / or swelling of the wattles.
- ✓ **Diagnosis:** based on history of the disease, clinical sign and gross lesion are helpful. Isolation of the organisms
- ✓ **Treatment:** trimethoprim or sulphadiazine preparation in the feed and water Injection of affected individual with long acting oxy tetracycline.
- ✓ **Control and prevention:** to eradicate infection, it is necessary to depopulate, to cleanse and disinfect building and equipments. Vaccination

2.4.2 Brucella

Brucella is a genus of Gram-negative bacteria. They are small (0.5 to 0.7 by 0.6 to 1.5 μm), nonmotile, nonperforming, rod-shaped (coccobacilli) bacteria. They are facultative intracellular parasites, capable of surviving inside macrophages.

- **Brucellosis**

- ✓ **Synonyms:** Bang's disease

Brucellosis is an acute or chronic contagious disease of domestic animals that causes placentitis and abortion. It is occupational disease in human being and may be acquired from animals.

- ✓ **Etiology:** - caused by *Brucella* spp. i.e. caused by *Br. abortus* (for cattle), *Br. suis* (for swine) and *Br. melitensis* (for sheep and goat) but they can produce infection in other species of animals and human being. It is a gram-negative bacterium
- ✓ **Occurrence:** -Brucellosis is widespread and of major economic importance in most countries of the world, particularly amongst dairy cattle. Infection occurs in cattle of all ages but persists most commonly in sexually mature animals. Pregnant cattle are highly susceptible.
- ✓ **Transmission:** -the organism has a greatest concentration in the contents of the pregnant uterus, the fetus and the fetal membranes, and these must be considered as major sources of infection. The disease is transmitted through ingestion of

contaminated feed and water, penetration of intact skin and conjunctive and contamination of the udder during milking. Artificial insemination by infected semen also spreads the disease. Organisms are shed before and after abortion.

- ✓ **Pathogenesis** : Once the organism get in to the body of the animal they picked up by the blood and lymph and are carried to the regional lymph node then taken by the cells of reticuloendothelial system and carried to the predilection site some of which are the uterus ,testicles and in to some joints. Then Brucella organism start to multiply when animal become pregnant because the fetus produce Erythritol, a substance capable of stimulating the growth of brucella organism, This substance occurs naturally in greatest concentration in the placental and fetal fluids and is probably responsible for localization of infection in these tissues result in **placentitis** and **abortion** occurs principally in the last three months of pregnancy.
- ✓ **Clinical Findings**: abortion occurs after 5th month of pregnancy with retention of placenta and metritis in cow. Metritis may be acute with septicemia and following death or chronic leading to sterility. In the bull, orchitis and epididymitis occur occasionally. One or both scrotal sacs may be affected with acute, painful swellings twice-normal size. The seminal vesicles become enlarged and can be detected through rectal palpation. Hygromatous swelling, especially on the knees, should be viewed with suspicion.
- ✓ **Diagnosis**: Abortion in last trimester of pregnancy. Blood samples for serological tests (Rose Bengal plate test, CFT, ELISA, milk ring test, etc). Examine uterine fluids and the contents of the fetal abomasums.
- ✓ **Differential Diagnosis**: Trichomonosis, Leptospirosis, Listeriosis, vibriosis, mycotic abortion,
- ✓ **Treatment**: Treatment is not usually under taken. For cow long acting Oxytetracycline at 20-mg/kg body weight. IM for 5 days treatments in combination with streptomycin at 25mg/kg bwt. IM or IV daily for 7 consecutive days may be partially successful
- ✓ **Control and prevention**:
 - 🚦 Test and slaughter method will be the most rational approach. All abortion should be suspected of brucellosis, unless otherwise proved.

- ✚ Hygienic disposal of uterine discharges, foetus, and foetal membranes should be done strictly and clean and disinfect contaminated premises.
- ✚ Restrict movement of infected animals.
- ✚ Vaccination with a single 5ml dose of *Br. abortus* strain 19 vaccine given subcutaneously for 2 to 6 months of age confers adequate immunity against abortion for 5 or more subsequent lactations. It is live vaccine.
- ✚ Strain 45/20 vaccines in adjuvant are used for those beyond 8 to 9 months of age. This is a killed (inactivates) vaccine.

2.4.3 Moraxella

Moraxella is a genus of gram-negative bacteria found in the mucous membranes, respiratory tract, and conjunctiva of animals. The genus includes several species, including *Moraxella bovis*, which is commonly associated with infectious keratoconjunctivitis in cattle. In cattle, infectious keratoconjunctivitis or "pinkeye" is a common disease that results in inflammation of the conjunctiva and cornea of the eye. The disease can lead to decreased weight gain, reduced milk production, and blindness if left untreated. *Moraxella bovis* is one of the most common causative agents of infectious keratoconjunctivitis in cattle.

- **Infectious keratitis**

✓ **Synonyms:** pink eye.

It is an infectious disease of cattle characterized by conjunctivitis, lacrymation and varying degree of corneal opacity and ulceration.

- ✓ **Etiology:** caused by *Moraxella bovis*. It is gram-negative bacteria.
- ✓ **Occurrence:** The disease is worldwide in distribution and is most common in summer and autumn. In these seasons the disease reaches epizootic proportions when flies and dust are abundant and grass is long.
- ✓ **Species Affected:** Cattle are chiefly affected and the young being most susceptible.
- ✓ **Transmission:** Infection is transmitted by direct contact with ocular and nasal discharges or indirectly by mechanical transmission with insect vectors.
- ✓ **Clinical Findings:** The disease has an incubation period of 2-3 days. One or both eyes can be involved. The signs are more severe in young than adult cattle.

- ✚ There is a profuse watery lacrymation accompanied by thick eye discharges at the canthus within 24 hours. Gradually the condition turns worse with severe conjunctivitis,
- ✚ Profuse watery or mucopurulent eye discharges. This is followed by photophobia, swelling of the eyelids, matting of eyelashes with smoky or diffuse corneal opacity.
- ✚ There is central or marginal keratitis and corneal ulceration and complete blindness is common in goats.



Figure 2.7 pink eye

Source: https://www.google.com/search?q=contagius+ectema+0r+pink+eye+in+bovine+symptoms&tbm=isch&ved=2ahukewi0yp6wgob_ahxqkicchalzagsq2-

- ✓ **Diagnosis:** is based on the following.
 - ✚ Characteristic clinical manifestation. Bacteriological culture of the organism.
 - ✚ Serological tests; serum agglutination test,
- ✓ **Differential Diagnosis:** The disease has to be differentiated from, Traumatic Conjunctivitis, IBR and keratitis due to bovine MCF, Eye worm
- ✓ **Treatment:** Treatment with ophthalmic ointment containing chloramphenicol, oxytetracycline, penicillin and streptomycin has been suggested. Anti-inflammatory drug in conjunction with local treatment with antibiotic ophthalmic preparation has been used successfully.
- ✓ **Control:** is not possible to eradicate the disease because of fly population. Cattle showing excessive lacrymation should be isolated and treated. Recovered cattle remain immune up to 12months.

- **Glanders**

✓ **Synonym:** Farcy

Glanders is a contagious disease of equines, occurring in either acute or chronic forms and characterized by formation of nodules and ulcers involving upper air passage, lungs and cutaneous tissues. The disease is highly fatal and of major importance in any affected horse population. The disease is also transmissible to human beings.

✓ **Etiology:** caused by *Pseudomonas mallei*. It is a gram negative and non-spore forming bacteria. The organisms show very less resistance to the environmental influences. They are readily destroyed by direct sunlight and by most of the commonly used disinfectants.

✓ **Occurrence:** the disease was prevalent in most of the horse rearing countries. It is still reported in Eastern Europe, Asia and Africa.

✓ **Species affected:** Horse, mule and donkeys are highly susceptible. Man is susceptible and the prognosis of the disease always ends fatally.

✓ **Transmission:** Spread mostly occurs by ingestion of contaminated feed and water contaminated by nasal discharge or sputum. Rarely the cutaneous form appears to arise through contamination of skin abrasions by direct contact or from harness or grooming tools. Spread by inhalation can also occur but this mode of infection is rare under natural conditions.

✓ **Clinical Findings:** In the acute form there is a high fever, cough and nasal discharge with rapidly spreading ulcers appearing on the nasal mucosa and nodules on the skin of the lower limbs or abdomen. Death due to septicemia occurs in a few days.

✚ In the chronic form, the signs may be related to the lesions, which occur in one or more of the predilection sites. When the localization is pulmonary, there is a chronic cough, frequent epistaxis and laboured respiration. Nasal lesions appear on the lower part of the nasal septum. They start as nodules that may ulcerate. In the early stages there is a serious nasal discharge, which may be unilateral and which later becomes purulent and blood stained. Enlargement of the sub maxillary lymph node is a common occurrence.

- ✚ The affected animal gradually loses its condition with poor hair coat. There is edema of the hind limbs down to the hock accompanied with discharge from lymph nodes.
- ✚ The skin form is characterized by the appearance of subcutaneous nodules (1-2 cm in diameter), which ulcerate and discharge pus. Lymph nodes draining the area also involved and may discharge to the exterior. The lesions mostly occur in the medial aspect of the hock but they can occur on any part of the body.
- ✓ **Diagnosis:** based on clinical signs and post mortem findings.
- ✓ **Differential Diagnosis:** it should be differentiated from epizootic lymphangitis, ulcerative lymphangitis, strangles and other pneumonia of horse.
- ✓ **Treatment:** Sodium sulfadiazine has been highly effective.
- ✓ **Control:** For control and eradication of glanders every animal with clinical evidence of glanders and those that react positively must be regarded as diseased and should be destroyed.
- ✚ The carcass of animals must be burned or buried
- ✚ Infected premises should be vigorously disinfected
- ✚ Manure, beddings, feed residue should be burned and buried.

2.5. Diseases caused by Enterobacteriaceae

Enterobacteria are a group of Gram-negative bacteria that are commonly found in the intestinal tracts of both animals and humans. These bacteria can also be found in the environment, such as soil and water. Enterobacteria belong to the family Enterobacteriaceae, which includes several genera such as *Escherichia*, *Salmonella*, and *Shigella*. In animals, enterobacteria can cause a wide range of infections, including gastrointestinal infections, septicemia, and urinary tract infections. Enterobacterial infections are often associated with poor sanitation or hygiene practices.

2.5.1 Salmonella

Salmonella is Gram-negative rods, Motile with flagella, Facultative anaerobes - Often found in the intestines of animals and can cause illness when ingested it can cause salmonellosis, a disease that affects the intestinal tract and can lead to fever, diarrhea, vomiting, and dehydration. In animals, salmonella infection can cause a range of symptoms, including mild illness or potentially fatal infections such as septicemia.

- **Salmonellosis**

It is an acute or chronic contagious disease of all species of animals and human being characterized by gastro-enteritis. The disease has got economic importance as it causes heavy mortality and high morbidity.

- ✓ **Etiology:** The disease is caused by different species of salmonella. Some of the important salmonella species are: *S. typhi* (man): *S. typhimarium* (cattle, sheep, goat, pig, and horse); *S. dublin* (cattle); and *S. galinarum* (birds). The organisms are gram-negative and rod shaped.
- ✓ **Occurrence:** The disease is worldwide in distribution and has been recorded from all the tropical countries.
- ✓ **Species affected:** The disease may occur in cattle, sheep, horse, man, fowl and pigs.
- ✓ **Transmission:** Contaminated water is the important source of transmission. Cattle may acquire the infection from infected cattle or from infected pig, horse, sheep, dog or poultry. Animals may contract the infection from human sources. Introduction of the disease into a new farm is through carrier animals.
- ✓ **Clinical Findings:** Animals may remain as a symptomatic carrier without exhibiting any clinical manifestations. The disease may appear as acute or chronic one. From clinical standpoint, it has been classified as enteric form and septicemic form.
- 🚩 **Enteric Form:** this form is most commonly observed in adults characterized by anorexia depression, high temperature and profuse watery diarrhoea. Initially, animals may pass semisolid faeces, which are followed by diarrhoea with blood clot and mucus. Abdominal pain and anemia may occur following frequent defecation and extensive haemorrhage, respectively. Gradually the initial temperature drops to subnormal level and death occurs. Pregnant cow may abort.

✚ **Septicaemic form:** this form is observed in young animals characterized by high-rise of temperature, dullness, ataxia, and recumbency. The animal may die within 1-2 days, but in some cases there is involvement of musculo-skeletal system resulting to enlargement of joints. Signs of pneumonia may appear (pneumoenteritis). The recovered animals continue to excrete the organisms throughout its life.

- ✓ **Diagnosis :**It can be made based on history, clinical signs, and necropsy findings. Confirmation is by isolation and identification of the organism by culture.
- ✓ **Differential Diagnosis:** Coccidiosis, pasteurellosis, colibacillosis, paratuberculosis and liver fluke infestation.
- ✓ **Treatment:** early treatment with broad-spectrum antibiotics and with sulphonamides is highly effective. Delays in treatment mean loss of the integrity of intestinal mucosa to the point where repair couldn't occur. Drugs commonly used for the treatment of salmonellosis are: Chloramphenicol, Trimethoprim and Sulfadoxine mixture, and Nitrofurazone (orally).

Note: Parental or oral treatment alone may be sufficient if cases are detected early, but in all circumstances parental and oral treatment combined is superior and is recommended.

✓ **Control**

- ✚ Calving should be made in a hygienic place.
- ✚ Calves should be allowed to have access to colostrums following their birth.
- ✚ Calves should not be exposed to adverse weather (too hot or too cold environment).
- ✚ Animals should not be allowed to drink water from sewage or stagnant pool or pond.
- ✚ Affected animals should be brought under immediate treatment.

• **Pullorum Disease**

- ✓ **Synonym:** bacillary white diarrhoea
- ✓ **Etiology:** Caused by *salmonella pulorum*. It usually cause high mortality in young chicken and occasionally in adult chicken.

- ✓ **Transmission:** is chiefly directly through the egg but also occurs by direct or indirect contact. Infection transmitted via egg or hatchery usually result in death during the first few days of life and up to 2-3 weeks of age.
 - ✓ **Clinical finding:** predominantly seen in chicken less than 3 weeks of age. And the first indication is excessive number of dead-in-shell chicks and death shortly after hatching. Affected bird shows depression, respiratory distress, anorexia and white viscous dropping that adhere to the feather around the vent. A sub acute form with lameness and swollen hock joints may be seen in growing birds and result in poor growth rate. Reduced egg production with lowered fertility and hatchability.
 - ✓ **Diagnosis:** based on post mortem finding, and isolation of the organism.
 - ✓ **Treatment:** no antibacterial are currently approved for the treatment of infected flock
 - ✓ **Control:** routine testing of breeding flock assure freedom from infection.
- **Fowl Typhoid**
 - ✓ **Etiology:** Caused by *S. gallinarum*. It is egg-transmitted and produce lesion in chicks. Mortality at all age usually is high.
 - ✓ **Transmission:** by ingestion of contaminated food and water. Recovered birds remain carrier for long period of time.
 - ✓ **Clinical sign:** In cute cases, anorexia, drop egg production, depression, ruffled feather and their eyes closed, is a common finding. Respiratory distress with rapid breathing can occur but the most characteristic sign is a watery to mucoid yellow diarrhoea. In chronic progressive loss of condition and intense anaemia develops which produce shrunken, pale combs and wattle
 - ✓ **Diagnosis:** by isolation and identification of the organism
 - ✓ **Treatment:** furazolidone continuously in the feed for 10 days. Recently, Enrofloxacin has been used
 - ✓ **Control:** vaccination.

2.5.2 Escherichia coli

Escherichia coli are a type of bacteria commonly found in the intestines of humans and other animals. While many strains of E. coli are harmless and even beneficial, some strains can cause illness they are Gram-negative rods - motile with flagella and Facultative anaerobes.

Page 53 of 132	Ministry of Labor and Skills Author/Copyright	Animal Health Level -3	Version -1
			May, 2023

- **Colibacillosis**

- ✓ **Synonyms:** Calf scour, white diarrhoea

Colibacillosis is one of the diseases of new borne animals (calves, piglets, lambs and foals) characterized by marked prostrations, profuse diarrhoea and septicemia.

- ✓ **Etiology:** *E. coli* is the causative agent. The agent produces endotoxin.

- ✓ **Occurrence:** The disease is common in farm animals under 3 days of age but it may occur as early as 12-18 hours after birth and occasionally occurs in calves up to several days of age when there is mixed infection with viral pathogens. There is high mortality in calves (10-50%) and piglets.

- ✓ **Species affected:** Incidence of colibacillosis is higher in dairy herds (calves) but also affect piglets, foals, lambs& kids.

- ✓ **Transmission:** The disease is transmitted through contaminated feed and water. In most species the major primary source of infection is the faeces of infected animals. In addition calves obtain the organism from contaminated bedding and calf pans, diarrheic calves, mastitic milk (coliform mastitis) and from the skin of the perineum and udder of the cow, and from the animal attendant.

- ✓ **Clinical findings:** The disease appears as acute, per acute or chronic form.

- ✚ In per acute infection there is marked diarrhoea, prostration and death within 12 hours.

- ✚ Acute signs appear when the newborn animal is 1-2 days old. The usual signs are depression, in appetite, stiffness, sunken eyes, tucked up abdomen, rough hair coat, elevation of temperature and rapid pulse and respiratory rate. There are profuse loose faeces (yellowish brown to greyish white) with rapid peristalsis. There is straining in evacuation of faeces, which has a fetid odor. There is soiling of the anal, and hindquarter region. There is rapid dehydration and ultimately death.

- ✚ Chronic cases show joint ill, navel ill and pneumonia.

- ✓ **Diagnosis:** This is based on history, clinical findings, lesions, isolation of organism in culture and serological tests.

- ✓ **Treatment:** The consideration for treatment of acute neonatal diarrhoea includes.

- ✚ Fluid & electrolyte replacement to prevent dehydration, acidosis and electrolyte imbalance. Give by parenteral or oral route. Glucose, sodium bicarbonate, sodium chloride; potassium chloride and water.
- ✚ Chloroamphenicol 25-50 mg/kg day body wt.
- ✚ Ampicillin 7-10 mg/kg for 5-7 days may be given orally or parentally.
- ✚ Streptomycin and tetracycline may also be used.
- ✓ **Control** - The following three principles are important in control of colibacillosis
 - ✚ Reduce the degree of exposure of new borne calves and piglets to the infectious agent.
 - ✚ Provide maximum non-specific resistance with adequate colostrums and optimum animal husbandry.
 - ✚ Increase the specific resistance of the newborn by vaccination of the dam or the new born by *E. coli* K⁹⁹ antigen.
 - ✚ Isolate affected animals from other susceptible calves and piglets.

2.5.3 Yersinia

Yersinia is a bacterial genus of the family Enterobacteriaceae that causes various diseases in both humans and animals. One of the economically important diseases in animals caused by Yersinia is Yersiniosis.

- **Yersiniosis**

- ✓ **Etiology:** Yersiniosis is caused by bacteria of the Yersinia genus, specifically Yersinia enterocolitica or Yersinia pseudotuberculosis.
- ✓ **Transmission:** The bacteria can spread from infected animals through direct contact or by contaminated food or water. Ingestion of contaminated feces can also lead to infection.
- ✓ **Host preference:** Yersinia enterocolitica typically affects pigs and other animals, while Yersinia pseudotuberculosis primarily affects rodents and rabbits. However, both species can also infect other animals including cattle, sheep, dogs, and cats.
- ✓ **Clinical signs:** Symptoms of Yersiniosis in animals include fever, diarrhea, abdominal pain, vomiting, and dehydration. Severely infected animals may also develop septicemia.

- ✓ **Diagnosis:** Diagnosis can be made through fecal or blood culture or by polymerase chain reaction (PCR) testing.
- ✓ **Treatment:** Antibiotics such as tetracycline, chloramphenicol, or fluoroquinolones can be used to treat Yersiniosis in animals. Supportive care such as fluids and electrolytes can also help to manage symptoms.
- ✓ **Prevention/control measures:** To prevent Yersiniosis in animals, it's important to practice good hygiene and sanitation practices such as regular cleaning and disinfection. Minimizing exposure to contaminated environments and properly cooking food can also reduce the risk of infection. Vaccines are available for some animal species but they are not commonly used.

2.6. Diseases caused by Mycobacterium

Mycobacterium are aerobic, nonmotile, acid-fast bacteria. They are classified into two groups: the fast-growing mycobacterium and the slow-growing mycobacterium. The slow-growing mycobacterium include the Mycobacterium tuberculosis complex (*M. tuberculosis*, *M. bovis*, *M. africanum*, and *M. microti*), *Mycobacterium leprae*, and *Mycobacterium avium* complex (MAC). The fast-growing mycobacterium include species such as *Mycobacterium fortuitum* and *Mycobacterium chelonae*.

- **Bovine tuberculosis**

It is a chronic contagious disease of man and animals caused by certain pathogenic organisms of the genus *Mycobacterium*, characterized by development of tubercles with resultant caseation and calcification in any of the organs.

- ✓ **Etiology:** *Mycobacterium bovis* is the common cause of tuberculosis in cattle. However, avian strain (*M. avium*) and human strain (*M. tuberculosis*) can cause infection in cattle and other species of animals. The organism is acid-fast, non-sporulated and non-motile in nature. The tubercle bacilli are enveloped in a waxy or fatty capsule, which protect and make them resistant to external influences.
- ✓ **Occurrence:** The disease has a worldwide distribution.

- ✓ **Species Affected:** The disease can occur in all species including man. But of all domestic animals, cattle are most susceptible and bovine strain is most pathogenic over the other strains.
- ✓ **Transmission:** Age, malnutrition, concurrent infection, hygiene and management are important factors that affect the transmission as well as establishment of the disease. Susceptible animals contract infection directly through inhalation of infected droplets and ingestion of contaminated feed and water by sputum, faeces, uterine and vaginal discharges, semen, milk, lymph and wound discharges. Calf may contract the infection via infected milk or as a result of uterine infection. Congenital infection of calf occurs due to tuberculous metritis of dam. Cattle may be infected while they come in contact with tuberculous people or when exposed to secretion or excretion of infected human sewage. Tuberculous cattle are the major sources of milk-borne disease in human particularly in children.
- ✓ **Clinical Findings:** The disease is a chronic insidious in nature and therefore may not be apparent clinically in the initial stage of the disease. .
 - ✚ The infected animals lose body weight slowly but continuously in spite of provision of optimum nutrition.
 - ✚ The respiratory system is mostly affected. There is sign of dyspnoea, increased rate of respiration and deep, persistent, painful, dry and hacking cough. Signs of bronchopneumonia is evident. Tuberculous pleurisy may occur.
 - ✚ Percussion will produce dull sound on the affected consolidated areas.
 - ✚ The involvement of the digestive tract is characterized by abdominal pain, persistent diarrhoea, chronic bloat and emaciation.
 - ✚ When the uterus is involved there is purulent caseous discharge tinged with blood. The affected animal may also show irregular oestrus cycle, abortion and sterility.
 - ✚ Tuberculous mastitis is a major importance because of the danger to public health and spread of the disease to calves.
- ✓ **Necropsy Findings:** The organs most commonly affected include lungs, liver, pleura, peritoneum, kidney, spleen and regional lymph nodes. In some cases bones and joints are involved. Male and female genital organs can also affect. Tuberculous lesions (granulomatus lesions) are the main lesions. The lesions may be large and involve the

entire lungs or liver or small and numerous; such condition is referred as **milliary tuberculosis**. In bovine tuberculosis lungs and associated lymph nodes are usually affected. Tuberculous lesions may persist in animals' body for the entire lifetime.

✓ **Diagnosis:** This is based on the following:

- ✚ Clinical signs: characterized by gradual emaciation, stunted growth, enlargement of lymph nodes, diurnal variation of temperature and characteristic coughing.
- ✚ Necropsy finding: clinical lesions are usually recognized during post-mortem examination of a slaughtered animal or the one that has died of the effect of the disease.
- ✚ Laboratory diagnosis: laboratory diagnosis is made by direct microscopic examination (sputum and milk) stained with ziehl - Nielsen may demonstrate acid-fast rods and culture (The samples which can be used to culture acid fast organisms are sputum, faeces, nasal discharge, urine, uterine discharge, milk and discharges from lymph nodes.)
- ✚ Radiology: radiological examination is helpful in small-domesticated animal (pet animals).
- ✚ Single Intradermal tuberculin test: In this case 0.1ml tuberculin or purified protein derivative (PPD) injected intradermally in the middle portion of neck of bovine, a site preferred due to maximum sensitivity. Prior to tuberculin injection, the skin should be cleaned, shaved and thickness of skin fold measured using caliper. The reaction can be read after 72 hr. of injection; the positive reaction is characterized by a hot, red and painful swelling with an increase of 4mm or more in the skin; less than 2mm is negative and between 2-4mm is doubtful.

✓ **Treatment:** treatment of tuberculosis in animals is generally attempted more or less in the same line as of human treatment. Since treatment takes long duration, in animals chemotherapy is not recommended because of economic reason and transmission of the disease while the animal is under chemotherapy. But if necessary, isoniazid and Streptomycin can be used against tuberculosis in animals.

✓ **Control:** The test and slaughter policy is the only method by which effective eradication of the disease can be achieved. Hygienic measures to prevent the spread of infection should be instituted as soon as the first group of reactors is removed. 5%

phenol can be used to disinfect all feeding and watering troughs. It is most important that calves being reared as herd replacements be fed on tuberculosis-free milk, either from known free animals or pasteurized. Avoid or prevent communal use of watering facilities or pasture.

- **Paratuberculosis**

- ✓ **Synonym:** Johne's disease

Johne's disease is specific infectious enteritis of cattle, sheep and goats. It is characterized by progressive emaciation in all species affected, and in cattle by chronic diarrhoea and a thickening and corrugation of the wall of the intestine.

- ✓ **Etiology:** the disease is caused by *Mycobacterium Paratuberculosis*. The organism is a small, acid-fast bacillus.
- ✓ **Occurrence:** The disease occurs throughout the world. Infection occurs in animals at a very early age, usually under 30 days of age, and clinical disease does not occur until 3-5 years of age. However calves reared on infected nurse cows may develop clinical disease at 13-18 months of age. Adult animals exposed for the first time may either develop clinical disease or they may become carriers of the organism without manifesting clinical signs
- ✓ **Species Affected:** Johne's disease is essentially a bovine disease but other ruminants like sheep and goats are also susceptible.
- ✓ **Transmission:** under field conditions the disease is transmitted principally by ingestion of feed and water contaminated by the faeces of infected animals. Calves may acquire the infection in their intra-uterine lives and/or through the way of alimentary tract by consuming milk from cows with clinical Johne's disease.
- ✓ **Clinical Findings:** The incubation period of the disease is long (15-18 month). In cattle, clinical signs do not appear before 2 years of age
 - ✚ Emaciation, sub-mandibular edema and absence of fever. The edema has a tendency to disappear as diarrhoea develops.
 - ✚ The animal eats normally but thirst is excessive. The faeces are soft and thin and without offensive odor (no blood, epithelial, debris and mucus)
 - ✚ Diarrhoea may be continuous or intermittent and the disease always terminates in severe dehydration, emaciation and weakness

- ✚ Cases occur only sporadically because of the slow rate of spread of the disease.
- ✚ In sheep and goats the disease is manifested by emaciation and less severe diarrhoea (faeces lose its usual pelleted form). Sheep may shed wool, lose weight and goats may show depression and dyspnoea..
- ✓ **Diagnosis:** The characteristic features of clinical Johne's disease includes the following:
 - ✚ Clinical signs like progressive weight loss and emaciation in single animal and chronic diarrhoea, which does not respond to therapy.
 - ✚ Definitive diagnosis can be obtained by using a combination of serological tests, faecal culture and biopsy of intestine.
 - ✚ Cutaneous delayed hypersensitivity tests may be used but bovine or avian tuberculosis and animals vaccinated against Johne's disease may give suspicious or positive reaction.
 - ✚ Rectal scrapings or rectal pinch biopsy are used. A positive finding is clumps of acid-fast bacilli in epithelial cells or macrophages.
- ✓ **Differential diagnosis:** In cattle: Salmonellosis, coccidiosis, parasitism, and malnutrition. In sheep and goats: caseous lymphadenitis, GI parasitism, Internal abscesses, ovine progressive pneumonia, etc
- ✓ **Treatment:** Not recommended due to adverse course of the disease. However, streptomycin 50mg/kg body weight per day may result transient improvement in clinical signs.
- ✓ **Control:** The lack of accurate tests and the long incubation period of the disease make Johne's disease difficult to control. On herd basis control depends upon eradication of infected animals Hygiene to prevent further spread .In some instances, vaccination to increase the resistance of the residual population. In cattle vaccination is carried out only in calves less than

2.7. Diseases caused by *Mycoplasma* and *Rickettsia*

2.7.1 *Mycoplasma*

Are the smallest prokaryotes capable of self-replication- (too small to seen under light microscope). Do not have cell wall- (Don't stain with a Gram's stain) completely resistant to penicillin and cephalosporin and vancomycin. It has a high content of sterols to prevent osmotic lysis. Part of normal flora of human genital tract or oral cavity of healthy adults. Grow on media enriched with serum (need cholesterol) gives Fried egg colonies on agar media.

- **Contagious Bovine Pleuropneumonia (CBPP)**

It is an acute, sub-acute, or chronic disease characterized by formation of massive pathological changes in the lungs and pleura.

- ✓ **Etiology:** *Mycoplasma mycoides* Var. *Mycoides* (small colony type) is the cause of the disease in cattle. The organism is extremely plemorphic in nature and sensitive to all environmental influences including disinfectants, heat and drying.
- ✓ **Occurrence:** The disease is one of the major plagues in cattle causing heavy losses in many parts of the world.
- ✓ **Species Affected:** This is primarily a disease of cattle, although the organism can infect sheep and goat.
- ✓ **Transmission:** The principal method of spread of this disease is by inhalation of infective droplets from active or carrier cases of the disease. Because of the method of spread, outbreaks tend to be more extensive in housed animals and in those in transit by train. A recovered cattle act as **a carrier**. They contain the organism in sequestra or in necrotic areas enclosed by connective tissue capsule of previously affected lungs. Such cattle are called '*Lungers*'. The sequestra may act as potential source of infection for a period as long as three years. Conditions of stress due to starvation, exhaustion or inter current disease can cause the sequestrum to break down and convert the animal into an active case.
- ✓ **Pathogenesis:** CBPP is an acute lobar pneumonia and pleurisy developing by localization from initial septicemia. An essential part of the pathogenesis of the disease is thrombosis in pulmonary vessels, probably prior to the development of pneumonic lesions. Death results from anoxia and presumably from toxemia. Approximately 50%

of animals that become infected go through mild form of the disease and often recognized as clinical cases.

✓ **Clinical Findings:** incubation period of the disease is 3-6 weeks

- ✚ Sudden onset of high fever (40°C), a fall in milk yield, anorexia, and cessation of rumination.
- ✚ There is severe depression and the animals stand apart or lag behind a travelling group.
- ✚ Coughing, first only on exercise, and chest pains are evident and disinclined to move.
- ✚ Standing with the elbows abducted, back arched and head extended,
- ✚ Respirations are shallow, rapid and accompanied by expiratory grunting.
- ✚ Auscultation reveals pleuritic frictional sounds in early stages of acute inflammation and dullness, fluid sounds and moist gurgling in the later stages of infection.
- ✚ Dullness of area of the lung may be detectable on percussion.
- ✚ Dilated nostrils, Drooling of saliva, nasal discharge and abortion of pregnant cow are also observed.

✓ **Diagnosis:** This is based on the following consideration.

- ✚ History - Prolonged incubation period, history of contact with infected animals.
- ✚ Clinical Findings-Typical signs of respiratory involvement
- ✚ Necropsy-Classical marbled appearance of the lung.
- ✚ Culture of the organism in special media. Pleomorphic organisms are seen in culture with predominant filamentous form.

✓ **Differential Diagnosis:** Pasteurellosis and parasitic pneumonia

✓ **Treatment** is recommended only in endemic areas because elimination of the organism may not be possible, and carriers may develop. Antibiotic treatment is generally not effective as it can result in extensive tissue damage and sequestration of the organism, although tylosin has been reported to be moderately effective. As soon as an outbreak is suspected, slaughter and necropsy of a suspect animal is advisable. Tylosin (10mg/kg every 12hrs for six injection), Spiramycin (10-50mg/kg for 3 days) are effective in the treatment of clinical cases

✓ **Control and prevention**

- ✚ Any measure which brings animals together should be avoided
- ✚ Strict quarantine of infected and in contact animals must be maintained
- ✚ infected animals should be removed as soon as possible
- ✚ Vaccination is effective procedure in the control of CBPP
- ✚ calves are vaccinated when 2 month old
- ✚ Eradication by test and slaughter of reactors

• **Contagious Caprine Pleuropneumonia (CCPP)**

This is a contagious disease of goats having resemblance with CBPP. CCPP has many similarities clinically and at necropsy to CBPP, but it is not transmissible to cattle.

- ✓ **Etiology:** *Mycoplasma mycoides* var. *caprae* is the causative organism.
- ✓ **Occurrence:** The disease is wide spread in tropical countries
- ✓ **Transmission:** The disease is readily transmitted by inhalation, but the organism does not survive for long time outside the animal body, so that a carrier animal brings infection into the flock.
- ✓ **Clinical Findings:** This is a highly fatal disease of sheep and goat. Mortality varies from 60-100%. The incubation period is about 4 days. The symptoms are comprised of anorexia, dullness, depression, nasal discharge, coughing, dyspnoea, fever (40.5-41.5°C) and in the terminal stages mouth breathing, tongue protrusion and frothy salivation with death in 2 or more days.
- ✓ **Diagnosis:** similar to CBPP
- ✓ **Treatment:** Treatment of cases of CCPP with tylosine tartrate (10mg/kg) or oxytetracycline (15mg/kg body weight) is highly successful
- ✓ **Control and prevention:** It depends on: Isolation of affected flocks, Vaccination

• **Chronic Respiratory Disease (CRD)**

- ✓ **Synonyms:** infectious sinusitis

CRD affects all parts of the respiratory tract of chickens and turkeys, and is characterized by being extremely responsive to stress factors.

- ✓ **Etiology:** *Mycoplasma gallisepticum*.
- ✓ **Occurrence:** It is prevalent in almost every tropical country.

- ✓ **Species Affected:** The disease is economically important in chickens and turkeys.
- ✓ **Transmission:** The major route is via the egg and the disease continues from generation to generation by embryonic transmission.. Air born transmission is also common and can occur rapidly among pen mats or between pens. The infection may be dormant in the infected chick for days to months, but when it is stressed, aerosol transmission occurs rapidly and infection spreads through the flock. In addition, the infection may be carried by person, by fomites or by infected birds introduced into a clean flock.
- ✓ **Clinical Findings:** Affected birds have varying degrees of respiratory distress, slight to marked rales, difficulty in breathing, coughing and sneezing. Morbidity is high and mortality low in uncomplicated cases. Feed efficiency and weight gain are reduced. In laying flocks, birds fail to reach peak egg production and the production rate is lower than normal.
- ✓ **Necropsy Findings:** uncomplicated CRD infection in chicken result in relatively mild sinusitis, trachitis and air sacculitis. *E. Coli* infection is concurrent and results in severe air sac thickening and turbidity, with exudative accumulations, fibrino-purulent pericarditis and per hepatitis.
- ✓ **Diagnosis:** Because of many poultry diseases that produce lesions in the respiratory tract, accurate diagnosis of CRD from lesions alone cannot be made. The disease can be confirmed by isolation and identification of the organism.
- ✓ **Differential Diagnosis:** Infectious coryza, laryngo trachitis, etc...
- ✓ **Treatment:** In the field many cases of CRD infection are complicated by other pathogenic bacteria, thus, effective treatment must also attack the secondary invader. Most strains of the organism are sensitive to a number of antibiotics (such as oxytetracycline, streptomycin, etc) and tylosin.
- ✓ **Control:** Avoid stress of any kind. Introduction of birds from endemic farms or areas has to be prevented. Workers and other personnel do not move from flock to flock on the same day. Vaccination is used to control the disease but not fully reliable.

2.7.2 Rickettsia

Is an obligate intracellular and grow in cytoplasm of eukaryotic cells. Pleomorphic, small gram-negative rod to coccoid. Stain poorly with gram stain but can be visualized with Giemsa method. Reservoirs are animal and arthropod vector. Zoonotic spread to human by arthropod.

- **Cowdriosis**

- ✓ **Synonym:** heart water

This is a tick-borne disease of cattle, sheep and goats; characterized by fever, nervous signs, edema of the body cavities and diarrhoea.

- ✓ **Etiology:** *Cawdria ruminantum* is the cause of the disease. It is the rickettsial disease

- ✓ **Occurrence:** Heart water affects imported breeds of cattle, sheep and goats and it is limited in its occurrence to Africa, Madagascar and some of the West Indian countries. Generally, young animals have an innate resistance against rickettsial diseases.

- ✓ **Transmission:** transmitted by ticks of the genus *Amblyomma*, especially *A. Variegatum*.

- ✓ **Clinical Findings:** The incubation period is 1-3 weeks.

- ✚ Per acute cases show only high fever, prostration and death with terminal convulsions.

- ✚ Acute cases have a course of about 6 days and show the nervous syndrome, which is characteristic of heart water. It comprises ataxia, chewing movements, twitching of the eyelids, circling, aggression, apparent blindness, recumbence, convulsion and death. In less severe cases the principal sign may be diarrhoea. The mortality rate in per acute cases is 100%, in acute cases 50-90% and in mild cases most animals recover.

- ✓ **Necropsy Findings:** Standard lesions are ascites, hydrothorax, and hydro pericardium. There may be pulmonary edema and sub-serosal haemorrhages in most cavities and there is splenomegaly and lymphadenopathy.

- ✓ **Diagnosis:** Identification of the rickettsial organisms in a stained, squash preparation of tissue. Serological tests like ELISA

- ✓ **Treatment:** Tetracycline is the standard treatment and may be used as a prophylactic measure.

✓ **Control and prevention:** eradication of the tick population is not usually a practical option but regular dipping of animals 3-7 days interval is a common practice in enzootic areas. Vaccination -this is possible through artificial exposure of animals followed by tetracycline administration. Artificial exposure can be affected in several ways, the most common of which is the intravenous, injection of heart water infected blood or ground up infected ticks.

- **Q Fever**

- ✓ **Etiology:** *Coxiella burnetii*, species of rickettsia.
- ✓ **Transmission:** Q fever is most commonly contracted from inhaling contaminated dust particles, usually from soil, animal waste, or unpasteurized dairy products.
- ✓ **Host preference:** Many different animal species, including cows, sheep, and goats can carry the bacteria and transmit them to humans. –
- ✓ **Clinical signs:** Symptoms can vary widely from mild flu-like symptoms to severe pneumonia or hepatitis. Chronic Q fever can lead to more serious complications such as endocarditis (inflammation of the heart lining).
- ✓ **Diagnosis:** Diagnosis is usually made through blood tests that detect antibodies to the bacteria in the patient's blood. Treatment: Antibiotics such as doxycycline are typically used to treat Q fever.
- ✓ **Prevention/Control Measures:** Prevention measures include proper disposal of animal waste, pasteurization of dairy products, and vaccination for individuals at high risk for exposure. Personal protective equipment such as gloves and masks may also be recommended for those who work with animals.

2.8. Identifying and handling cases of mastitis

The term mastitis refers to inflammation of the mammary gland regardless of the cause. It is characterized by physical, chemical and usually bacteriological changes in the milk and by pathological changes in the glandular tissue. The most important changes in the milk include discoloration, the presence of clots and the presence of large numbers of leukocytes.

- ✓ **Etiology:** Mastitis can be categorized in to three based on etiology.

- ✚ **Specific mastitis:** caused by true pathogens that do not require predisposing factors. Such mastitis usually takes place through haematogenous route. Ex- Tuberculosis, Brucellosis, leptospirosis.
- ✚ **Non -specific Mastitis:** This is caused by agents that are normally found on the surface of the skin and teat. Most commonly staphylococcus and streptococcus are responsible for this type of mastitis. These are not true pathogens and they require predisposing factors to produce mastitis.
- ✚ **Environmental Mastitis:** This is usually associated with the number of micro organisms in the environment. Ex-E. coli (Coliform mastitis)
- ✓ **Species Affected:** All breeds of dairy cows, goats, sheep, pig, horse are susceptible. High yielding dairy cows are more commonly affected than low yielder. Infection rate is more in successive lactation than the first lactation. Exotic and cross breed cows are more prone to mastitis than our local cattle.
- ✓ **Transmission:** There are two sources of infection. These are the udder (organisms that are normal inhabitant in the udder) and contaminated environment. The cutaneous surface of the cow may have many organisms as resident population and from where the organism may have the chance to invade through contamination by handlers. The contamination of milkers' hands, clothes, and milking machine cup by milk from the affected quarter may lead to the spread of the disease to other non-infected teats of cow.
- ✓ **Clinical Findings:** Mastitis may clinically be classified as:
 - ✚ **Per acute mastitis:** This is the most serious form of mastitis, which most often endangers the life of the animal. The affected animal shows a very high temperature, remain off feed and show respiratory distress. The udder is swollen and extremely painful. There is cessation of milk secretion and exudates are often blood stained.
 - ✚ **Acute Mastitis:** In this case systemic reactions are slight to moderate. The udder becomes swelled and there is change in the milk.
 - ✚ **Sub-acute Mastitis:** There are variable changes in the milk but practically no changes in the udder tissues. Culture of milk will show presence of pathogenic bacteria.

✚ **Chronic Mastitis:** This is the terminal stage of the disease. Udder becomes hard due to fibrosis.

✓ **Diagnosis:** This is based on the following consideration.

✚ Physical examination of the udder; Shape, size and consistency of the udder should be viewed properly. Detailed examination of the teat and teat orifices should be made.

✚ Signs of inflammation (hot, swelling, pain, redness, and loss of function) should be clinically assessed.

✚ Tests for milk abnormalities: there are different testes used to indicate the presence of abnormalities in the milk. Ex- California mastitis test (CMT), White side test etc

✚ Direct test: Identification and isolation of the organisms from suspected milk.

✚ Cultural examination

✓ **Treatment:** Isolation of the affected animal should be made from rest of the herd.

✚ Healthy quarter should be milked first before milking of affected quarter.

✚ Removal of secretions as much as possible should be attempted before administration of treatment.

✚ Intra-mammary antibiotic preparations should be used for local treatment.

✚ Systemic antibiotic therapy may be necessary where systemic reactions are evident.

✚ Supportive treatment may be required. It includes parenteral injection of large quantities of isotonic fluid containing glucose and corticosteroids.

✚ Drying off Quarters: When the quarter does not respond to treatment the animal is required to be made dried off by using chemical agents. This is done by infusion of 30-60 ml of 3% silver nitrate solution or 20ml of 5% copper sulphate.

✓ **Control:**

A) Reduction of duration of infection. This may be achieved by:

✚ Treatment of all quarters of all cows at drying off

✚ Treatment of clinical cases as soon as possible

✚ Culling of chronic or clinically non-responsive cases

B) Reduction of new infection rate. This may be achieved by:

✚ Dipping of all teats following milking

- ✚ Washing of udder and hand of the milker before and after milking
- ✚ Milking order I.e. the healthy, non-infected cows should be milked first and known infected cows should be milked at least.

Self-Check 2	Written test
--------------	--------------

Name _____ ID _____ signature _____ date _____

Part I. Write **True** if the statement is correct and **False** if the statement is incorrect (1pt each)

1. Paratuberculosis affect young stocks and highly nourished rapidly growing animals
2. *Actinomyces bovis* is a common inhabitant of bovine oral cavity.
- 3 Botulism result in spastic paralysis caused by ingestion of preformed toxin

Part II: choices the correct answer (1pt each)

1. How do bacteria reproduce?
 - A. Sexual reproduction
 - B. Horizontal gene transfer
 - C. Binary fission
 - D. Mitosis
2. Which one of the following is soil-borne disease?
 - A. Foot and Mouth Diseases
 - B. Pasturellosis
 - C. Tuberculosis
 - D. Anthrax
3. Disease of new borne animals caused by *Escherichia coli* is called
 - A. Colibacillosis
 - B. Lumpy Skin Diseases
 - C. Actinomycosis
 - D. Salmonellosis
4. Is acute or chronic contagious disease of domestic animals that causes placentitis, abortion and persists most in sexually mature animals.
 - A. CBPP
 - B. Brucellosis
 - C. Glanders
 - D. Salmonellosis
5. Characterized by development of tubercles with resultant caseation and calcification in any of the organs.
 - A. Tuberculosis
 - B. CBPP
 - C. Brucellosis
 - D. Salmonellosis

6. Which of the following diseases characterize by protrusion of third eyelid and spastic paralysis? A. Rabies B.blackleg C.PPR D. Tetanus

Part II: Short Answer Questions (2pts each)

1. How to diagnose tuberculosis
2. List diseases caused by Mycolasma
3. Explain Disease caused by Enterobacteriaceae
4. What are general characteristics of bacteria?

Note: Satisfactory rating –17 points

Unsatisfactory – 17 below points

LG #3	LO3 # - Identify and handle viral disease of animals
--------------	---

Instruction sheet 3

This learning guide is developed to provide you the necessary information regarding the following content coverage and topics:

- Unique feature and replication of viruses
- Etiology, host, pathogenesis, clinical sign, diagnosis, control and prevention of DNA viral diseases
- Etiology, host, pathogenesis, clinical sign, diagnosis, control and prevention of RNA viral diseases

This guide will also assist you to attain the learning outcomes stated in the cover page. Specifically, upon completion of this learning guide, you will be able to:

- Identify Unique feature and replication of viruses
- Identify and implement Etiology, transmission, host preference, clinical sign, diagnosis, treatment and prevention/control measures of economically important and prevalent diseases of animals' caused by DNA viruses
- Identify and implement Etiology, transmission, host preference, clinical sign, diagnosis, treatment and prevention/control measures of economically important and prevalent diseases of animals' caused by RNA viruses

Learning Instructions:

1. Read the specific objectives of this Learning Guide.
2. Follow the instructions described below.
3. Read the information written in the information Sheets
4. Accomplish the Self-checks

Information Sheet 3

3.1. Unique feature and replication of viruses

Viruses are small, obligate intracellular parasites, which can replicate only in living susceptible cells. They differ fundamentally from other classes of microorganisms in that, they possess a central core of nucleic acid enclosed within a protein - rich outer coat. The genome nucleic acid is either DNA or RNA. They lack enzyme systems (no independent growth and multiplication). They are not susceptible to antibiotic or other agents that act against the metabolic pathways of microorganisms.

Viruses are usually termed as “at the edge of life” because they lack any functional organelles except nucleic acid. They possess attributes of both aliveness and inanimate things.

Living Characteristics

Able to reproduce by their NA & carry forward their next generation. Once inside their prospective host, viruses replicate at an extremely fast rate. Individual virus can undergo mutation under specific conditions.

Non-Living Characteristics

Do not grow or replicate by their own. Do not perform any metabolic activities, unless being triggered by the respective host. Acellular (no cells) = devoid of cytoplasm & cell organelles

The various properties of viruses are listed as follows

- Obligate intracellular Microorganism
- Smallest in the size (150-200 nm but poxvirus reaches 350nm)
- Lack a genetic material for the synthesis of enzymes responsible for energy metabolism.
- Depends on a living host cell to get energy, precursors, enzymes, & ribosomes
- Consists single type of nucleic acid (NA), either DNA or RNA but never both.
- All viruses consist of a protective protein coat (capsid)

- Sometimes nucleic acid (NA) is surrounded by envelope.
- Virus reproduces solely from nucleic acid but not by division
- Viruses are not sensitive to antibiotics
- Can infect unicellular MOs (mycoplasmas, bacteria, and algae), all higher plants & animals.
- Morphologically most viruses are spherical or roughly spherical with the exception of
 - ✓ Rhabdovirus- bullet or rod shaped
 - ✓ Poxviruses - brick shaped

Virus structure

- ✚ The virion is a gene delivery system; it contains the virus genome,
- ✚ The genome is packaged in a protein structure known as a capsid.
- ✚ Nucleocapsid is the protein-nucleic acid complex
- ✚ Many viruses also have a lipid component, generally present at the surface of the virion forming an envelope,
- ✚ The envelope also contains proteins (spikes) with roles in aiding entry into host cells.

Viral envelope

- It is a lipid or lipoprotein enclosed the capsid
- Most virions acquire envelopes during maturation by budding from cellular membrane.
- Lipid component derived directly from host cell but protein component is virus-coded.

Viral replication

Is the formation of biological viruses during the infection process in the target host cells. Viruses must first get into the cell before viral replication can occur. Through the generation of abundant copies of its genome and packaging these copies, the virus continues infecting new hosts. Replication between viruses is greatly varied and depends on the type of genes involved in them. Most DNA viruses assemble in the nucleus while most RNA viruses develop solely in cytoplasm.

Viruses multiply only in living cells. The host cell must provide the energy and synthetic machinery and the low- molecular-weight precursors for the synthesis of viral proteins and nucleic acids.

The virus replication occurs in seven stages, namely;

1. Attachment
2. Entry,
3. Uncoating,
4. Transcription / mRNA production,
5. Synthesis of virus components,
6. Virion assembly and
7. Release (Liberation Stage).

Attachment: It is the first step of viral replication. The virus attaches to the cell membrane of the host cell. It then injects its DNA or RNA into the host to initiate infection. In animal cells these viruses get into the cell through the process of endocytosis which works through fusing of the virus and fusing of the viral envelope with the cell membrane of the animal cell and in plant cells it enters through the process of pinocytosis which works on pinching of the viruses.

Entry: The cell membrane of the host cell invaginates the virus particle, enclosing it in a pinocytotic vacuole. This protects the cell from antibodies like in the case of the HIV virus.

Uncoating: Cell enzymes (from lysosomes) strip off the virus protein coat. This releases or renders accessible the virus nucleic acid or genome.

Transcription / mRNA production: For some RNA viruses, the infecting RNA produces messenger RNA (mRNA), which can translate the genome into protein products. For viruses with negative stranded RNA, or DNA, viruses are produced by transcription then translation. The mRNA is used to instruct the host cell to make virus components. The virus takes advantage of the existing cell structures to replicate itself.

Synthesis of virus components: The components are manufactured by the virus using the host's existing organelles . Viral proteins: Viral mRNA is translated on cellular ribosomes into two types of viral protein:

Structural: proteins which make up the virus particle. **Nonstructural:** proteins not found in the virus particle, mainly enzymes for virus genome replication.

Viral nucleic acid (genome replication): New viral genomes are synthesized; templates are either the parental genome or newly formed complementary strands, in the case of single-stranded genomes. These genomes are made by either a viral polymerase or (in some DNA viruses) a cellular enzyme, particularly in rapidly dividing cells.

Virion assembly: A virion is simply an active or intact virus particle. In this stage, newly synthesized genome (nucleic acid), and proteins are assembled to form new virus particles. This may take place in the cell's nucleus, cytoplasm, or at plasma membrane for most developed viruses.

Release: The viruses, now being matured are released by either sudden rupture of the cell, or gradual extrusion (force out) of enveloped viruses through the cell membrane. The new viruses may invade or attack other cells, or remain dormant in the cell. In the case of bacterial viruses, the release of progeny virions takes place by lysis of the infected bacterium. However, in the case of animal viruses, release usually occurs without cell lysis.

Diagnostic and Control methods of viral Infections

In general, the procedures that are available for the diagnosis of viral diseases fall into two main categories. Isolation and identification of the causative virus from clinical specimens

Serological tests for the detection and measurement of the specific antibodies that develop during the course of illness.

Before attempting a diagnosis, a good clinical history is desirable. This should include the nature of infection, clinical signs, lesions and other abnormalities, the stages of the illness, details of the geographical location, breed, age, sex of the animals affected, movements on and off the farm, previous vaccinations and so on.

Provisional diagnosis made based on the clinical history and detail of any post-mortem examination, together with the results of microscopical and histological examination. Confirmatory diagnosis can be obtained by isolation of the causal virus (Animal injection, cell

and tissue cultures), identification of the virus (cytopathic effects, fluorescent antibody staining, electron microscopy) and demonstration of a rise in titre of specific antibodies.

Methods of immunization of animals against viral diseases.

There are several methods of immunization animal against viral diseases including the following.

1. Inoculation of live/ non-attenuated virus.
2. Attenuated vaccines and
3. Killed vaccines

Live vaccines stimulate a stronger immunity over a longer period of time but there is possibility of inoculated virus multiplying in the animal body. Inactivated vaccines provide shorter immune reaction but are the safest types of vaccines to use.

3.2. DNA viral diseases

DNA viruses are viruses that have DNA as their genetic material. These viruses can infect a wide range of organisms, including animals, humans and plants. Some of the most common DNA viruses that infect animals include Pox virus, herpes virus and Parvo virus. These viruses can cause a wide range of diseases, including Lumpy skin diseases, sheep and goat pox, and other diseases. Treatment for DNA virus infections varies depending on the specific virus and the severity of the infection. Prevention measures include good hygiene practices and vaccination.

3.2.1 Pox virus

The family *Poxviridae* includes numerous viruses of veterinary and/or medical importance.

Poxviruses are large DNA viruses that are capable of infecting both invertebrates and vertebrates. Poxvirus diseases occur in most animal species, and are of considerable economic importance in some regions of the world. A characteristic of many of these viruses is their common ability to induce characteristic “pox” (pockmark) lesions in the skin of affected animals.

- **Lumpy Skin Disease**

It is a highly infectious skin disease of cattle that characterized by sudden appearance of nodules on all parts of the skin.

- ✓ **Etiology:** *Neethling poxvirus*. The *Neethling virus* has similar cultural characteristics to sheep pox and goat poxviruses. All the three are classified as Capri poxvirus. Moreover, prior to infection with sheep poxvirus confers immunity against the Neethling virus infection.
- ✓ **Species Affected:** All ages and breeds of cattle are susceptible to the causative virus except animals recently recovered from an attack, in which case there is a solid immunity lasting for about 3 months.
- ✓ **Occurrence:** Different African Countries such as South Africa, Kenya, Sudan, Chad, Niger, Central Africa and Ethiopia are now known to suffer from the disease.
- ✓ **Transmission:** The exact mode of transmission is not known, however, the rapid spread of the disease occur by insect vector.
- ✓ **Pathogenesis** In the generalized disease there is viremia accompanied by a febrile reaction. Localization in the skin occurs with development of inflammatory nodules. In the experimental disease following ID inoculation, local lesions can develop at the site of challenge without viremia and generalization of the infection.
- ✓ **Clinical Findings:** incubation period 2 to 4 weeks.
 - ✚ In severe cases there is an initial rise of temperature accompanied by lacrimation, nasal discharge, salivation and lameness.
 - ✚ Multiple nodules appear suddenly about a week later. They are round and firm, varying from 1 to 4 cm in diameter. They vary in number from a few to hundreds and in most cases are confined to the skin area.
 - ✚ In severe cases the lesions may also be present in the nostrils, causing respiratory obstruction and snoring. They may also be in the mouth. Nodules may develop on the conjunctiva, causing severe lacrimation.
 - ✚ Lymph nodes draining the affected area become enlarged and there may be local edema. Pregnant cows may abort.



Figure 3.1. Multiple nodules on skin

- ✓ **Necropsy Findings:** Granulomatous cutaneous nodules in the affected part of the skin. Similar lesions are present in the mouth, pharynx, trachea, bronchi and stomach and there may be accompanying pneumonia.
- ✓ **Diagnosis:** The rapid spread of the disease and the sudden appearance of lumps in the skin after an initial fever make this disease quite unlike any other disease in cattle. Serology
- ✓ **Treatment:** No specific treatment is available, but prevention of secondary infection is essential. The use of antibiotics or sulphonamides is recommended.
- ✓ **Control:** A safe vaccine against the Neethling virus is effective. It is administered to all animals over 6 months of age. A freeze-dried, live attenuated virus vaccine is also available. Vaccination of cattle with attenuated sheep poxvirus is effective in preventing infection with Neethling virus.

Note: Although the mortality rate is low (less than 10%), the economic loss caused is high due to loss of milk production, damage to hides and loss of body condition during the long course of the disease.

- **Sheep Pox and Goat Pox**

It is viral infectious disease sheep and goat characterized by fever and generalized pox lesion on the skin. They are the most serious of all pox diseases of domestic animals characterized by

Fever, Edema of the eyelids, Mucous discharge from nose, Appearing of cutaneous nodules all over the body, most obvious in the area of skin where the wool or hair is short.

- ✓ **Etiology:** *Capri poxvirus* (family *poxviridae*) affects sheep and goats and animals of all age are susceptible.
- ✓ **Occurrence:** These diseases are distributed in different parts of the world including Ethiopia. Sheep pox is the most serious of all the pox diseases in animals, often causing death in 50% of affected animals. Major losses may occur in each new crop of lambs. Goat pox in sheep is more severe than sheep pox, and lesions occur on the lips and oral mucosa, the teat and udder. The goat poxvirus affords solid protection in sheep against both sheep pox and goat pox but sheep pox vaccine does not protect goats against goat pox.
- ✓ **Species Affected:** Sheep pox affects sheep whereas goat pox can affect both sheep and goats. In all cases younger are more susceptible than adults.
- ✓ **Transmission:** The usual mode of transmission is contact with the infected animal. The virus may gain entrance through skin wounds and abrasions. The virus is present in the skin lesions and while the affected animals rub their body on other animals, the virus is passed directly to susceptible animals. Aerosol or droplet infection is quite possible.
- ✓ **Pathogenesis:** During an initial viremia, the virus is deposited in most tissues, including the skin. The virus is present in greatest quantities between the 7th and 14th day after inoculation. Circulating antibody limits spread of infection, but does not prevent replication of virus at the site of inoculation. Since the virus is *epitheliotropic*, it will infest the epithelium tissues of the organism. Generally, pox infections follow five stages while producing the lesions.
- ✓ **Clinical Findings:** There is an incubation period of 2-14 days. The disease may appear in three clinical forms.
 - 🚩 **Malignant Form:** This form is more common in lambs and kids. There is marked depression and prostration, a high fever and discharge from the eyes and nose. Affected animal may die during this stage before typical pox lesions develop. They lesions commence as papules, then become pustule, vesicular, and finally scabs. Skin lesions appear in unwooled skin and on the buccal,

respiratory, digestive and urogenital tract mucosa. The mortality rate in this form may reach 50%.

🚦 **Mild (Benign) form:** This is the common form in adults. Only skin lesions occur, particularly under the tail, and there are no systemic signs. The mortality rate is low, usually about 5%.

🚦 **Abortive form:** In this form generalization is rare. The mortality is low and affected pregnant animal may abort and the fetus may show pox lesions. Lactating ewe may show the signs of mastitis due to lesions in the udder.



Figure 3.2. Sheep and goat pox clinical sign

Source: https://www.google.com/search?q=goat+pox+and+sheep+pox+symptoms&tbm=isch&ved=2ahukewi55pqbgbyb_ahuzpcchaunc2kq2

- ✓ **Necropsy Findings:** Characteristic papules, pustules, vesicles and scabs are noted on cutaneous surfaces. Lesions may be observed in the mucosa of respiratory and alimentary tract especially on trachea.
- ✓ **Diagnosis:** based on the clinical sign (characteristic of a pox disease) and endemic occurrence of the disease. Serology like CFT
- ✓ **Differential Diagnosis:** Bluetongue and contagious ecthyma (orf).
- ✓ **Treatment:** Antiseptic or antibiotic ointments or lotions may be applied to control secondary bacteria complication.

- ✓ **Control and Prevention:** Strict sanitary measures are to be adopted. Sick animals are to be isolated from healthy one. Vaccination

- **Contagious Ectyima**

- ✓ **Synonyms:** contagious pustular dermatitis, orf

Contagious ecthyma is a viral disease of mainly sheep and goat characterized by formation of **nodules, vesicles, pustules** and **scabs** on mucous membrane oral cavity, lips, distal part of limbs, udder, reproductive organs and rarely on other parts of affected animals.

- ✓ **Etiology:** caused by virus belonging to family *pox viridae*. The virus is quiet resistance to many environmental conditions and persists year to year in contaminated premises
- ✓ **Occurrence:** has worldwide distribution.
- ✓ **Species affected:** it is a disease of sheep and goats but the disease has been recorded in human being.
- ✓ **Transmission:** the virus enters the hosts through abrasion on the skin of the lips and face. It also transmitted to the udder of goats and ewe by clinically affected suckling kids or lambs.
- ✓ **Pathogenesis:** Damage to the skin is essential for the establishment of orf infection and the development of typical lesions. After infection of the orf virus replicate in epithelium cells of skin cloves, lips and reproductive organs especially in vulva and prepuce. This causes proliferation and degenerative process in affected epithelial cells and form nodules with red colour. And there is formation of cytoplasmic inclusions and exudation. As the result the nodules are changed to vesicles then to pustules. Lastly the thick crust forms which can fall to the ground. What it makes different from **sheep pox and goat pox** is it never forms scar in affected epithelium tissues. In unfavorable condition the pustules may join together and are
- ✓ **Clinical findings:** the initial sign of the disease are dullness, depression, anorexia and rising of body temperature. There is profuse salivation, lacrimation accompanied with mucopurulent nasal discharge. Congestion of nasal and buccal mucosa with appearance of nodular eruption on the oral commissure, lips, mouth and nostrils. The lesions (proliferative lesions) also found in the tongue and gum. Lesions are followed by papules, vesicles, pustules and ulcer in 3-4 days. In mild cases, the disease is

febrile and self-limiting scab is formed within a week and peel off the skin within a month.

- ✓ **Diagnosis:** based on clinical sign and serology like, CFT
- ✓ **Differential diagnosis:** sheep pox and FMD
- ✓ **Treatment:** treatment is symptomatic. Apply 5% copper sulphate solution on the lesion.
- ✓ **Control and Prevention :** affected animal should be isolated from the herd

Vaccination of animals.

- **Camel Pox**

- ✓ **Etiology:** Camel pox virus (CMLV)
- ✓ **Transmission:** Camel pox is mainly transmitted through contact with infected camels or their bodily fluids. It can also be spread through contaminated bedding or equipment.
- ✓ **Host preference:** As the name suggests, camel pox primarily affects camels but can also infect humans in close contact with infected animals.
- ✓ **Clinical signs:** Clinical signs in camels include fever, skin lesions, and scabs on various parts of the body. In humans, symptoms are similar to those of smallpox but milder.



Figure 3.3. Camel pox

Source: <https://www.google.com/search?q=camel+pox&client=firefox-b-d&>

- ✓ **Diagnosis:** Camel pox can be diagnosed based on clinical signs and confirmed with laboratory tests.

- ✓ **Treatment:** There is no specific treatment for camel pox. Supportive care is used to manage symptoms in both camels and humans.
- ✓ **Prevention/Control Measures:** Preventative measures include vaccination of camels and the use of protective equipment when handling infected animals. Isolation of infected animals and enhanced hygiene measures can also help prevent the spread of the virus.
- **Fowl Pox**
 - ✓ **Etiology** caused by virus belonging to family *pox viridae*, fowl pox virus
 - ✓ **Species affected:** Most poultry chickens, turkeys, pheasants, quail, ducks, psittacine, and ratites of all ages are susceptible.
 - ✓ **Transmission:** Fowl pox is transmitted by direct contact between infected and susceptible birds or by mosquitoes. Virus-containing scabs also can be sloughed from affected birds and serve as a source of infection. The virus can enter the blood stream through the eye, skin wounds, or respiratory tract. Mosquitoes become infected from feeding on birds with fowl pox in their blood stream. There is some evidence that the mosquito remains infective for life. Mosquitoes are the primary reservoir and spreaders of fowl pox on poultry ranges. Several species of mosquito can transmit fowl pox. Often mosquitoes winter-over in poultry houses so, outbreaks can occur during winter and early spring.
 - ✓ **Clinical signs :** There are two forms of fowl pox.
 - 🌈 **The dry form** is characterized by raised, wart-like lesions on unfeathered areas (head, legs, vent, etc.). The lesions heal in about 2 weeks. If the scab is removed before healing is complete, the surface beneath is raw and bleeding. Unthriftiness and retarded growth are typical symptoms of fowl pox. In laying hens, infection results in a transient decline in egg production. In the wet form there are canker-like lesions in the mouth, pharynx, larynx, and trachea.
 - 🌈 **The wet form** may cause respiratory distress by obstructing the upper air passages. Chickens may be affected with either or both forms of fowl pox at one time.



Figure 3.4. Fowl pox

Source: <https://www.google.com/search?Q=fowl+pox&client=firefox-b-> (accessed on may 19/2023).

- ✓ **Treatment:** No treatment is available. However, fowl pox is relatively slow-spreading. Thus, it is possible to vaccinate to stop an outbreak. The wing-web vaccination method is used for chickens and the thigh-stick method for turkeys older than 8 weeks.
- ✓ **Control and Prevention:** Fowl pox outbreaks in poultry confined to houses can be controlled by spraying to kill mosquitoes. However, if fowl pox is endemic in the area, vaccination is recommended.

3.2.2 Herpes viruses

Herpesvirus virions are enveloped and include a core, capsid, & tegument. The core consists of the viral genome packaged as a single, linear dsDNA molecule within the protein capsid that in human herpesviruses has an external diameter of ~125 nm and is composed of 162 hollow capsomers—150 hexons and 12 pentons. The DNA genome is wrapped around a fibrous spool-like core, which has the shape of a torus and appears to be suspended by fibrils that are anchored to the inner side of the surrounding capsid and pass through the hole of the torus. Surrounding the capsid is a layer of globular material, known as the tegument, which is enclosed by a typical lipoprotein envelope with numerous small glycoprotein spikes.

- **Marek's Disease**

Marek's disease ('MD' or 'fowl paralysis') is **a very common disease of chickens caused by a herpes virus**. Marek's disease affects both commercial and backyard poultry and may result in death or severe production loss. The disease causes changes in many of the nerves and may cause tumours in major internal organs. It is economically important neoplasms in poultry.

- ✓ **Etiology:** it is caused by a *herpes virus*. Three genotypes are recognized for this cell-associated herpes and virus serotype 1 is the virulent chicken isolate.
- ✓ **Occurrence:** Marek's disease is one of the most ubiquitous avian infections (world wide)
- ✓ **Species Affected:** Chickens are the only important natural host
- ✓ **Transmission:** Marek's disease is highly contagious and is readily transmitted among chickens. The virus is released into environment from the epithelium of the feather follicle. It may survive for months or years in poultry house litter or dust. Infection usually occurs through aerosol exposure. Once infected chickens continues to be carriers for long periods and act as a source of infection.
- ✓ **Clinical Findings:** Depression is noted prior to death. Sometimes transient paralysis is also noted. Chicken becomes ataxic for periods of several days, followed by recovery. This syndrome (transient paralysis syndrome) is rare in immunized birds,



Figure 3.5. Transient paralysis

Source: <https://www.google.com/search?q=mareks+disease&client=firefox-b-> (accessed on may 19/2023).

- ✓ **Necropsy Finding:** peripheral virus particularly the vagus, brachial and sciatic nerve become enlarged and lose their striations.
 - ✚ Diffuse or nodular lymphoid tumors may be seen in various organs, particularly in the liver, spleen, heart, lung, kidney, muscle, proventriculur and gonad.
 - ✚ The bursa is frequently atrophic.
 - ✚ Enlarged feather follicles (skin leukosis) may be noted in broilers.
- ✓ **Diagnosis:** the disease occurs at any age over 3 weeks. Confirmation of a diagnosis may be made histologically (a mixture of tumor cells and reactive inflammatory cells) or by demonstrating the tumor-associated surface antigen on some of the individual cells by immuno fluorescence.
- ✓ **Differential Diagnosis:** Lymphoid leukosis.
- ✓ **Treatment:** No effective treatment
- ✓ **Control:** Vaccination is the principal method of control.
- **Infectious Bovine Rhinotrachitis (IBR)**
 - ✓ **Synonyms:** red nose

It is an acute highly contagious disease of cattle characterized by fever, rhinitis, dyspnoea, meningo encephalitis, keratoconjunctivitis and pustular vulvo-vaginitis.

- ✓ **Etiology:** *bovine herpes virus*.
- ✓ **Occurrence:** The disease is widely prevalent in all parts of the world.
- ✓ **Species Affected:** Cattle of all ages are affected. Dairy and beef cattle are equally susceptible.
- ✓ **Transmission:** the main sources of infection are the nasal discharge, genital secretions, semen and fatal fluids and tissues. The virus is usually transmitted through infected feed and water, infected droplet. Venereal transmission and transmission through semen via artificial. insemination is possible.
- ✓ **Clinical Findings:** They have been grouped into five. These are: -
 - ✚ **Respiratory Form:** fever, reduced appetite, rapid respiration and dyspnoea. Serous to mucopurulent nasal discharges, hyperaemia of the upper respiratory tract mucosa like pharynx and larynx with mucopurulent exudation causing dyspnoea. The mortality rate is comparatively less unless there is superimposed bacterial or viral infection. Animals may show signs of bronchitis and pneumonia.
 - ✚ **Vulvo vaginal form:** This form is characterized by sharp fall in milk yield and appearance of erythematous and pustular lesions on vulval and vaginal mucosa. Vulva may remain swelled and animal may urinate at frequent intervals. There may be mucopurulent discharge from vulva and vagina.
 - ✚ **Ocular form:** This form may appear along with respiratory form. There is conjunctivitis. However, in some cases, severe Conjunctivitis and ocular discharges may be noted without respiratory involvement. Sometimes there may be corneal opacity.
 - ✚ **Encephalomyelitis form:** it comprises of high-rise of temperature, incoordination, tremor, circling, falling, coma and death. Death occurs within 4 days following appearance of neurological disorders.
 - ✚ **Abortive form:** The pregnant cattle may abort following infection. Abortion is usually at 4th month of gestation.

- ✓ **Diagnosis:** Diagnosis is based on clinical findings and characteristic lesions. But, confirmation is by laboratory tests, virus isolation, FAT, ELISA
- ✓ **Differential Diagnosis:** BVD, Rinderpest, MCF, Pneumonic pasteurellosis. The nervous form may be confused with rabies.
- ✓ **Treatment:** There is no effective treatment. Superimposed bacterial infection can be treated by broad-spectrum antibiotic or sulphonamide. Symptomatic treatments include correction of fluid and electrolyte imbalance and provision of adequate ration.
- ✓ **Control:** -Strict isolation of affected animals and Vaccination by using live attenuated vaccine

- **Malignant Catarrhal Fever (Bovine Malignant Catarrha)**

It is an acute highly fatal disease of cattle characterized by development of an erosive stomatitis and gastroenteritis and erosions in the upper respiratory tract, keratoconjunctivitis, encephalitis, and lymph node enlargement.

- ✓ **Etiology:** *herpes virus 1*. The virus is susceptible to external influences.
- ✓ **Occurrence:** The disease may occur sporadically or in explosive outbreaks. It occurs in most countries but is probably of most importance in Africa.
- ✓ **Species affected:** the clinical disease occurs only in cattle & buffalo, but sheep & goats develop inapparent infections.
- ✓ **Transmission:** the method of spread of MCF is uncertain. All age, races and breeds of cattle are equally susceptible
- ✓ **Clinical Findings:** The incubation period of MCF is 3-8 weeks. The disease occurs in a number of forms- the alimentary tract form and head and eye form.

✚ In the head and eye form: there is depression, anorexia, and fever (41-41.5°C), rapid pulse, profuse mucopurulent nasal discharge, severe dyspnoea, ocular discharge with variable degrees of edema of the eyelids and blepharospasm. There are painful erosive lesions in the oral and nasal mucosa. There is involvement of the skin of the muzzle, teats, vulva, scrotum and nervous signs and consistent occurrence of superficial lymph node enlargement are noted.

✚ In the intestinal form: the consistency of the faeces varies from constipation to profuse diarrhoea with dysentery. Persistence fever throughout the course of

the disease is characteristic and death occurs 7-10 days later of acute encephalitis.

✚ Opacity of the cornea is always present in some degree and in some cases there is gross hematuria.

- ✓ **Diagnosis:** Diagnosis is based on clinical findings, oral, nasal and ocular lesions along with persistent high temperature and enlarged lymph nodes. Confirmation is based on pathognomonic histopathological lesions (thrombosis, infiltration of tissue with dividing lymphocytes and macrophages)
- ✓ **Differential Diagnosis:** Rinderpest, mucosal disease and infections stomatitis
- ✓ **Treatment:** There is no effective treatment against this disease.
- ✓ **Control and prevention:** Isolation of affected cattle is usually recommended. Vaccination of animals. Recovered animals are immune further infection for 4-8 months

- **Herpes simplex and varicella zoster in humans:**

- ✓ **Etiology:** These diseases are caused by different members of the herpesvirus family including varicella zoster virus (VZV) and herpes simplex virus 1 and 2 (HSV-1 and HSV-2).
- ✓ **Transmission:** Contact with infected bodily fluids such as saliva, semen, and vaginal secretions; skin-to-skin contact; and contact with contaminated objects
- ✓ **Host preference:** Humans
- ✓ **Clinical signs:** Herpes simplex can cause oral or genital herpes while varicella zoster virus causes chicken pox and shingles which are characterized by a skin rash and flu-like symptoms
Diagnosis: Liquid PCR, serological examination for antibodies against the virus.
- ✓ **Treatment:** Antiviral medications such as acyclovir, valacyclovir, and famciclovir may reduce symptoms and shorten the length of the illness or suppress recurrences, pain relief ointments or medicine like ibuprofen can help in reducing symptoms.
- ✓ **Prevention/control measures:** Proper hygiene measures can reduce transmission. Vaccines are also available for varicella and herpes zoster.

- **Avian infectious laryngotracheitis**

- ✓ **Etiology:** Caused by Gallid herpesvirus 1 (GaHV-1) which belongs to the alphaherpesvirus genus.
- ✓ **Transmission:** Through contact with mucus or respiratory secretions and through dust particles in the environment
- ✓ **Host preference:** Chickens especially in backyard poultry farms.
- ✓ **Clinical signs:** Respiratory disease which includes coughing, nasal discharge, gasping, and tracheal rales. Additionally ocular discharge with keratoconjunctivitis can be observed in birds.
- ✓ **Diagnosis:** Detection of virus through PCR or virus isolation.
- ✓ **Treatment:** Symptomatic treatment with antibiotics and anti-inflammatory medications may help in reducing respiratory distress.
- ✓ **Prevention/control measures:** Vaccination with proper management practices including cleaning and disinfection of the farm facilities.

- **Equine herpes virus (EHV)**

- ✓ **Etiology:** EHV-1 and EHV-4 belong to the alphaherpesvirus genus.
- ✓ **Transmission:** Through respiratory secretions or fomites.
- ✓ **Host preference:** Horses
- ✓ **Clinical signs:** EHV-1 can cause neurological symptoms such as ataxia and urinary incontinence, as well as respiratory symptoms. EHV-4 can cause respiratory infection and abortion in pregnant mares.
- ✓ **Diagnosis:** Detection of virus through PCR or virus isolation.
- ✓ **Treatment:** Symptomatic treatment with anti-inflammatory medications such as flunixin meglumine and antiviral medications like acyclovir.
- ✓ **Prevention/control measures:** Vaccination and biosecurity measures like isolation of infected horses.

3.2.3 Parvo virus

Parvoviruses are small, nonenveloped, single-stranded DNA viruses. They are among the smallest viruses and are 18-26 nm in diameter. The parvovirus genome is linear and single-

stranded and contains approximately 5 kb of DNA. Parvoviruses are highly resistant to inactivation by physical and chemical agents.

- **Canine Parvo virus**

- ✓ **Etiology:** Canine parvovirus which is a highly contagious and deadly viral disease that affects dogs.
- ✓ **Transmission:** The virus is spread through contact with infected feces, contaminated surfaces, or direct contact with an infected dog. The virus can survive for a long time in the environment, making it very difficult to control.
- ✓ **Clinical sign:** The virus mainly infects the cells lining the small intestine, causing severe diarrhea and vomiting which can lead to dehydration, shock, and death in untreated cases. Puppies are particularly vulnerable to infection and can die within days of being infected.
- ✓ **Diagnosis** of canine parvovirus is based on clinical signs and laboratory tests such as ELISA or PCR.
- ✓ **Treatment** involves aggressive supportive therapy including intravenous fluids, antibiotics, antiemetic, and analgesia. However, early detection and intervention are crucial for the successful outcome of treatment.
- ✓ **Prevention/control measures** include strict sanitation practices, vaccination of puppies at 6-8 weeks of age and boosters every 3-4 weeks until the age of 16 weeks, and avoiding contact with infected dogs or contaminated environments. It is important to isolate infected dogs to prevent the spread of the virus to other animals.

3.3. RNA viral diseases

RNA viruses are viruses that have RNA as their genetic material. These viruses can infect a wide range of organisms, including humans, animals, and plants. Some of the most common RNA viruses that infect animals include Foot and Mouth diseases, Pest des petites (PPR), Rabies, New castle disease, African horse sickness, Blue tongue and others.

3.3.1 Foot and Mouth Disease (FMD)

- **Synonym:** Aphthous fever

It is an extremely contagious acute disease of all cloven-footed animals that characterized by fever and vesicular eruption in the epithelium of buccal cavity, tongue, muzzle, feet, teat and udder.

- **Etiology:** caused by *picorna virus* group (genus *Aphtho virus*). At least 7 immunologically distinct serotypes of the virus have been identified. These are A, O, C, SAT-1, SAT-2, SAT-3 and Asia-1. The virus is resistant to various external agents including common disinfectant. NaOH, formalin (1-2%), Na₂CO₃ (4%) have ability to destroy the virus within few minutes.
- **Occurrence:** The disease is endemic in South America, central Europe and central African countries notably in Ethiopia and Tanzania.
- **Species Affected:** Cattle are more susceptible. A wide range of hosts like sheep, goat, pig and camel are susceptible to FMD. Dog, cat and man may occasionally be infected after close contact with infected host.
- **Transmission:** The disease spread at an extremely rapid rate through direct contact with infected animals. In tropical countries, spread is through ingestion of contaminated feed and water and inhalation of infected droplets. All the fomites like clothes, harness, beddings, straws, hay etc may get infected and therefore act as a source of infection. All the secretion and excretions like urine, milk, feces and saliva remain infective. Cattle may remain carrier following recovery.
- **Clinical Findings:** the morbidity is 100% but mortality is less in indigenous cattle and comparatively more in cross and pure breed cattle. The disease has an incubation period of 2-8 days. The signs are drooling and vesicles on the nares, in the buccal cavity and between the claws. Before the visible sign of illness, animals may show dullness, inappetence, fever and shivering followed by smacking of the lips, drooling and shaking or kicking of the feet. After vesicle formation there is pronounced salivation and lameness. Pregnant animals may abort. Hoof deformation may result in permanent lameness. Mammary gland involvement may result in mastitis and permanent impairment of milk production.



Figure 3.6. clinical sign of foot and mouth diseases

Source: <https://www.google.com/search?Q=foot+and+mouth+diseases+in+animal> (accessed on may 19/2023).

- **Necropsy Findings:** The diagnostic lesions are vesicles or blisters. They may be found on the tongue, dental pad, gum, cheek, hard and soft palate, lips nostrils, muzzle, coronary bands, teat and udder, as well as in the myocardium (degenerative change) and in skeletal muscles.
- **Differential Diagnosis:** MCF, BVD, Rinderpest, vesicular stomatitis, blue tongue.

- **Treatment:** There is no specific treatment. Symptomatic treatments may be rendered depending on clinical manifestations. Antiseptic solution like potassium permanganate or sodium bicarbonate may be applied over mouth lesions. Similarly antiseptic or antibiotics may be given on feet lesions and lesions on mammary tissue.
- **Control:** restrict animal movements
 - ✓ Vaccination of all animals of an area/village is to be done at one time.
 - ✓ A footbath or truck bath may be made at the entrance of village or farm.
 - ✓ Always purchase fodder from a place where FMD has not been recorded for a period of 6 months or so.
 - ✓ Try to isolate and confine the affected animals immediately after detection.
 - ✓ Calves should not be allowed to suckle affected mothers and they should not be fed with milk from affected animals.

In endemic areas

- ✚ Vaccination and quarantine are the bases for control.
- ✚ The vaccine must be type specific.
- ✚ Killed trivalent (O, A and C) vaccines are in general use.
- ✚ Immunity lasts only for 6 - 8 months.

NB. In Ethiopia, there is bivalent vaccine contains O and C strain which is absorbed into Aluminum Hydroxide gel ($\text{Al}(\text{OH})_3$ concentrated and inactivated with 0.3% of formaldehyde.

Recognising Foot and Mouth Disease for veterinarians and animal health officers

<https://www.youtube.com/watch?v=V1HbJ8xlQXY> (accessed on May 19/2023)

3.3.2 Newcastle Disease (NCD)

An acute, rapidly spreading, viral disease of poultry in which the respiratory signs (coughing, sneezing and rales) are often accompanied or followed by nervous manifestations and infection with some strain, by diarrhoea and swelling of the head.

- **Etiology:** caused by *paramyxovirus*. It has three strains: high virulent (Velogenic), intermediate (Mesogenic) and less virulent (Lentogenic).
- **Occurrence:** The disease occurs worldwide in a variety of domestic and wild birds.

- **Transmission:** Virus is shed during incubation, during the clinical stage and limited period during convalescence. Virus is present in exhaled air, in respiratory discharges, in faeces, in eggs laid during clinical disease and in all parts of the carcass during acute infection. Chickens are readily infected by aerosols and by ingesting water or food contaminated with the virus.
- **Clinical Findings:** Respiratory or nervous signs or both occur in the most widespread forms of the disease. Signs appear almost simultaneously throughout the flock 2-15 days after exposure.
 - ✓ Respiratory signs are gasping and coughing.
 - ✓ Nervous signs include dropping wings, dragging legs, twisting of head and neck, circling, walking back ward (particularly after drinking water), depression, in appetite and complete paralysis.
 - ✓ Laying flocks may have partial or complete cessation of production. Eggs become abnormal in colour, shape or surface and with watery albumen are produced.
 - ✓ Viscerotropic signs are predominant in the peracute disease includes watery and greenish diarrhoea and swelling of the tissue around the eyes and in the neck.



Figure 3.7. Torticollis (twisting of head)

Source: https://www.google.com/search?Q=newcastle+diseases&client=firefox-b-d&source=lnms&tbm=isch&sa=X&ved=2ahukewig5fhdiy7_ahuuqvedhrcxbz4q_auoanoecaeqba&biw=1366&bih=643&dpr=1#imgsrc=pjippos3twvh3m (accessed on May 19/2023)

- **Necropsy Findings:** Petechiation may be seen on the serous membranes.
 - ✚ Haemorrhages of the proventricular mucosa and the intestinal serosa occur and are accompanied by necrotic areas on the mucosal surface.
 - ✚ Congestion and mucoid exudates may be seen in the respiratory tract, with opacity and thickening of the air sacs.
- **Diagnosis:** Tentative diagnosis is based on history of rapidly spreading disease and signs associated with involvement of both the respiratory and nervous system. Confirmation by serology.
- **Differential Diagnosis:** Avian influenza.
- **Control:** Live virus vaccine are widely used. Lentogenic strains, chiefly B1 and Lasota are administered in drinking water, nose or eye drops. Healthy chicks are vaccinated as early as the 4th or even the first day of life.

3.3.3 Peste Des Petitis Ruminants (PPR)

- **Synonym:** Goat plague

It is an acute or sub acute viral disease of goat and sheep that characterized by fever, gastroenteritis, necrotic stomatitis, and pneumonia.

- **Etiology:** *Morbilli* virus of the family *paramyxoviridae*
- **Occurrence:** prevalent in Africa and Middle East
- **Species affected:** sheep and goats are highly susceptible to the disease. Sheep are less susceptible than goats; cattle are only subclinically infected .The mortality and morbidity rate is very high in young animals.
- **Transmission:** Secretion and excretions of sick animals are the source of infection. Close contact is important for the spread of the disease. Entry into the animal body occurs through inhalation and ingestion of contaminated feed and water.
- **Clinical sign:** fever (40-41⁰c), dull coat, dry muzzle, congested mucous membrane and depressed appetite. Initially serous nasal discharge latter changed in to mucopurulent nasal discharge. Necrotic stomatitis affects the lower lip, gum, tongue and cheek. Profuse diarrhoea followed by dehydration, hypothermia and death. Broncho pneumonia may also develop.

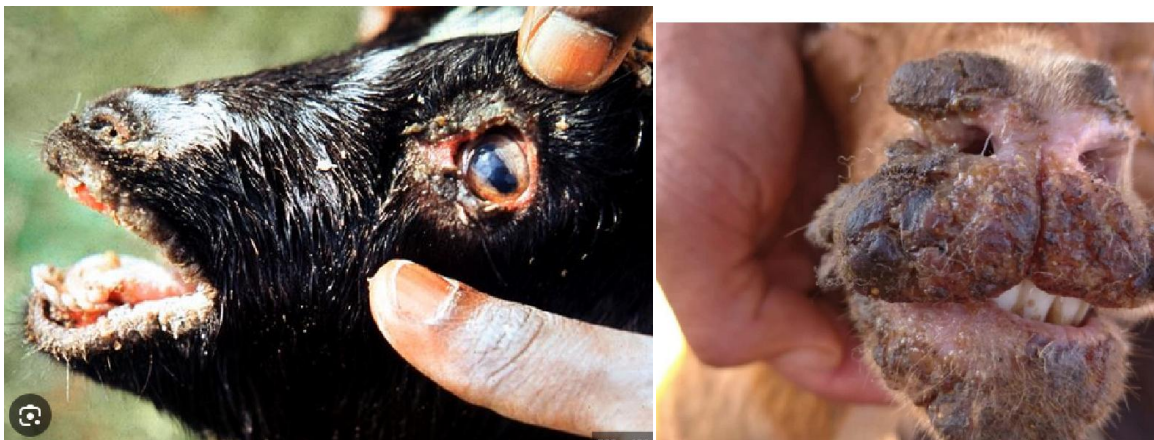


Figure 3. 8. PPR clinical sign

Source: <https://www.google.com/search?q=ppr+disease+symptoms&tbm=isch&ved=2ahukewi6>
(accessed on May 19/2023)

- **Differentia diagnosis;** Rinderpest, CCPP, blue tongue, coccidiosis, mineral poisoning
- **Treatment:** no specific treatment. Administration of antibacterial and anti parasitic drug to reduce secondary complication.
- **Control and prevention:** Vaccination, isolation

3.3.4 Avian Influenza

- **Synonyms:** AI, flu, influenza, fowl plague
- **Species affected:** Avian influenza can occur in most, if not all, species of birds.
- **Clinical signs:** Avian influenza is categorized as **mild or highly** pathogenic.
 - ✓ The mild form produces listlessness, loss of appetite, respiratory distress, diarrhea, transient drops in egg production, and low mortality.
 - ✓ The highly pathogenic form produces facial swelling, blue comb and wattles, and dehydration with respiratory distress. Dark red/white spots develop in the legs and combs of chickens. There can be blood-tinged discharge from the nostrils. Mortality can range from low to near 100 percent. Sudden exertion adds to the total mortality. Egg production and hatchability decreases. There can be an increase in production of soft-shelled and shell-less eggs.
- **Transmission:** The avian influenza virus can remain viable for long periods of time at moderate temperatures and can live indefinitely in frozen material. As a result, the

disease can be spread through improper disposal of infected carcasses and manure. Avian influenza can be spread by contaminated shoes, clothing, crates, and other equipment. Insects and rodents may mechanically carry the virus from infected to susceptible poultry.

- **Treatment:** There is no effective treatment for avian influenza. With the mild form of the disease, good husbandry, proper nutrition, and broad spectrum antibiotics may reduce losses from secondary infections. Recovered flocks continue to shed the virus. Vaccines may only be used with special permit.
- **Control and Prevention:** A vaccination program used in conjunction with a strict quarantine has been used to control mild forms of the disease. With the more lethal forms, strict quarantine and rapid destruction of all infected flocks remains the only effective method of stopping an avian influenza outbreak. If you suspect you may have Avian Influenza in your flock, even the mild form, you must report it to the state veterinarian's office. A proper diagnosis of avian influenza is essential. Aggressive action is recommended even for milder infections as this virus has the ability to readily mutate to a more pathogenic form.

3.3.5 Rabies

- **Synonyms:** lyssa; madness

This is an acute viral infection in man and other warm-blooded animals characterized by signs of abnormal behaviour, nervous disturbance (such as increased excitability and irritability), impairment of consciousness, ascending paralysis and death.

- **Etiology:** caused by a virus belonging to the family *Rhabdoviridae* and genus *lyssa virus*. The virus is truly neurotropic and causes lesions only in nervous tissue. It is one of the larger viruses and is relatively fragile.
- **Occurrence:** Rabies occurs in most countries of the world except the island countries that are able to exclude it by rigid quarantine measures or prohibition of the entry of dogs.
- **Species Affected:** All warm-blooded animals are susceptible and there is no difference in susceptibility with relation to age of the animals. Susceptibility varies in different species. Animals like dog, fox, cat, wolf, rat and vampire bat are extremely susceptible in the

tropical areas of the world. Cattle, goat and sheep are moderately susceptible. Equine are also susceptible for the disease and they are considered as dead end host for the rabies virus. Man and birds are susceptible to the disease. In general, rabies in animals exist in two epidemiological types: **Urban type:** where the transmission takes place through dogs. **Sylvatic type:** where the disease is transmitted through wild life ex-fox, wolf, hyena and vampire bat.

- **Transmission:** Transmission of rabies under natural condition is commonly by bite of rabid animals usually carnivorous animals. Rabies virus can penetrate the mucous membrane of eyes, nose and mouth and thus aerosol infection is possible in a rabies laboratory by accident. The ability of the virus to reach the central nervous system depends on various factors.
 - ✓ Distance of bite. I.e. The distance of the bite from the CNS is an important factor in reaching the virus in the brain.
 - ✓ Virulence of the virus
 - ✓ Concentration of the virus
- **Clinical Findings:** Rabies has got two types of clinical manifestations

i) Furious form

ii) Dumb (paralytic) form

The incubation period of the disease varies considerably from weeks to several months depending on the site and severity of wound inflicted by rabid animals and the amount of virus deposited at the site.

i) Furious Form: It refers to syndrome in that excitation is the predominant changes and it can be divided into stage of melancholy and stage of excitation.

- ✓ **Stage of Melancholy:** Here, there is change in the behaviours of the animal. The dog may show the tendency to bite either inanimate or animate objects. Dogs those are under restraint bite nearby objects. But, set free dogs show the tendency to bite other animals and human beings that comes across their way. Infected dog remains in unusual alert condition. They respond very sharply to any stimulus. They may show imaginary fly catching stance. They will bark with no cause behind that. There is no deviation in appetite in the initial phase, but later on it becomes changed. They will have the urge to chew or bite or eat non-edible substances like

stone, mud, grass or even their own faces. Gradually, signs of excitability increases. The pupils get dilated and there is altered facial expression. Appetite is entirely suppressed but the animal may lick water and attempt to drink water but due to paralysis of pharyngeal and laryngeal muscles does not succeed to drink water. Saliva hangs from mouth in long strand. The period lasts for 1-3 days.

- ✓ **Stage of excitement:** In this stage the excitability and irritability increases and dogs become very much aggressive. This period may last for 1-7 days. At the onset dog may hide in dark place due to photophobia. There is development of change in the bark-which is very characteristic. The animal is unable to swallow. There is drooling of saliva. The change in voice is due to paralysis of the vocal cord. If these animals are not confined they will wonder aimlessly and bite animals or persons who will fall in their way. After some time the affected animals will show emaciation and exhaustion. There will be incoordination and muscle tremors. The bitch may show the sign of heat and accept the male. The dog will lose its ability to bark; the lower jaw will hang, tongue will protrude and head will drop down. The dog will develop dyspnoea, ascending paralysis, coma and death. The total episode may last as long as 10 days.

ii) Dumb (paralytic) Form: In this form, there is paralysis of the lower jaw, tongue, larynx and hind-quarters. The dogs are not capable to bite but their saliva remains infective, throat muscles are paralysed. There is hanging of the jaw due to paralysis of the jaw muscle and the dog is unable to close his mouth. In the terminal stage of the disease dogs show progressive weakness and paralysis that causes them to stagger or fall. Ultimately there is coma and death.

- **Necropsy Findings:** There is no gross lesion at necropsy. However, during histopathological preparation of nervous tissue (hippocampus) reveals the presence of cytoplasmic inclusion bodies, which are called **Negri bodies**.
- **Diagnosis:** The diagnosis of rabies is one of the most difficult and important duties that a veterinarian is called upon to perform. Since in most cases there is a probability of human exposure, failure to diagnose the disease may place human life in jeopardy. Confirmation of a diagnosis of rabies depends on careful laboratory examination of fresh brain.

- **Differential Diagnosis:** Acute and sub acute lead poisoning in cattle, Listeriosis- in sheep and cattle, Enterotoxaemia- In sheep
- **Treatment:** No specific treatment but immediately after exposure irrigation of the wound with antiseptic solution (soap) may prevent the establishment of infection. Post-exposure vaccination is unlikely to be of value in animals, as death usually occurs before appreciable immunity had to develop.
- **Control:** reduce the population of dogs by destruction of street dogs, castration and spaying vaccination of all cats and dogs and keeping farm animals indoors.

3.3.6 Rota virus, AHS virus, Blue tongue virus

- **Rota virus**

Rotaviruses are a significant cause of viral gastroenteritis in young animals

- ✓ **Etiology:** Rotaviruses are a group of non-enveloped, double-stranded RNA viruses that belong to the family Reoviridae. They are known to cause severe diarrhea in a wide range of young animals, including calves, lambs, piglets, and foals.
- ✓ **Transmission:** Rotaviruses are spread via the fecal-oral route, with ingestion of contaminated feed, water, or feces being the primary mode of transmission. Infected animals can shed large quantities of virus in their feces for several days, which can contaminate the environment and infect other susceptible animals.
- ✓ **Host preference:** Rotaviruses have a broad host range and can cause disease in many species of domestic and wild animals, including cattle, sheep, pigs, horses, dogs, cats, and birds.
- ✓ **Clinical signs:** The primary clinical sign of rotavirus infection is acute diarrhea, which can be profuse and watery in nature. Infected animals can also exhibit vomiting and dehydration. **Diagnosis:** Diagnosis is based on clinical signs and laboratory tests such as enzyme-linked immunosorbent assay (ELISA) or reverse-transcription polymerase chain reaction (RT-PCR) tests on fecal samples.
- ✓ **Treatment:** There is no specific treatment for rotavirus infections. Treatment is generally supportive and includes rehydration with fluids and electrolyte solutions.

- ✓ **Prevention/control measures:** Prevention measures include good hygiene practices such as cleaning and disinfecting contaminated environments. Vaccines are also available for the prevention of rotavirus infections in some animal species. Early detection and prompt isolation of infected animals can help prevent the spread of disease to other susceptible animals.

- **Blue Tongue**

Bluetongue is a disease of sheep and occasionally cattle, characterized by catarrhal stomatitis, rhinitis and enteritis and lameness due to inflammation of the coronary bands and sensitive lamina of the feet.

- ✓ **Etiology:** caused by *orbi virus* of the family *Reo viridae*. The virus is resistant to decomposition, desiccation and against antiseptic agents. The virus is destroyed by 3% NaOH solution.
- ✓ **Occurrence:** Bluetongue is arthropod borne viral disease. The infection is widespread on the African continent and also to many other parts of the world.
- ✓ **Species Affected:** It is basically a disease of sheep but amongst sheep susceptibility varies in age; young sheep within the age group of one year are more prone to infection. Suckling lambs are relatively resistant. Goats are relatively resistant
- ✓ **Transmission:** spread is through blood sucking midges of the genus *culicoides*. Ticks and mosquitoes also transmit the virus. The disease is more prevalent in late summer and early autumn that provide conducive environment for the multiplication of the vectors.
- ✓ **Clinical Findings:** Incubation period varies from 1-10 days. The disease is characterized by high fever persisting for 5-6 days with progressive reddening of the buccal mucous membrane. It has acute, sub acute and abortive form.
 - ✚ **Acute form:** is characterized by fever, nasal discharge, salivation and lacrymation. Swelling of the tongue, gum and lips followed by ulceration is evident. There may be extensive necrosis of the dental pad. Skin of other areas like flank, neck, back, etc may crack. There is cyanotic and bluish appearance of the tongue. Udder may be swollen and teat may show signs of ulceration. Coronary band may become swollen and it may crack. Hoof may become separated leading to lameness, Rapid shallow respiration, pneumonia & muscular weakness may be seen. Abortive form is

characterized by abortion and deformities. Deformed calves may remain normal after birth.

✚ **Sub acute:** noted in cattle and generally passed unnoticed.

- ✓ **Necropsy Findings:** Characteristic lesions are observed in the mouth and tongue. On the mouth there is hyperaemia, oedema, cyanosis, erosion, haemorrhage and ulceration of the oral mucosa. The tongue becomes blue and gangrenous. Signs of hyperaemia and necrosis on the skeletal and cardiac muscles and pneumonic changes on the lungs are seen.
- ✓ **Diagnosis:** Presumptive diagnose is based on history, clinical findings (edematous swelling and cyanosis), seasonal occurrence, absence of spread by direct contact. Confirmative diagnosis is based on animal inoculation, isolation of the virus and serological tests (CFT, AGID.)
- ✓ **Differential Diagnosis:** FMD, MCF, photosensitization.
- ✓ **Treatment:** There is no treatment against the virus. Localized lesions may be treated with topical antibiotic solution to accelerate rate of healing. Affected animals should be housed and protected from hot sun.
- ✓ **Control:** Recovered animals (sheep) are immune for 6 months.
 - ✚ Reduce vector population by applying insecticides and good water
 - ✚ Managements required.
 - ✚ Infected animals should be isolated.
 - ✚ Vaccination (attenuated live vaccine)

- **African Horse Sickness**

It is a highly fatal, infectious disease of horses, mules and donkeys.

- ✓ **Etiology:** *orbivirus* (family *reoviridae*). The virus has 9 antigenically distinct serotypes and there is no cross-reaction or protection between the different strains.
- ✓ **Occurrence:** The disease is distributed throughout sub-Saharan African countries and Middle East. The incidence of the disease is seasonal because of the seasonal variation in the number of arthropod vector present, particularly *culicoides spp*. Areas most commonly affected are low lying and swampy areas and most cases occur after mid-summer.

- ✓ **Species Affected:** natural infection occurs in equine species. The degree of susceptibility are horses > mules > donkeys. Zebras are highly resistant to the disease and serve as reservoir of infection for susceptible hosts.
- ✓ **Transmission:** by biting of arthropod insects especially *Culicoides* spp. There is no transmission of the disease through contact with infected animals
- ✓ **Clinical Findings:** The incubation period is less than 10 days and the disease has three clinical forms. In all forms of the disease an intermittent fever of 40-41°C is a characteristic finding. The three clinical forms are:
 - ✚ **Acute (pulmonary) form:** the most common form in acute outbreaks in susceptible animals. Initially there is fever followed by very laboured breathing and severe paroxysm of coughing. There is a profuse nasal discharge of yellowish serous fluid. Profuse sweating and the horse become very weak, develops a staggy gait and becomes recumbent. Death follows within a few hours
 - ✚ **Sub acute (Cardiac) form:** this most common in horses in endemic areas. This term is characterized by fever but develops more slowly and persists for long period. The most obvious sign is edema in the head region, particularly in the supra orbital fossa, eyelids and the lips, and this may spread to the chest. Auscultation of the heart and lung reveals evidence of hydro pericardium, endocarditis and pulmonary edema. Paralysis of the oesophagus with inability to swallow and regurgitation of food and water through the nose.
 - ✚ **Horse sickness Fever:** this presents no diagnostic signs and may unrecognise except that it usually occurs in endemic areas. Most commonly occur when an existing immunity is partially overcome. The temperature rises to 40.5°C over a period of 1-3 days but return to normal about three days later.
- ✓ **Diagnosis:** based on Clinical signs Serological tests: CFT, Serum neutralization, ELISA
- ✓ **Differential diagnosis:** Equine infectious pneumonia, Babesiosis, equine infectious anaemia.
- ✓ **Treatment:** no treatment has been shown to have any effect on the course of the disease but careful nursing and symptomatic treatment are recommended.

✓ **Control and prevention:** It is very difficult because of the mode of transmission of the disease.

✚ Restriction of introduction of horses.

✚ **Vaccination:** Polyvalent vaccine containing all 9 strains or sometimes only 7 strains is effective in the control of the disease. Immunity following vaccination is solid for one year. Therefore, annual vaccination is recommended.

3.3.7 infectious Bursal disease (IBD)

• **Synonyms:** Gumboro, IBD, infectious bursitis, infectious avian nephrosis

Infectious bursal disease (IBD) is caused by a virus that is a member of the genus Avibirnavirus of the family Birnaviridae. Although turkeys, ducks, guinea fowl and ostriches may be infected, clinical disease occurs solely in chickens. Only young birds are clinically affected. Severe acute disease of 3–6-week-old birds is associated with high mortality, but a less acute or subclinical disease is common in 0–3-week-old birds. This can cause secondary problems due to the effect of the virus on the bursa of Fabricius. IBD virus (IBDV) causes lymphoid depletion of the bursa, and if this occurs in the first 2 weeks of life, significant depression of the humoral antibody response may result.

• **Species affected:** chickens

• **Transmission:** The virus is spread by bird-to-bird contact, as well as by contact with contaminated people and equipment. The virus is shed in the bird droppings and can be spread by air on dust particles. Dead birds are a source of the virus and should be incinerated.

• **Clinical signs:** In affected chickens greater than 3 weeks of age, there is usually a rapid onset of the disease with a sudden drop in feed and water consumption, watery droppings leading to soiling of feathers around the vent, and vent pecking. Feathers appear ruffled. Chicks are listless and sit in a hunched position. Chickens infected when less than 3 weeks of age do not develop clinical disease, but become severely and permanently immunosuppressed

• **Treatment:** There is no specific treatment. Antibiotics, sulfonamides, and nitrofurans have little or no effect. Vitamin-electrolyte therapy is helpful. High levels of tetracyclines are

contraindicated because they tie up calcium, thereby producing rickets. Surviving chicks remain Unthrifty and more susceptible to secondary infections because of immunosuppression.

- **Prevention:** vaccination

3.3.8 Vesicular Stomatitis

A viral disease characterized by a febrile response accompanied by vesicles on the mucous membranes of the mouth, epithelium of the tongue, teats, soles of the feet, coronary band and occasionally, in other parts of the body.

- **Etiology:** *vesiculo* virus belonging to *Rabdo virus* group.
- **Species Affected:** Vesicular stomatitis is found in all breeds and sexes of cattle, horse, pig and donkeys. Outbreak is most common in cattle.
- **Transmission:** The exact mode of transmission is not clearly understood. The disease may be transmitted though direct contact of the diseased animals. The infected saliva or saliva contaminated foods, or water or feedlot equipments may help in transmission. It is also suggested that the spread of the disease is not always through immediate contact. Mechanical transmission through insect, fly, mosquitoes has been reported.
- **Clinical Findings:** Following an incubation period of 2-5 days, there is high-rise of temperature. There is formation of vesicles predominantly on the mucous membrane of the mouth and tongue. As a result of mouth lesions animals are unable to eat. Following the rupture of the vesicles there is appearance of eroded areas and irritation, which lead to production of ropy salivation. There is reduction in milk yield due to appearance of lesions in the teat and udder. This may lead to mastitis. Foot lesions are less as compared with FMD.
- **Diagnosis:** It is based on clinical findings and characteristic lesions. Serology like CFT Isolation of the agent
- **Differential Diagnosis:** FMD, swine vesicular disease
- **Treatment:** There is no specific treatment against the virus. Symptomatic treatments may be rendered to the diseased animal. These include topical application of anesthetic ointment having antibiotic or sulphonamide. This will relief pain and restore appetite. To reduce

inflammatory reaction systemic antibiotic and corticosteroid may be used. Animals should be provided with adequate clean water and non-fibrous soft feed.

- **Control:** strict hygienic and sanitation measures are to be adopted.
 - ✓ Premises should be treated with disinfectants (1% formalin)
 - ✓ Strict quarantine measures should be followed
 - ✓ Isolation of affected animals, Recovered animals may have immunity for more than 6 months

3.3.9 Rift valley fever (RVF)

It is an acute, febrile disease of cattle, sheep and man characterized by hepatitis and high mortality in lambs and calves, in adult sheep and in cattle abortion and in man influenza like disease.

- **Etiology:** *phlebo virus* (family *Bunyeviridae*).
- **Species affected:** Cattle, sheep, camels, buffalo, monkeys and man are highly susceptible and goats moderately susceptible.
- **Occurrence:** is conferred to Africa continent but it has great potential for spread to other countries. Losses are due mainly to deaths in young lambs and calves, although there may be a high incidence of abortions, and some deaths, in adult sheep and cattle.
- **Transmission:** is by biting flies, chiefly mosquitoes
- **Clinical findings:** In lambs and calves after an incubation period of about 12 hours there is sudden onset of high fever and incoordination followed by collapse and sudden death within 36 hours. In adult sheep and cattle, abortion is the common sign. In fatal cases sudden death is preceded by high fever for 1-2 days. Goats show a febrile reaction but few other clinical signs.
- **Diagnosis:** The hepatic lesions are characteristic; severe leukopenia is a common finding. Serological tests like ELISA, hemagglutination, CFT and serum neutralization are used.
- **Differential Diagnoses:** Blue tongue (in sheep), Ephemeral fever (in cattle), Enterotoxaemia and other causes of abortion.
- **Treatment:** no specific treatment.
- **Control:** - Prevent introduction of infected animals and human beings to free areas.

- ✓ In endemic areas mosquito control is important to reduce the spread of disease
- ✓ Both killed and living attenuated virus vaccines are available.

3.3.10 Bovine Viral Diarrhoea (BVD) or (Mucosal Disease)

It is a sub acute, acute or in apparent contagious disease that characterized by high rise of body temperature, diarrhoea and erosion on the mouth, oesophagus, rumen, abomasums and intestines.

- **Etiology:** caused by a virus belongs to the genus *pest virus* and family *Toga viridae*.
- **Species Affected:** Cattle are the only species that develop mucosal disease but pig and sheep can also be infected
- **Occurrence:** The disease has nearly a worldwide distribution
- **Transmission:** Virus is present in secretions and excretions of affected cattle. The disease often rapidly spread by direct or indirect contact with infected animal. In crowded feedlot and transport vehicles, virus from nasal and oral secretions easily spread from affected cattle to susceptible animals. Contaminated feed and water are the important agents of transmission of virus. Urine and nasal discharges may act as a source of infection.
- **Clinical Findings:** The clinical manifestation may be grouped as
 - ✓ **Acute (epidemic) form:** This form is serious in nature and characterized by high fever, anorexia, depression, polypnea, tachycardia , and polydypsia. Nasal discharges are mucoid to mucopurulent in nature. Erosion may be noted under the crest. There may be signs of conjunctivitis accompanied with mucopurulent ocular discharges. Profuse foul smelling watery faces and may contains mucus and blood. Oral lesions appear 1 to 3 days following diarrhoea. There is hyperemia and ulceration of oral mucosa, tongue, palate and gum. Perfuse salivation will be noted at this stage. There is abortion. Fetus is abnormal and malformed. Finally, animals may die due to septicemia and severe dehydration.
 - ✓ **Mild form:** characterized by fever of short duration, temporary loss of milk yield, transient diarrhoea and infrequent mouth lesions. Anorexia, nasal discharge and enlargement of superficial lymph nodes may be the clinical outcome.

- ✓ **Sub acute (chronic) form:** There is retarded growth, loss of body weight, emaciation, rough skin coat and intermittent diarrhoea. Chronic buccal lesions and chronic bloat may be noted.
- ✓ **Diagnosis:** made based on typical clinical findings and characteristic gross and microscopic lesions. High morbidity and low mortality rate and leucopenia are the other aids in diagnosis. Confirmatory diagnosis is by isolation of the virus and serology like ELISA, CFT etc.
- ✓ **Differential Diagnosis:** MCF, Rinderpest, IBR, Johne's disease
- ✓ **Treatment:** There is no treatment against the virus. But supportive therapy is indicated to safeguard the life of the animal. To correct dehydration balanced electrolyte and fluid therapy should be restored. Broad-spectrum antibiotics to control secondary bacterial complication
- ✓ **Control:** All the affected animals should be isolated during an outbreak. All sorts of hygienic and sanitary measures are to be adopted to inhibit further spread of the disease. Vaccination of animals by using modified live virus vaccine after 6 months of age

3.3.11 Swine Vesicular Disease

A transient disease of pigs in which vesicular lesions appear in the mouth and on the feet. The lesions are similar to those of FMD, but the pig does not lose condition and the lesions heal rapidly. The disease is of minor economic importance but requires differentiating from other vesicular diseases, especially FMD, although in laboratory workers have occurred, the pig is the only animal naturally affected.

- **Etiology:** The causal agent is a *porcine enterovirus* of the *picornaviridae* family.
- **Transmission:** It is transmitted by direct or indirect contact or by the feeding of infected pork or pork products.
- **Clinical Findings:** The primary signs are vesicular lesions in the mouth, on the lips or snout, and on the feet, especially the coronary band. The lesions may be mild or inapparent, especially where the pigs are kept on soft beddings.

- **Diagnosis:** The differentiation of swine vesicular disease from other vesicular disease relies on laboratory examination and virus identification.
- **Differential Diagnosis:** FMD, vesicular stomatitis.
- **Control:** Banning the import of pork and pork products. Any suspected outbreaks should be reported to the proper authorities. If disease does appear, important measures are the effective sterilization of all garbage fed to swine and control of pig movement. Disinfection of premises, trucks and equipment must be thorough

3.3.12 Rinderpest (Cattle plague)

Rinderpest is an acute, highly contagious disease of ruminants and swine, characterized by high fever and focal, erosive lesions on the mucosa of the alimentary tract

- **Etiology:** *Morbilli virus* of the family *paramyxo viridae*. The virus has some antigenic relationship to the virus of distemper, measles and peste des petitis ruminants (PPR)
- **Occurrence:** it has been among the most devastating diseases of cattle. In outbreaks, which occur in a highly susceptible population, morbidity rates 100% and mortality rates 50%(25-90%) are to be anticipated.
- **Species affected:** All ruminants and pig are susceptible to infection with rinderpest virus. Wild ruminates are also affected and are a common source of infection and a very great hindrance to an eradication programme.
- **Transmission:** Close contact with infected animals is usually necessary for spread of the disease to occur. Entry into animals occurs principally by inhalation and ingestion of contaminated feed and water.
- **Clinical Findings:** Incubation period of the disease is 6-8 days.
 - ✓ High fever (40.5-41.5°C) followed by anorexia, reduced milk yield and lacrimation
 - ✓ Bubbly salivation stained with blood.
 - ✓ Ocular and nasal discharges.
 - ✓ Discrete, necrotic lesions develop on the inside part of lower lip, the adjacent gum, on the cheek mucosa at the commissures and the lower surface of the tongue. Similar lesions are common on the nasal, vulvae and vaginal mucosa.

- ✓ Severe diarrhoea and sometimes dysentery with tenesmus appear as lesions developing in the abomasum and intestine.
- ✓ For the last 3 to 5 days, there is a sudden fall in temperature accompanied by exacerbation of the mucosal lesions, dyspnoea, cough, severe dehydration and sometimes-abdominal pain.
- ✓ **Diagnosis:** provisional diagnosis based on history of outbreak, clinical findings, and necropsy findings. Confirmatory diagnosis requires isolation and serological identification of the virus.
- ✓ **Differential Diagnosis:** FMD, MCF, Mucosal disease, Hemorrhagic septicemia, PPR (in goats), Bluetongue, Sheep and goat pox.
- ✓ **Treatment:** There is no effective treatment
- ✓ **Control:** It is important to apply a national eradication programme, i.e. Mass vaccination. Complete Prohibition of imports of domestic ruminants; pigs and animal Products from enzootic areas are required. If out breaks of disease occurs, it has to be controlled by
 - ✚ Slaughter of the infected animals
 - ✚ By restriction of stock movements
 - ✚ Vaccination of all neighboring animals.

3.3.13 Papillomatosis

It has also another name called **warts**. Papillomatosis affects cattle, horses, sheep and goats.

It is cutaneous **benign tumors** induced by host-specific papillomaviruses. The viruses infect epithelial cells causing hyper proliferative lesions that are benign, self-limiting. In most case papillomatosis in animals are spontaneously regress. Differentiation of types is based on

The histological features of the lesion and DNA identification by hybridization or PCR. There has been little research on the papilloma virus of horses or small ruminants.

- **Pathogenesis** Papiloma virus infects the basal keratinocytes, replicating its genome in the differentiating spinous and granular layers causing the excessive growth that is characteristic of wart formation. The tumor contains epithelial and connective tissues and can be a papilloma or a fibropapilloma. Papiloma or fibropapilloma formation depends on the relative proportions of epithelial and connective tissue present in the wart

Papillomas contain little connective tissue and fibropapillomas are mostly fibrous tissue with very little epithelial tissue.

- **Clinical findings:** It is known that warts are solid outgrowths of epidermis. Structurally they can be Sessile, Pedunculated. Clinically warts have different forms in different species of animals.
- **Diagnosis:** Diagnose of external warts mostly depends on clinical findings. However laboratory methods help us to confirm the disease. There are no specific changes in the haemogram or blood chemistry. But cattle with papilloma have lower numbers of CD4 lymphocyte subpopulations. Biopsy of a lesion can be used as specimen, used to observe the histopathological changes and to identify the virus. Microscopically it is easy to differentiate the true papillomas from fibropapillomas. Because papillomas consist of a hyperplastic epidermis with scant dermal tissue.
- **Treatment** Warts can be removed by surgery or cryosurgery. However, there is a tendency for spontaneous recovery. So that surgical removal followed by vaccination with an autogenous vaccine give a good result. Since surgical intervention and even vaccination, in the early stages of wart development may increase the size of residual warts and prolong the course of the disease. This indicates treatment of warts must not begin in the early development of warts. Vaccination in cattle is done by autogenous vaccines prepared from wart tissues of the affected animal. Commercially the vaccine are available but may be less efficacious
- **Prevention and control measures:** Specific control procedures are usually not instituted or warranted. Because of the unpredictable nature of the disease and its minor economic importance. However the following prevention and control measures may reduce the occurrence of warts in animals. Vaccination of animals with vaccine which has all serotypes of the papillomavirus because they are very type-specific. Avoidance of close contact between infected and uninfected animals. The use of communal equipment between affected and unaffected animals should be avoided.

Self-Check – 3	Written test
----------------	--------------

Name_____ ID_____ signature_____ date_____

Directions: Answer all the questions below according to the instruction

Part I: choice the correct answer from the given alternative (2pt each)

1. A disease transmitted by biting of arthropod insects especially culicoides spp is
A. African Horse sickness B. Ulcerative lymphangitis C. Tetanus D. Epizootic lymphangitis
2. Extremely contagious acute disease of all cloven-footed animals characterized by fever and vesicular eruption in the epithelium of buccal cavity
A. Rabies B. FMD C. LSD D. PPR

Part II: Short Answer Questions (5pts each)

1. List DNA viral diseases with their control and prevention method
2. List major RNA viral diseases with their control and prevention method

Note: Satisfactory rating -14 points, Unsatisfactory 14 below

You can ask you teacher for the copy of the correct answers.

LG #4	LO4 # Identify and handle fungal disease of animals
--------------	--

Instruction sheet 4

This learning guide is developed to provide you the necessary information regarding the following content coverage and topics:

- Biology and classification of funguses
- Aetiology, host, clinical sign, diagnosis, treatment and prevention of fungal diseases

This guide will also assist you to attain the learning outcomes stated in the cover page. Specifically, upon completion of this learning guide, you will be able to:

- Explain Biology and classification of funguses
- Identify Aetiology, transmission, host preference, clinical sign, diagnosis, treatment and prevention/control measures of economically important diseases of animals' caused by fungal organisms

Learning Instructions:

1. Read the specific objectives of this Learning Guide.
2. Follow the instructions described below.
3. Read the information written in the information Sheets
4. Accomplish the Self-checks

Information sheet 4

4.1 Biology and classification of fungi

Mycology is the study of fungi. Fungi have several features that distinguish them from other organisms. They have a filamentous branching of system of cells with apical growth, lateral branching and heterotrophic nutrition. They are characterized by a life-cycle that begins with germination from spore or a resting structure, followed by a period of growth the substrate is exploited to produce a biomass. Finally, there is a period of sporulation where propagules are formed that can be disseminated from the parent mycelium.

Importance of Fungi

They are vital to the biosphere for many reasons not the least, for their decomposing activities on dead substrate that ensure the release of nutrients like carbon, minerals and nitrogen back into the atmosphere (i.e. act as decomposers of complex organic materials in the environment)

They are major cause of diseases in plant animals and man. Fungi, especially the yeasts are essential to many industrial processes involving fermentation e.g. bread, wine and beer making.

They are important in the manufacture of many antibiotics. Fungi are important research tool in the study of fundamentals of biological process.

Sexual Reproduction in Fungi

Involves the union of compatible nuclei. Some fungal are self-fertilizing and produce asexually compatible gametes on the same mycelium- homothallic. Other species require out crossing between different but sexually compatible mycelia heterothallic.

Sexual reproduction yields spores. For example in the zygomycetes the zygote develops a zygospore, in the ascomycetes, an ascospore, in the basidiomycetes a basidiospore.

Fungal spores are important for several reasons:

- The size, shape, colour and number are useful in the identification of fungal species.
- For fungal dissemination

Fungi metabolism

Fungi prefer moist habitat and are largely mesophyllic preferring temperature between 15-35⁰C. The carbon needs of fungi for energy metabolism and biosynthesis has to be met heterotrophically by one of three life styles.

- Parasitism of plants and animals- causing diseases.
- Saprophytism- growing on dead animal, plant or microbial biomass.
- Symbiosis- growing together with algae, plants or insects.

Structure

- ✓ The body or vegetative structure of fungus is called a thallus (thalli).
- ✓ It varies in complexity and size ranging from single cell microscopic yeasts to multicellura molds, macroscopic puffballs and mushrooms.
- ✓ The fungal cell is enclosed in a cell wall of chitin.

Classification of fungi

- Fungi are eukaryotic organisms that do not contain chlorophyll, but have cell walls, filamentous structure, and produce spores.
- These organisms grow as saprophytes and decompose dead organic matter. There are between 100,000 to 200,000 species depending on how they are classified. About 300 species are presently known to be pathogenic for man and animals
- The taxonomy of the Kingdom Fungi is evolving and is controversial. Formerly based on gross and light microscopic morphology, studies of ultra structure, biochemistry and molecular biology provide new evidence on which to base the taxonomy positions.

Medically important Fungi are in four phyla.

- ✓ Ascomycota: Sexual reproduction in a sack called an ascus with the production of ascospores
- ✓ Basidiomycota: Sexual reproduction in a sack called basidium with the production of basidiospores
- ✓ Zygomycota: Sexual reproduction by gametes and asexual reproduction with formation of zygosporangia
- ✓ Mitosporic Fungi (Fungi imperfecti): no recognizable form of sexual reproduction, includes most pathogenic Fungi.

Morphology

The two main morphological fungal forms are moulds and yeasts. Moulds grow as branching filaments called hyphae whereas the unicellular yeasts have an oval or spherical appearance. A mass of hyphae is called mycelia. Yeasts are unicellular organisms and mycelia are multicellular filamentous structures, constituted by tubular cells with cell wall. The yeasts reproduce by budding. The mycelia forms branch and the pattern of branching is an aid to the morphological identification. If the mycelia do not have septa, they are called coenocytic (nonseptate). The terms “hypha” and “mycelium” are frequently used interchangeably. Some fungi occur in both the yeast and mycelial forms. These are called dimorphic fungi.

- The yeast form usually seen in tissue, in exudates, or cultured in an incubator at 37⁰C.
- The mycelium form observed in nature or when cultured at 25⁰C. Conversion to the yeast form appears to be essential for pathogenicity.

Pathogenic fungi that cause infections:

- ✓ Most common pathogenic fungi do not produce toxins but they do show physiologic modifications during a parasitic infection. They are also thermo tolerant, and can resist the effects of the active oxygen radicals released during the respiratory burst of phagocytes. Thus, fungi are able to withstand many host defenses.
- ✓ Fungi are ubiquitous in nature and most animals are exposed to them.
- ✓ The establishment of mycotic infection usually depends on the size of the inoculums and on the resistance of the host. The severity of the infection seems to depend mostly on the immunologic status of the host.

Most mycotic agents are soil saprophytes and mycotic diseases are generally not communicable (occasional exceptions: candida and some dermatophytes). Outbreaks of disease may occur, but these are due to a common environmental exposure, not communicability. Most of the fungi which cause systemic infections have a peculiar characteristic ecologic niche in nature. In this environment, the normally saprophytic organisms proliferate and develop. This habitat is also the source of fungal elements and/or spores, where man and animals, incidental hosts, are exposed to the infectious particles. It is important to be aware of these associations to diagnose mycotic diseases.

4.2 Fungal diseases

Since fungi are larger than bacteria, they were recognized as the agent of infectious diseases earlier than other infectious agents (bacteria and virus).

Nowadays, even though there are over 250, 000 fungal species, only 150 species are pathogen for animals and human beings. Infectious diseases which are caused by fungi are called **mycoses**(singular **mycosis**).

4.2.1 Dermatophytosis

It is a fungal disease of large animal characterized by invasion of cutaneous keratinized epithelial cells and hair fibers by dermatophytes.

- **Etiology** Ringworm is caused by groups of fungi imperfect belonging to genus *Microsporum*, *Trichophyton*, *Epidermophyton*. The most important genera in veterinary medicine are *Microsporum* and *Trichophyton*. Genus *Epidermophyton* is represented by single species *E. floccosum* and affects only human being.
- **Clinical findings:** The clinical findings are almost similar in different species of large animals and have the following features. The typical lesion is a heavy, gray-white crust raised perceptibly above the skin. The lesions are roughly circular and about 3 cm in diameter. In the early stages the surface below the crust is moist. However, in older lesions the scab becomes detached and **pityriasis** and **alopecia** may be the only obvious abnormalities. Lesions are most commonly found on the neck, head and perineum. But a general distribution over the entire body may occur particularly in calves. In severe cases the lesions may coalesce together. Itching does not occur and secondary acne is unusual.
- **Diagnosis** Diagnosis of ringworm in animals is done based on Epidemiology of the disease. Clinical signs of the disease (the appearance of characteristic lesions), Result of laboratory findings. Diagnostic confirmation is done in laboratory by demonstration of fungal elements in a scraping or biopsy which include Demonstration of spores and mycelia in skin scrapings and in culture. Isolation of culture of fungi on/in artificial media. Specimens for fungi isolation and identification in dermatophytosis is skin scrap that are taken peripherally from animals those were not treated.

- **Treatment:** Treatment is widely practiced and recommended. Because it greatly reduces the contamination of the environment by infected animals. Depending on the size of the lesion treatment of ringworm can be
 - ✓ **Local treatment:** It can be Local application, local spray or washes, local application is topical application of medicaments. Local application is widely used when the lesions of the skin are localized. Topical treatment is probably of greater value in the early stages of an outbreak when the lesions are small and few in numbers. The crusts should be removed by scraping or brushing with a soft wire brush and burned. The medicament should be brushed or rubbed in vigorously. The suitable medicaments that are used for topical applications include a weak solution of iodine, Whitfield's ointment, 10 % ammoniated mercury ointment
 - ✓ **Systemic treatment:** Systemic treatments recommended for use in farm animals include the intravenous injection of 10% sodium iodide at the dose of 1 g/14 kg body weight repeated on several occasions. If the cost of the drug is not problem oral administration of griseofulvin has high efficacy.
- **Control and prevention measures:** Control and prevention measures of ringworm can be achieved by Hygiene, Vaccination, Nutrition. Failure to control an outbreak of ringworm is usually due to the widespread contamination of the environment before treatment is attempted. The following practices help to control ringworm
 - ✓ Isolation and treatment of infected animal
 - ✓ The provision of separate grooming tools, horse blankets and feeding utensils
 - ✓ Disinfection of these items after use on affected animals
 - ✓ The following chemicals are used to clean and to disinfect the fungi
 - ✓ Commercial detergent or a strong solution (2.5-5 %) of phenolic

4.2.2 Aspergilosis

- **Synonyms:** Aspergillomycosis, Bronchomycosis, Brooder pneumonia, Pneumonomycosis.

It is an infectious fungal saproozoonosis.

- **Etiology:** Principally caused by *Aspergillus fumigatus*; rarely other species such as *A. deflexus*, *A. flavus*, *A. nidulans*, *A. niger*, *A. terreus* are also found implicated with

disease. These fungi are ubiquitous in environment and grow as saprophytes. On decaying vegetation, compost pile, litter, hay, grains, dead leaves, Soil and other organic matters.

- **Host:** cattle, Avians, buffalo, camel, sheep, man
- **Transmission:** Man and animals (including birds) acquire the infection through respiratory route by inhaling large number of infectious spores of *Aspergillus* from the saprobic reservoirs. Accidentally, infection can be introduced following a trauma to the skin by a contaminated object. Nosocomial infections
- **Symptoms:**
 - ✓ **Man:** Low grade fever, productive cough, breathlessness, haemoptysis, malaise, prostration, weight loss; otomycosis, sinusitis, cellulitis, onychomycosis, keratitis, myocarditis, endocarditis and fungus ball (aspergilloma) in lung; invasive disease occurs in severely immunocompromised person.
 - ✓ **Animals:** Mammals: Pyrexia, nasal discharge, moist cough dyspnoea, sinusitis, rhinitis, otitis, corneal ulcer, endophthalmitis, pneumonia, enteritis; epistaxis, dysphagia, necrosis and haemorrhage in guttural pouch (horse), abortion, placentitis, mastitis, cystitis and diskospondylitis (dog).
 - ✓ **Avians:** Dullness, anorexia, dyspnoea. Rise in temperature, nasal discharge, oedema of eyelid, lachrymation, foetid diarrhoea, torticollis, ataxia, incoordination, convulsion and death.
- **Diagnosis:**
 - ✓ Direct microscopic examination of bronchial aspirate, sputum, pus, corneal scrapings, tissue biopsies etc. in 10-15 KOH for the presence of 2-4 μ m septate and branched hyphae.
 - ✓ Histopathological demonstration of dichotomously branched and septate hyphae by GMS, H and E, GMS-H and E, Brown-Hopps methods.
 - ✓ Repeated isolation of fungus in pure and luxuriant growth from clinical specimens on SDA with antibiotics at 37⁰C. Serological tests like AGD, CIE and ELISA for detecting antibodies against *Aspergillus* infection
- **Treatment:**

- ✓ Surgical removal of fungus ball of the lung by lobectomy.
- ✓ Combined therapy with Amphotericin B (0.2-1.0 mg/kg IV) and Flucytosine (25-100 mg/kg PO) in nonneurotraptic patients with invasive disease.
- ✓ Intraconazole (200 mg 12h PO) may help in less immuno-compromised patients with slowly progressive invasive aspergillosis.
- ✓ Posaconazole and Voriconazole are recently introduced drugs that may be tried in clinical cases of aspergillosis

• **Control and production:**

- ✓ Persons working in poultry industries should use face mask. Do not handle mouldy feed or bedding.
- ✓ Proper disposal of *Aspergillus* contaminated litter.
- ✓ Periodical spraying of 1% copper sulphate solution in poultry pen, incubator, brooder, feed hopper, water utensils and other equipment to kill fungal spores.

4.2.3 Epizootic Lymphangitis

- **Synonym:** pseudo glanders

It is a chronic contagious disease of horses characterized by supportive lymphangitis, lymphadenitis and ulcers of the skin

- **Etiology:** *Histoplasma farciminosum* is the cause of the disease. It is fungal disease.
- **Occurrence:** The disease occurs chiefly in Asia, Africa and the Mediterranean. It occurs in out breaks rather than as an endemic disease.
- **Species affected** -Horses and rarely cattle and man are the species affected. Horses under 6 years of age are the most susceptible,
- **Transmission:** from infected animals by direct contact or on inanimate objects such as bedding, grooming utensils, horse blankets or harness, and gain entry through cutaneous abrasions.
- **Clinical Findings:** ulcer develops at the portal of entry, lymphatic vessels leaving the ulcer become thickened and enlarged and develop nodules along their course. The nodules erupt and discharging thick creamy pus. Local lymph nodes also enlarge and may rupture. Thickening of the skin in the area and general swelling of the whole limb are common. The lesions are quite painless. In most cases the lesions develop on the

limbs particularly about the hocks but may also be present on the back, sides, neck, vulva and scrotum. Occasionally, lesions appear on the nasal mucosa but are situated inside the nostrils and not involved in the nasal septum. The disease is chronic, persisting for 3-12 months and affected animals lose body condition and cannot be worked. Spontaneous recovery occurs and immunity is solid after an attack but many animals are destroyed because of the chronic nature of the diseases.

- **Necropsy Findings:** Lesions are usually confined to the skin, subcutaneous tissues and lymph vessels and lymph nodes
- **Diagnosis:** microscopic examination from the discharge reveals Gram-positive yeast cells, Culturing. Serological tests like FAT can be used.
- **Differential Diagnosis:** Glanders, ulcerative lymphangitis.
- **Treatment:** Early cases can be cured by extensive excision of affected parts followed by frequent local application of silver nitrate or tincture of iodine
- **Control** - Strict hygienic precautions must be observed.
 - ✓ In endemic areas severe cases should be destroyed and less severe cases kept in strict quarantine while undergoing treatment.
 - ✓ All contaminated bedding, harness and utensils should be destroyed or vigorously disinfected.

4.2.4 Candidiasis

- **Etiology:** Candidiasis is caused by various species of the fungal genus *Candida*, including *Candida albicans* and *Candida glabrata*. These fungi are commonly found on the skin and mucous membranes of healthy animals and humans but can cause disease in susceptible individuals.
- **Transmission:** Candidiasis transmission occurs through direct contact with infected individuals or environments contaminated with the fungus. Overgrowth of yeast can be caused by various factors including prolonged antibiotic therapy, immune suppression, or alteration in normal skin or mucosal flora.
- **Host preference:** Candidiasis has been reported in a variety of animal species, including dogs, cats, horses, birds, reptiles, rodents, and primates.

- **Clinical signs:** Clinical signs of candidiasis in animals depend on the location of infection and severity of disease. Skin infections can present as redness, scaling, and crusting. Oral infections can present as white, patchy plaques on mucous membranes. Infections involving the digestive tract may cause diarrhea, abdominal pain or vomiting.
- **Diagnosis:** Diagnosis of candidiasis is typically based on clinical presentation and characteristic microscopic appearance of the yeast cells. Scrapings or smears from affected sites can often confirm diagnosis.
- **Treatment:** Treatment is dependent on the site and severity of infection. Treatment typically involves topical or systemic antifungal medications and addressing any underlying predisposing factors.
- **Prevention/control measures:** Minimizing exposure to candidal organisms in high-risk situations, such as immuno-compromised animals or those undergoing antibiotic therapy, can be achieved with good hygiene practices, such as effective disinfection protocols and taking preventative measures against predisposing factors.

4.2.5 Histoplasmosis

Histoplasmosis is a fungal infection caused by the *Histoplasma* species.

- **Etiology:** *Histoplasma* is a dimorphic fungus that grows in soil and bird droppings. The fungus can exist in two forms: as a mold in the soil and as yeast in the host organism. When soil containing the fungus is disturbed, spores are released into the air, and when inhaled, can cause infection.
- **Transmission:** Histoplasmosis infection occurs when animals inhale fungal spores or dust particles contaminated with spores. The spores can also enter the body through mucous membranes or broken skin.
- **Host preference:** Histoplasmosis has been reported in a wide variety of animal species, including dogs, cats, horses, cattle, birds, rodents, bats, and primates.
- **Clinical signs:** The clinical signs of histoplasmosis can vary depending on the severity of the infection and the species affected. Acute infections may cause fever, anorexia, lethargy, and coughing. Chronic infections may present with weight loss, anemia, and respiratory problems.

- **Diagnosis:** Diagnosis of histoplasmosis is based on clinical signs and laboratory tests such as radiography and blood tests. Fungal culture is considered the gold standard for diagnosis.
- **Treatment:** Treatment depends on the extent of infection and clinical signs. In some cases, antifungal medications may be necessary to treat severe infections. Supportive care such as fluid therapy and hospitalization may be necessary for severely affected animals.
- **Prevention/control measures:** The most effective means of prevention is to minimize exposure to contaminated environments. This can be achieved through good hygiene practices such as proper waste disposal. In addition, antifungal medications can prevent dissemination of the fungus from infected individuals to healthy individuals in specific situations such as primate colonies.

Self-Check – 4	Written test
----------------	--------------

Name _____ ID _____ signature _____ date _____

Directions: Answer all the questions below according to the instruction

Part I: Multiple choices (1pt each)

1. Chronic contagious disease of horses characterized by supportive lymphangitis, lymphadenitis and ulcers of the skin

- A. Candidiasis
- B. Histoplasmosis
- C. Epizootic Lymphangitis
- D. Aspergilosis

2. Fungal disease of large animal characterized by invasion of cutaneous keratinized epithelial cells and hair fibers by.

- A. Dermatophytosis
- B. Histoplasmosis
- C. Epizootic Lymphangitis
- D. Aspergilosis

Part II: Short Answer Questions (4pts)

1. Write general characteristics of Fungi
2. List diseases caused by fungi with their causative agent and Prevention/control measures

Note: Satisfactory rating -10 points, Unsatisfactory -10 below

Reference Materials

Books

- Andrews A.H., R.W. Blowey, H. Boyd and R.G. Eddy (2004). *Bovine Medicine Diseases and Husbandry of Cattle* 2nd edition
- Carlile, M. J., Watkinson, S. C., & Gooday, G. W. (2001). *The fungi*. Gulf Professional Publishing.
- Drug Administration and control authority in Ethiopia (2006). *Standard Veterinary Treatment guide line for veterinary clinics* 1st edition.
- Fenner, F. J., McAuslan, B. R., & Mims, C. A. (2013). *The biology of animal viruses*. Elsevier.
- Fields, B. N., Knipe, D. M., Howley, P. M., Everiss, K. D., & Kung, H. J. (1996). *Fundamental virology* (pp. 401-402). Philadelphia^ ePA PA: Lippincott-Raven.
- Fischetti, V. A., Novick, R. P., Ferretti, J. J., Portnoy, D. A., & Rood, J. J. (2000). *Gram-positive pathogens*. ASM Press, American Society for Microbiology.
- Gomes, F., & Henriques, M. (2016). Control of bovine mastitis: old and recent therapeutic approaches. *Current microbiology*, 72, 377-382.
- Gyles, C. L., Prescott, J. F., Songer, J. G., & Thoen, C. O. (2010). Pathogenesis of bacterial infections in animals 4th edn. *USA: Office. p*, 267-285.
- Merck, S., & Dohme, C. (2009). *The Merck veterinary manual*. A subsidiary of Merck & Co. Inc. Whitehouse Station NJ, USA.
- Pattison, M., McMullin, P., Bradbury, J. M., & Alexander, D. (Eds.). (2007). *Poultry diseases*. Elsevier Health Sciences.
- Payne, S. (2022). *Viruses: from understanding to investigation*. Elsevier.
- Radostits, O., Blood, C., Hinchclif, K., & Constable, P. (2007). *Veterinary medicine: a text book of disease of cattle, horse, sheep, pig and goat*. Saunders, London.
- Sellon, D. C., & Long, M. T. (2013). *Equine infectious diseases E-book*. Elsevier Health Sciences.

Songer, J. G., & Post, K. W. (2004). *Veterinary microbiology-E-book: bacterial and fungal agents of animal disease*. Elsevier Health Sciences. Thomas J. Divers and Simon F. (2008). *Diseases of dairy cattle*, Elsevier Inc 2nd edition

Victoria, J. G., Kapoor, A., Dupuis, K., Schnurr, D. P., & Delwart, E. L. (2008). Rapid identification of known and new RNA viruses from animal tissues. *PLoS pathogens*, 4(9), e1000163.

World Health Organization (2008). *Anthrax in humans and animals*.

Web address

Bacterial cell structure available at; Bacteria cell structure 365530 Vector Art at Vecteezy (accessed on May 19/2023).

Candidiasis available at; Merck Veterinary Manual (merckvetmanual.com) Candidiasis in Animals - Generalized Conditions - MSD Veterinary Manual (msdvetmanual.com) (accessed on may 19/2023).

Anthrax in animals available at Anthrax factsheet | Anthrax in animals | Important animal diseases | Animal diseases | Biosecurity | Agriculture Victoria (accessed on may 24/2023 <https://www.google.com/search?q=actinomycosis+in+bovine&client=firefox-b-d&source=lnms&tbn=isch&sa=X&ved=2ahUKEwicm> (accessed on may 19/2023)).

AKNOWLEDGEMENT

Ministry of Labor and Skills wish to extend thanks and appreciation to the many representatives of TVET instructors and respective industry experts who donated their time and expertise to the development of this Teaching, Training and Learning Materials (TTLM).

The experts who developed the learning guide

No	Name	Qualification	Educational background	Region	Phone number	E-mail
1	Dr. Zenebe Tekle	DVM, MSc	Veterinary Medicine	Alage	0962230227	teklezenedvm@gmail.com
2	Dr. Yeshiwas Tarekegn	DVM, MSc	Veterinary Medicine	Alage	0910749374	yeshiwastarekegn@gmail.com
3	Dr. Sileshi Aregahegn	DVM, MSc	Veterinary Medicine	Kombolcha	0920480599	kochasile@gmail.com
4	Dr. Hirpa Bubaso	DVM, MSc	Veterinary Medicine	Alage	0991452566	Hirpabobaso17@gmail.com
5	Dr. Aynalem Shibabaw	DVM	Veterinary Medicine	Alage	0928580943	ayenalemshibabaw@yahoo.com
6	Dr. Abate Worku	DVM, MSc	Veterinary Medicine	Holeta	0923382973	Abeworku48@gmail.com
7	Dr. Bereket Fekadu	BVS, MSc candidate	Animal Health	Wolaita	0916706332	berfeka21@gmail.com
8	Dawit Shewarega	BVsc	IT	Alage	0913672115	dshewa22@gmail.com